

INSTANTON AND PILLOWCASE HOMOLOGY OF THE $(-2, 3, q)$ PRETZEL KNOTS

BERND JOHANNES WUEBBEN

ABSTRACT. For every odd $q \geq 3$ the reduced singular instanton homology of the pretzel knot $P(-2, 3, q)$ is free abelian of rank $q + 2$, and for $q \geq 7$ we record its $\mathbb{Z}/4$ -graded ranks. We study two decompositions of these knots along a Conway sphere. In the first, the tangle that Hedden, Herald and Kirk use for the torus knot $T(3, 5)$ is reglued to the trivial tangle by $(q - 5)/2$ Dehn twists. For small holonomy perturbations and zero bounding cochain we compute the pillowcase Floer homology of this decomposition by hand and show that it is isomorphic to the instanton homology as a $\mathbb{Z}/4$ -graded vector space; the Floer differential is nonzero for $q \geq 9$. This verifies the conjecture of Hedden, Herald and Kirk for a family of hyperbolic knots outside the two-bridge family, and the same argument applies to the twisted torus knots $T(3, n; 2, m)$ with $n \equiv 1, 2 \pmod{6}$. In the second decomposition, Smith's Conway sum, the pairing with zero bounding cochain has the wrong rank. In the two-arc algebra of Kotelskiy, Watson and Zibrowius, where the Maurer–Cartan equation is a finite identity, we show that at $q = 7$ smoothings of a model of Smith's curve give Maurer–Cartan elements with the correct pairing, and that a second closure selects one homotopy class among them, conditionally on the instanton–pillowcase correspondence and a localization hypothesis. For $q = 5, 7$ the selected object is the sum of a rational arc and a closed curve with a two-dimensional local system, and one further term near a corner carries it to the Bar-Natan invariant of the tangle; no rational closure distinguishes the two. We ask whether the instanton object of the tangle is its Bar-Natan invariant, and we record observations at $q = 7$ and $q \geq 11$ that bear on the question.

1. INTRODUCTION

1.1. Two homologies for one knot. Let $K \subset S^3$ be a knot. On the gauge-theoretic side sits the reduced singular instanton knot homology $I^\natural(K)$ of Kronheimer and Mrowka [KM11a, KM11b]. Its rank is bounded below by the coefficient norm of the Alexander polynomial and above by the rank of reduced Khovanov homology (3.1). On the symplectic side sits the pillowcase program of Hedden, Herald and Kirk [HHK14, HHK18]. Write $K = (Y, T) \cup (D, U)$ as the union of a tangle and the trivial tangle along a Conway sphere. The traceless $\mathrm{SU}(2)$ character variety of the four-punctured sphere is the *pillowcase*, a two-sphere with four orbifold points, and after a holonomy perturbation each side determines an immersed curve in its smooth part: the tangle side a curve L_1 , and the trivial side, with an earring, the curve L_0 . Their Lagrangian Floer homology $H^\natural(Y, T, \pi)$ is defined when the pair is admissible, and Hedden, Herald and Kirk conjecture that for suitable perturbations it is isomorphic to $I^\natural(K; \mathbb{F}_2)$ as a $\mathbb{Z}/4$ -graded vector space [HHK18, Conj. 6.5] (Conjecture 2.5). For general decompositions figure-eight bubbles at the self-intersections of the curves deform the differential. Cazassus, Herald, Kirk and Kotelskiy conjecture that each tangle is assigned an immersed curve together with a bounding cochain, and that the deformed theory recovers $I^\natural(K)$ [CHKK22, Conj. D]; conditional on this conjecture,

Date: First version: July 27, 2026. This version: October 3, 2026.

2020 Mathematics Subject Classification. 57K18 (primary); 57R58, 53D40, 57K10 (secondary).

Key words and phrases. Singular instanton homology, pillowcase, Lagrangian Floer homology, pretzel knots, twisted torus knots, immersed curves, Khovanov homology, Maurer–Cartan elements.

Smith proves that the cochain on one Conway-sum tangle in $P(-2, 3, 5)$ cannot vanish [Smi24, proof of Thm. 5.9].

This paper studies both sides for the pretzel knots $K_q = P(-2, 3, q)$, $q \geq 3$ odd (2.8). The smallest members are the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$, whose instanton and pillowcase homology are treated in [Wue26a, Wue26b, Wue26c], and for $q \geq 7$ the knot K_q is hyperbolic. We use two decompositions of K_q . In the first, the tangle of Hedden, Herald and Kirk for $T(3, 5)$ is reglued to the trivial tangle by Dehn twists along the Conway sphere. There the curves need no bounding cochain, and we prove the conjecture of Hedden, Herald and Kirk for every K_q , and for a family of twisted torus knots (Theorems 1.2 and 1.3). The second is Smith's decomposition $K_q = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/q})$, with the earring on the -2 band. There the pairing with zero bounding cochain has the wrong rank. For $q = 5, 7$ we construct Maurer–Cartan elements in the two-arc algebra of Kotelskiy, Watson and Zibrowius [KWZ19] that correct it, and we identify the resulting objects with the Bar-Natan invariant of the tangle $Q_{1/3} + Q_{1/q}$ up to one further term (Theorems 1.4 and 1.5).

1.2. The instanton side.

Theorem 1.1. *For every odd $q \geq 3$, $I^\natural(K_q; \mathbb{Z}) \cong \mathbb{Z}^{q+2}$. For $q \geq 7$ the signature is $\sigma(K_q) = -(q+1)$, and in Kronheimer and Mrowka's absolute grading the graded ranks of $I^\natural(K_q)$ are*

$$(1 + N_3, N_1, N_1, N_3) \quad \text{in degrees } \text{gr} - \sigma(K_q) = 0, 1, 2, 3, \quad N_1 = \lceil \frac{q+1}{4} \rceil, \quad N_3 = \lfloor \frac{q+1}{4} \rfloor.$$

The rank over $\mathbb{Q}$ is due to Lobb and Zentner [LZ20, Prop. 4.7]. For $q \geq 7$ the freeness and the graded ranks follow from the computation of the irreducible instanton homology of K_q by Daemi and Scaduto [DS24b, Prop. 8.30] together with their integral spectral sequence (Proposition 3.6). Our proof of freeness is uniform in q : it runs Lobb and Zentner's squeeze with integer coefficients, using Manion's integral computation of Khovanov homology [Man18] and the integral spectral sequence of Kronheimer and Mrowka. For $q = 3, 5$ the graded ranks are those of the torus knots in [Wue26c]. Section 3 gives the proof and Remark 3.4 the remaining attributions.

1.3. The conjecture of Hedden, Herald and Kirk for the pretzel knots. Hedden, Herald and Kirk compute the pillowcase homology of $T(3, 5)$ from a tangle (Y, T) whose complement has free fundamental group of rank two, glued to the trivial tangle [HHK18, §§10–11]. Let t_e be the Dehn twist along the curve of the Conway sphere that separates two of the punctures from the other two. Regluing by t_e^{-m} adds m full twists on two strands, and the resulting knot is K_{5+2m} (Theorem 4.4). This uses the convention, shared with [HHK18] and [Wue26c], that the original gluing gives the positive torus knot $T(3, 5)$ (Remark 4.3). The holonomy perturbations on the tangle side are those of [Wue26c], with parameters $\varepsilon = (\varepsilon_A, \varepsilon_B)$ in an open cone $\mathcal{C}_a$ (§4.4).

Theorem 1.2. *Let $q \geq 7$ be odd, and let $K_q = (Y, T) \cup_{t_e^{-(q-5)/2}} (D, U)$ be the decomposition above. Fix $a > 0$. For every sufficiently small $\varepsilon \in \mathcal{C}_a \setminus \{0\}$ and every sufficiently small earring parameter:*

- (1) L_1 is a restricted immersed 1-manifold, and (L_0, L_1) is admissible;
- (2) the Floer complex has $q + 2 + 4N_q$ generators, where $N_q = \lfloor 5(q-6)/12 \rfloor - \lfloor (q-6)/12 \rfloor$;
- (3) the differential has rank $2N_q$, and it is nonzero for $q \geq 9$;
- (4) with the absolute grading normalized by $\text{gr}(r_+) = \sigma(K_q)$, $H^\natural(Y, T, \pi) \cong I^\natural(K_q; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces.

In particular, the conjecture of Hedden, Herald and Kirk [HHK18, Conj. 6.5] holds for every K_q .

The perturbation can resolve the double points of the curve in two ways, joined for $|\varepsilon_B| < \sqrt{3}|\varepsilon_A|$ and split for $|\varepsilon_B| > \sqrt{3}|\varepsilon_A|$; both occur in $\mathcal{C}_a$ and the theorem holds in both. For $q \geq 9$ the complex is not minimal. Hedden, Herald and Kirk verify the conclusion of their conjecture

for two-bridge knots [HHK18, §7], a family that contains hyperbolic knots, and for $T(3, 4)$ and $T(3, 5)$ [HHK18, §§10–11]; the torus knots $T(3, n)$ are treated in [Wue26c]. Fukumoto, Kirk and Pinzón-Caicedo describe the unperturbed character variety of the decomposition used here and count 9 generators for $P(-2, 3, 7)$ [FKP17, §5] (Remark 4.19). For $q \geq 7$ the knot K_q is hyperbolic and not two-bridge (Proposition 4.5). To our knowledge, these are the first hyperbolic knots outside the two-bridge family for which the conjecture is verified, with the $\mathbb{Z}/4$ gradings and nonzero Floer differentials.

The same regluing applies to the tangles of [HHK18, §10] for $T(3, n)$. The *twisted torus knot* $T(3, n; 2, m)$ is the closure of the braid $(\sigma_1 \sigma_2)^n \sigma_1^{2m}$.

Theorem 1.3. *Let $n \geq 7$ with $n \equiv 1$ or $2 \pmod{6}$, let $m \geq 1$, and let $K = T(3, n; 2, m)$. Then $\sigma(K) = \sigma(T(3, n)) - 2m$, and $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$. The tangle of [HHK18, §10] for $T(3, n)$, reglued by t_e^{-m} , is a decomposition of K , and for small holonomy perturbations and small earring parameters it satisfies the conclusion of [HHK18, Conj. 6.5]: $H^\natural \cong I^\natural(K; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces.*

Theorem 5.1 gives the precise form, with the graded ranks. The family of Theorem 1.3 contains torus knots, $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$, and by braid conjugacy $T(3, 6j + 2; 2, m) = T(3, 6j + 1; 2, m + 1)$ (Lemma 5.3). The other members, $T(3, 6j + 1; 2, M)$ with $j \geq 1$ and $M \geq 2$, are not torus knots, not alternating and not pretzel knots $P(-2, 3, q)$ (Lemma 5.6). We make no claim about their hyperbolicity.

1.4. Smith’s decomposition. In Smith’s decomposition the earring tangle is $\widehat{Q_{-1/2}}$ and the other tangle is $T_q = Q_{1/3} + Q_{1/q}$. We work in Smith’s coordinates on the pillowcase, with his holonomy perturbation for small negative values of its parameter t (Remark 6.2). Following Cazassus, Herald, Kirk and Kotelskiy [CHKK22, §11], we encode curves as twisted complexes over the two-arc algebra $\mathcal{B}$ of Kotelskiy, Watson and Zibrowius. The higher products of $\mathcal{B}$ vanish, so the Maurer–Cartan equation for a deformation b of a twisted complex (V, δ) is the finite identity $(\delta + b)^2 = 0$. We write N_q for the twisted complex of a piecewise-linear model of Smith’s curve for T_q and $\mathcal{E}$ for that of the earring curve.

For $q = 7$ the pairing of $\mathcal{E}$ with N_7 has dimension 7, not $9 = \dim I^\natural(K_7)$ (Proposition 6.7). Smoothing the model at any one of four of its self-intersections gives a Maurer–Cartan element b_s that raises the dimension to 9, and the four objects (N_7, b_s) fall into three homotopy classes (Corollary 6.9). Replacing $\widehat{Q_{-1/2}}$ by $\widehat{Q_{-3/4}}$ closes T_7 into a second knot, whose instanton homology is free of rank 31 by the same squeeze, and the pairings with the second earring separate the three classes (Proposition 6.8). Conditional on the conjecture of Cazassus, Herald, Kirk and Kotelskiy and on a localization hypothesis (Hypothesis 6.13), the two closures together select one class, that of $b_{S_{18}}$ and $b_{S_{25}}$ (Corollary 6.14). For positive t two of the four smoothings give 5 and 7 instead of 9, and the two-closure selection holds for both signs (Remark 6.17). We also show that Gao’s wrapped representability theorem [Gao17] cannot be applied to the Lagrangian correspondence induced by the line of $Q_{1/3}$, which is immersed with a triple point (Proposition 6.4).

Kotelskiy, Watson and Zibrowius assign to T_q its Bar-Natan invariant $\widetilde{\text{BN}}(T_q)$, an immersed multicurve with local systems in the four-punctured sphere [KWZ19]. Drawn in Smith’s coordinates it becomes an object BN_q of $\text{Tw}(\mathcal{B})$ (§7.1). For $r \in \mathbb{Q} \cup \{\infty\}$ let $K_r(q) = \text{num}(Q_r + T_q)$, and let E_r be the earring curve of $\widehat{Q_r}$, so that $K_{-1/2}(q) = K_q$. A compact curve in the pillowcase is *linear* if its lifts to the plane lie close to parallel straight lines and are graphs over them (Definition 7.4). For $q \in \{5, 7\}$ let b be one of the selected elements $b_{S_{18}}, b_{S_{25}}$ at $q = 7$, or at $q = 5$ the switch b_5 of a smoothing that raises the pairing with $E_{-1/2}$ from 5 to 7 (§7.2).

Theorem 1.4. *Let $q \in \{5, 7\}$.*

- (1) The object (N_q, b) is strictly isomorphic to $\alpha_q \oplus (c, J_2)$, where α_q is a linear arc between two corners and (c, J_2) is a linear simple closed curve c carrying the indecomposable two-dimensional local system J_2 .
- (2) There is a term $\kappa_q \in \text{End}(N_q)$, which replaces one arrow of the differential near the corner $(0, 0)$, such that $b + \kappa_q$ is a Maurer–Cartan element and $(N_q, b + \kappa_q)$ is strictly isomorphic to BN_q .

For every odd q with $5 \leq q \leq 21$, BN_q is an arc obtained from α_q by sliding one of its ends along c .

Theorem 1.5. *Let $5 \leq q \leq 21$ be odd, and let R be a linear curve parallel neither to α_q nor to c . Then the minimal intersection numbers satisfy $i(R, \text{BN}_q) = i(R, \alpha_q) + 2i(R, c)$, and if R is bigraded, $\dim HF(R, \text{BN}_q) = \dim HF(R, \alpha_q \oplus (c, J_2))$.*

Corollary 1.6. *For $q \in \{5, 7\}$ and every $r \in \mathbb{Q} \cup \{\infty\}$,*

$$\dim HF(E_r, (N_q, b)) = \dim HF(E_r, (N_q, b + \kappa_q)) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2).$$

Thus no rational closure distinguishes (N_q, b) from BN_q ; curves parallel to c do (Remark 7.8). Section 7 states these results in precise form (Theorems 7.1 and 7.5, Corollary 7.6).

Question 1.7. Let $q \geq 5$ be odd and prime to 3. If the tangle assignment of Cazassus, Herald, Kirk and Kotelskiy is defined for T_q with Smith’s perturbation for small $t < 0$, is the object of $\text{Tw}(\mathcal{B})$ that it determines homotopy equivalent to BN_q ?

The evidence for $q = 5, 7$ is Theorem 1.4 and Corollary 1.6. Two observations point the other way. First, BN_7 is not homotopy equivalent to any object allowed by Hypothesis 6.13 (Proposition 7.10). An affirmative answer for $q = 7$ would therefore show that the hypothesis fails, and no rational closure could detect the failure. Second, for $q = 11, 13$ the encoding of a numerical trace of Smith’s curve pairs with $E_{-1/2}$ to $q + 4$, and for $q = 17, 19$ to $q + 8$, against $\text{rank } \widetilde{Kh}(K_q) = q + 2$; in finite searches no deformation by a few switches reaches BN_q (Remark 7.11). These searches do not exclude a larger bounding cochain, and we pose the question without conjecturing an answer.

1.5. Outline of the arguments. *Theorem 1.2.* The regluing replaces L_1 by its image under the linear shear $(\gamma, \theta) \mapsto (\gamma, \theta + (q - 5)\gamma)$ of the pillowcase (Lemma 4.6). The unperturbed character variety of the $T(3, 5)$ tangle is the union of an arc and a central circle meeting it in two points. A word in the tangle group whose trace is constant on the central circle shows that its image in the pillowcase is a union of straight segments (Lemma 4.7 and Corollary 4.8). The perturbed curves are then those of [Wue26c], sheared, and on them we compute the intersection points with L_0 , their $\mathbb{Z}/4$ gradings, and the bigons by hand. The bigons come in zigzags along the strands of the earring, which gives the rank $2N_q$ of the differential (Lemma 4.16). The identification of the reglued knot uses only its fundamental group: the relator is that of Clay and Watson’s presentation of the twisted torus knots [CW13, Prop. 22] (Lemma 4.2), two-generator knots are prime [Nor82], knots are determined by their complements [GoLu89], and an induction on the signature fixes the chirality.

Theorem 1.3. For $n \equiv 1, 2 \pmod{6}$ the unperturbed variety is an arc with a straight image and $2\lfloor n/6 \rfloor$ circles that miss the edges of the pillowcase [Wue26c]. The sheared arc meets L_0 in $1 + 2M$ points and carries no bigon, and each circle contributes one dimension in each degree to the homology, by a theorem of Hedden, Herald and Kirk [HHK18, Thm. 5.4]. On the knot side, a closed formula for the Burau matrix of the braid gives the signature and the coefficient norm of the Alexander polynomial, and the instanton rank follows from the work of Daemi and Scaduto.

Theorems 1.4 and 1.5. All objects in the statements are reduced twisted complexes of loop type, and a strict isomorphism between two of them is decided by comparing the words of their components; $\widetilde{\text{BN}}(T_q)$ is computed with Zibrowius’s program [Zib]. For the pairing theorem we realize each curve by a geodesic in a metric of nonpositive curvature (Alexander–Berg–Bishop [AKP26]), where minimal intersection numbers are counted locally, and we use the pairing theorems of Kotelskiy, Watson and Zibrowius [KWZ19, Zib20] for bigraded curves.

1.6. Scope of claims. Theorem 1.1 is unconditional. Theorems 1.2 and 1.3 rest on the analysis of the perturbed curves of the torus-knot tangles in [Wue26c] and on the convention for the base case in Remark 4.3, which the knot group cannot detect. Numerical traces of the perturbed curves agree with them (Remark 4.21) but are not used in the proofs. Proposition 6.7, Theorems 1.4 and 1.5 and Corollary 1.6 are statements about explicitly presented objects of $\text{Tw}(\mathcal{B})$; they neither assume nor prove the instanton–pillowcase correspondence, and they do not identify the object that the conjecture of Cazassus, Herald, Kirk and Kotelskiy assigns to T_q . The piecewise-linear model represents Smith’s perturbed curve in the sense of Proposition 6.1. Corollary 6.14 is conditional on that conjecture and on Hypothesis 6.13. Statements labelled *Computation* are outputs of finite procedures and are recorded as data.

1.7. Organization. Section 2 recalls the pillowcase Floer theory of Hedden, Herald and Kirk, Smith’s coordinates and the two-arc algebra. Section 3 proves Theorem 1.1. Section 4 proves Theorem 1.2, and Section 5 proves Theorem 1.3. Section 6 treats Smith’s decomposition with zero bounding cochain and the Maurer–Cartan elements at $q = 7$, and Section 7 relates them to the Bar-Natan invariant. Section 8 discusses related work and open problems, and Appendix A prints the remaining data and describes the computations.

Acknowledgements. The pillowcase model and the geometry used here are due to Hedden, Herald and Kirk [HHK14, HHK18], Cazassus, Herald, Kirk and Kotelskiy [CHKK22], Herald and Kirk [HK24], Fukumoto, Kirk and Pinzón-Caicedo [FKP17], and Smith [Smi24]. The immersed-curve invariants are those of Kotelskiy, Watson and Zibrowius [KWZ19], computed with Zibrowius’s program [Zib]. The closed-form Khovanov input is Manion’s [Man18], the Alexander polynomials belong to Hironaka’s family [Hir01], the presentation of the twisted torus knot groups is Clay and Watson’s [CW13], the rational rank of the instanton homology is Lobb and Zentner’s [LZ20], and the integral and graded instanton computations for $q \geq 7$ and for twisted torus knots also follow from the work of Daemi and Scaduto [DS24b].

2. BACKGROUND

This section fixes notation and collects the results from the literature that are used later. §§2.1–2.4 recall the Lagrangian Floer theory of Hedden, Herald and Kirk in the pillowcase [HHK18], which is the setting of Sections 4 and 5; results of [HHK18] are cited by their numbering in arXiv:1501.00028v1. §§2.5 and 2.6 recall Smith’s description of tangle sums [Smi24] and the algebraic model of the wrapped Fukaya category of the pillowcase due to Kotelskiy, Watson and Zibrowius [KWZ19] and to Cazassus, Herald, Kirk and Kotelskiy [CHKK22], which are the setting of Sections 6 and 7. §2.7 fixes the conventions for the knots. We use $\text{SU}(2)$ as the group of unit quaternions, so that a traceless element is a purely imaginary unit quaternion. Coefficients are in $\mathbb{F}_2$ unless stated otherwise.

2.1. The pillowcase. Let $G \cong \mathbb{Z}^2 \rtimes \mathbb{Z}/2$ be the group of isometries of $\mathbb{R}^2$ generated by $(\gamma, \theta) \mapsto (\gamma + 2\pi, \theta)$, $(\gamma, \theta) \mapsto (\gamma, \theta + 2\pi)$ and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The *pillowcase* is the quotient $P = \mathbb{R}^2/G$, a topological 2-sphere. The quotient map $\mathbb{R}^2 \rightarrow P$ is a branched covering, branched over the images of the lattice points $(\pi\mathbb{Z})^2$, which are the four *corners* of P [HHK18, §3.1]. The rectangle

$[0, \pi] \times [0, 2\pi]$ is a fundamental domain; we call $[0, \pi] \times [0, \pi]$ the front face, $[0, \pi] \times [\pi, 2\pi]$ the back face, and the images of the lines $\gamma \in \pi\mathbb{Z}$ the left and right edges. Put $P^* = (\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2)/G$, oriented by $d\gamma \wedge d\theta$. The restriction $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2 \rightarrow P^*$ is a regular covering with deck group G , so paths, loops and immersed discs in P^* lift to $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; we work with such lifts throughout. The double cover $\mathbb{R}^2/(2\pi\mathbb{Z})^2$ of P is a torus, and P is its quotient by the elliptic involution $(\gamma, \theta) \mapsto (-\gamma, -\theta)$.

The pillowcase is the traceless character variety of the four-punctured sphere. With the presentation $\pi_1(S^2 \setminus \{a, b, c, d\}) = \langle a, b, c, d \mid ba = cd \rangle$ of [HHK14, HHK18], the assignment

$$a \mapsto \mathbf{i}, \quad b \mapsto e^{\gamma\mathbf{k}}\mathbf{i}, \quad c \mapsto e^{\theta\mathbf{k}}\mathbf{i}, \quad d \mapsto e^{(\theta-\gamma)\mathbf{k}}\mathbf{i} \quad (2.1)$$

induces a homeomorphism of P with the space $R(S^2, \{a, b, c, d\})$ of conjugacy classes of $\mathrm{SU}(2)$ representations sending each of a, b, c, d to a traceless element; the corners correspond to the abelian representations [HHK14, Prop. 3.1], [HHK18, Prop. 6.1]. We call (2.1) *HHK's coordinates*. Smith uses a different presentation of the boundary group, and hence different coordinates on the same space (§2.5). Sections 4 and 5 work entirely in HHK's coordinates and Sections 6 and 7 entirely in Smith's; the two coordinate systems are never compared.

On $\mathbb{R}^2$ we use the function

$$\varphi = \theta - \gamma.$$

Translations in G change φ by elements of $2\pi\mathbb{Z}$, and $(\gamma, \theta) \mapsto (-\gamma, -\theta)$ changes its sign. The *diagonal arc* $\Delta = \{(\gamma, \gamma) \mid \gamma \in [0, \pi]\} \subset P$ [HHK18, (12)] joins the corners $(0, 0)$ and (π, π) , and its preimage in $\mathbb{R}^2$ is the union of the lines

$$\Delta_k = \{\varphi = 2\pi k\}, \quad k \in \mathbb{Z}.$$

The lines $\{\varphi = \pi + 2\pi k\}$ are lifts of the arc of slope one on the back face, which is not part of Δ . The lattice points on Δ_k are the points $(j\pi, j\pi + 2\pi k)$, $j \in \mathbb{Z}$.

2.2. The earring curve. For $\epsilon > 0$ let $L_0 = L_0^{\epsilon, 0}: S^1 \rightarrow P^*$ be the immersion defined by

$$\tilde{L}_0(t) = \left(t + \epsilon \sin t + \frac{\pi}{2}, t - \epsilon \sin t + \frac{\pi}{2}\right), \quad t \in [0, 2\pi], \quad (2.2)$$

followed by the branched covering [HHK18, Def. 3.4, (13)]; as in [HHK18, §11] we take the perturbation function g of [HHK18, (13)] to be zero. It is the restriction to the boundary of the traceless character variety of the trivial tangle with an earring, after a small holonomy perturbation [HHK18, Thm. 6.2]. For small ϵ it is self-transverse with one double point, at $(\pi/2, \pi/2)$, it lies in an $O(\epsilon)$ -neighbourhood of Δ , and it passes around the corners $(0, 0)$ and (π, π) through the back face.

Since $\tilde{L}_0(t + 2\pi) = \tilde{L}_0(t) + (2\pi, 2\pi)$, the preimage of L_0 in $\mathbb{R}^2$ is the union of the curves

$$S_+^k = \tilde{L}_0 + (0, 2\pi k), \quad S_-^k = -\tilde{L}_0 + (0, 2\pi k), \quad k \in \mathbb{Z},$$

which we call the *strands* of L_0 . Each is a properly embedded line within $\sqrt{2}\epsilon$ of Δ_k , and a translation by $(2\pi a, 2\pi b)$ carries $S_\pm^k$ to $S_\pm^{k+b-a}$. The slope of S_+ at the point with parameter t is $(1 - \epsilon \cos t)/(1 + \epsilon \cos t)$, and that of S_- is its reciprocal. Hence on the front strip $0 < \gamma < \pi$ the strand S_+^k has slope less than 1 and S_-^k has slope greater than 1; in P these are the two branches of L_0 along the interior of Δ [HHK18, §5.1]. The two strands near Δ_k cross exactly at the points $(\frac{\pi}{2} + j\pi, \frac{\pi}{2} + j\pi + 2\pi k)$, and they pass each lattice point of Δ_k on opposite sides at distance $\sqrt{2}\epsilon$: for instance S_+^0 passes $(0, 0)$ through $(-\epsilon, \epsilon)$ and S_-^0 through $(\epsilon, -\epsilon)$, and the sides alternate from one lattice point of Δ_k to the next.

2.3. Restricted curves and the Floer complex. Let ℓ_1 be the line field of slope one on $\mathbb{R}^2$; it descends to P^* . For a path α immersed in P^* , the Maslov index $\mu(\alpha, \ell_1)$ is the signed number of times the tangent line of α passes ℓ_1 , counterclockwise passages counting $+1$, with the half-open endpoint convention of [HHK18, §2.3]. For lines ℓ_0, ℓ'_1, ℓ , pairwise transverse, the *triple index* $\tau(\ell_0, \ell'_1, \ell) \in \{0, 1\}$ is 1 exactly when the clockwise rotation of ℓ_0 to ℓ'_1 passes through ℓ [HHK18, Def. 2.5]. Let $z \in H^1(P^*; \mathbb{Z}/4)$ be the class taking the value 1 on a small counterclockwise loop around a corner [HHK18, Def. 3.1].

An immersed circle in P^* is *unobstructed* if every lift of it to the universal cover of P^* is a properly embedded line; by a lemma of Abouzaid [Abo08, Lemma 2.2] this holds if and only if it is homotopically essential and contains no *fishtail*, that is, no subinterval whose endpoints map to the same point and which maps to a nullhomotopic loop [HHK18, Def. 2.1]. A *proper immersion* of an interval is the image of a smooth immersion $I \rightarrow \mathbb{R}^2$ sending the endpoints to lattice points and the interior into $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$; its slopes at the endpoints are its *limiting slopes* [HHK18, Def. 3.3].

Definition 2.1 ([HHK18, Def. 3.6 and p. 33]). A *restricted immersed circle* is an unobstructed immersed circle $L: S^1 \rightarrow P^*$ with $\mu(L, \ell_1) + z(L) \equiv 0 \pmod{4}$. A *restricted immersed arc* is a proper immersion of an interval whose restriction to the complement of small discs around the corners contains no fishtail. A *restricted immersed 1-manifold* is a disjoint union of one restricted immersed arc and finitely many restricted immersed circles.

The earring curve L_0 is unobstructed with $\mu(L_0, \ell_1) = 0 = z(L_0)$, so it is a restricted immersed circle [HHK18, pp. 16, 18]. The image of a straight segment joining two lattice points whose interior misses the lattice is a restricted immersed arc [HHK18, p. 18].

Definition 2.2 ([HHK18, Def. 3.11]). A pair (L_0, L_1) of restricted immersed curves is *admissible* if at least one of them is a circle, if L_0 and L_1 intersect transversely, and if, whenever loops α_0 in the domain of L_0 and α_1 in the domain of L_1 have $L_0 \circ \alpha_0$ freely homotopic to $L_1 \circ \alpha_1$ in P^* , both α_0 and α_1 are nullhomotopic.

For an admissible pair, $C(L_0, L_1)$ is the $\mathbb{F}_2$ -vector space on $L_0 \cap L_1$. A *bigon* from p to q is an orientation-preserving immersion of a disc into P^* with two convex corners, at p and q , whose boundary, traversed counterclockwise, runs along L_0 from p to q and then along L_1 from q back to p ; equivalent bigons are identified [HHK18, Def. 2.10, (5)]. The differential $\partial p = \sum_q \# \mathcal{M}(p, q) q$ counts bigons from p to q modulo 2, and by [HHK18, Cor. 3.13] each $\mathcal{M}(p, q)$ contains at most one bigon. A bigon lifts to an immersed disc in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$, so its two corners lie on the same component of L_1 ; there is no differential between generators on different components [HHK18, p. 37]. We write $HF(L_0, L_1)$ for the homology.

The complex carries a relative $\mathbb{Z}/4$ -grading $\text{gr}(p, q) = \text{gr}(p) - \text{gr}(q)$, defined for generators on the same component of L_1 [HHK18, Def. 3.7]. The differential lowers it by one, and a bigon from p to q forces $\text{gr}(p, q) = 1$ [HHK18, Prop. 3.8]. If α_0 is a path in L_0 from p to q and α_1 a path in L_1 from q to p , then [HHK18, Lemma 3.10]

$$\text{gr}(p, q) = \mu(\alpha_0, \ell_1) + \mu(\alpha_1, \ell_1) + \tau(T_p L_0, T_p L_1, \ell_1) - \tau(T_q L_0, T_q L_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}. \quad (2.3)$$

Two consequences are recorded in [HHK18, §§5.1–5.2].

- (1) If L_1 meets the interior of Δ transversely at x , then for small ϵ there are exactly two generators near x : x^+ on the branch of L_0 of slope less than 1 and x^- on the branch of slope greater than 1. They satisfy $\text{gr}(x^+, x^-) = 1$.
- (2) If p_+, q_+ lie on the branch of slope less than 1 on the front face, α_0 runs along that branch, and L_1 is transverse to Δ at both points, then

$$\text{gr}(p_+, q_+) = \mu(\alpha_1, \ell_1) + z(\alpha_0 * \alpha_1) \pmod{4}, \quad (2.4)$$

and the same formula holds for two points on the branch of slope greater than 1.

In terms of lifts, $x^\pm$ are the intersections of a lift of L_1 with the strands $S_\pm^k$ near a point of Δ_k on the front strip.

Suppose L_1 is a restricted immersed arc with an endpoint at the corner $(0, 0)$ whose limiting slope is bounded away from 1. Then for small ϵ there is a unique intersection point r_+ of L_0 and L_1 converging to that corner as $\epsilon \rightarrow 0$ [HHK18, Lemma 5.1]. When L_1 comes from a knot K as in §2.4, the grading of the component containing the arc is made absolute by

$$\text{gr}(r_+) = \sigma(K) \pmod{4} \quad (2.5)$$

[HHK18, (19)]. Since $\sigma(K)$ is even, (2.5) does not depend on the sign convention for the signature. On circle components Hedden, Herald and Kirk fix the absolute grading by matching one generator with the corresponding generator of Kronheimer and Mrowka's complex [HHK18, §6.1], [KM11b, Prop. 4.4].

Theorem 2.3 ([HHK18, Thm. 5.4]). *Let $L_1: S^1 \rightarrow P^*$ be a restricted immersed circle that misses the left and right edges of the pillowcase, and let $\tilde{L}_1: [0, 2\pi] \rightarrow \mathbb{R}^2$ be a lift of $t \mapsto L_1(e^{it})$. Then its vertical degree $d = \frac{1}{2\pi}|\theta(\tilde{L}_1(2\pi)) - \theta(\tilde{L}_1(0))|$ is even, and $HF(L_0, L_1)$ has rank $d/2$ in each of the four gradings.*

Remark 2.4. By [HHK18, Thm. 4.1, Cor. 4.6], $HF(L_0, L_1)$ is independent of ϵ and, as a relatively graded group, depends only on the free homotopy classes of the circle components of L_1 and the homotopy class rel endpoints of its arc. The computations in Sections 4 and 5 are direct and do not use this invariance.

2.4. Tangle decompositions and the conjecture of Hedden, Herald and Kirk. Let $K \subset S^3$ be a knot, decomposed along a Conway sphere as $(S^3, K) = (Y, T) \cup (D, U)$, where (D, U) is the trivial 2-tangle in a ball and the common boundary is identified with $(S^2, \{a, b, c, d\})$ [HHK18, (17)]. Let $R(Y, T)$ be the space of conjugacy classes of traceless $SU(2)$ representations of $\pi_1(Y \setminus T)$, and $R_\pi(Y, T)$ its version perturbed by a holonomy perturbation π supported in the interior of Y . Restriction to the boundary gives $L_1: R_\pi(Y, T) \rightarrow P$, and the earring curve L_0 of §2.2 comes from the trivial side. When L_1 is a restricted immersed 1-manifold and (L_0, L_1) is admissible, Hedden, Herald and Kirk define

$$H^\natural(Y, T, \pi) = HF(L_0, L_1),$$

which is the direct sum of the contributions of the components of $R_\pi(Y, T)$ [HHK18, p. 34, §6.1]. The two abelian traceless representations of $\pi_1(Y \setminus T)$ are the endpoints of the arc component; the one that extends over the knot complement maps to the corner $(0, 0)$, and its perturbation gives the generator r_+ [HHK18, §6.1, §8]. The remaining generators come in pairs $x^\pm$, one pair for each transverse intersection point of L_1 with the interior of Δ . For a torus knot, with the tangle side unperturbed and a small perturbation on the side of the earring, these intersection points correspond to the irreducible traceless representations of the knot group, and there are $1 + |\sigma(K)|$ generators [HHK14, Thm. 11.1], [HHK18, §11.10].

Conjecture 2.5 ([HHK18, Conj. 6.5]). “Given a knot (X, K) in a homology 3-sphere, there exists a 2-tangle decomposition as in Equation (17), such that for suitably small generic perturbations π , $L_1: R_\pi(Y, T) \rightarrow P$ is a restricted immersed 1-manifold and $H^\natural(Y, T, \pi)$ is isomorphic to the reduced instanton homology $I^\natural(X, K)$.”

The conjecture asserts the existence of one good decomposition, and the answer does depend on the decomposition: different choices of the gluing data give different curves L_1 [HHK18, p. 55], and with the earring on the other tangle the pillowcase homology of $T(4, 5)$ has the wrong rank [CHKK22, §10.3]. Hedden, Herald and Kirk verify the conjecture for all two-bridge

knots [HHK18, §7]. For the torus knots $T(3, n)$ with $n \geq 4$ prime to 3, suitable decompositions from [HHK18, §§10–11] satisfy it, with the $\mathbb{Z}/4$ -gradings [Wue26c]. Section 4 obtains the decompositions used for $P(-2, 3, q)$ from these by regluing.

2.5. Smith’s coordinates and the sign of the perturbation. Sections 6 and 7 use the coordinates of Smith [Smi24, §2.4], in which the boundary meridians satisfy $ac^{-1} = bd^{-1}$ and are sent to

$$a \mapsto \mathbf{i}, \quad b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}, \quad c \mapsto e^{\theta \mathbf{k}} \mathbf{i}, \quad d \mapsto e^{(\gamma + \theta) \mathbf{k}} \mathbf{i}. \quad (2.6)$$

In these coordinates no holonomy-perturbed tangle character variety meets the corner (π, π) . This is [CHKK22, Prop. 11.1], which is stated in coordinates in which the excluded corner is $[\gamma, \theta] = [0, \pi]$. Its proof applies verbatim: an abelian traceless representation is determined by how the tangle joins the four boundary points, and in Smith’s coordinates the three possible joinings give the corner pairs $\{(0, 0), (0, \pi)\}$, $\{(0, 0), (\pi, 0)\}$ and $\{(\pi, 0), (0, \pi)\}$. We write Q_r for the rational tangle whose traceless character variety is the straight interval of slope r [Smi24, Prop. 2.27]. For $r = m/n$ in lowest terms, the line of Q_r joins $(\pi, 0)$ to $(0, \pi)$ if m and n are odd, $(0, 0)$ to $(0, \pi)$ if n is even, and $(0, 0)$ to $(\pi, 0)$ if m is even.

For tangles T_1, T_2 with character varieties in the pillowcase, Smith describes the character variety of the Conway sum $T_1 + T_2$ as a fiber product over γ : the θ -coordinates add, and over each *edge* $\gamma \in \{0, \pi\}$ a circle of representations appears when both summands are non-abelian there [Smi24, Thms. 3.15 and 3.26]. Smith calls such a circle *internal* when it misses the corners [Smi24, Def. 3.24]. He perturbs the sum by a holonomy perturbation in a three-boundary configuration C_3 : the summands are glued along two inner boundary spheres, with pillowcases P_1 and P_2 , and the sum is read off on the outer sphere, with pillowcase P_3 . The perturbation depends on a real parameter t . For small $t > 0$, after an arbitrarily small additional perturbation where needed, each internal circle is replaced by two arcs crossing once, joined crosswise to the rest of the curve [Smi24, Prop. 4.1, Thm. 4.22]; by the symmetry described next the same holds for small $t < 0$ (Proposition 6.1). We write $\mathcal{R}_t(T_1 + T_2)$ for the perturbed character variety of the sum, mapped to P_3 .

The two signs of t give different curves. Smith parametrizes the perturbed configuration by the images of the meridians a, b, c of the outer sphere and of a meridian x of the first inner sphere [Smi24, Prop. 4.1]. The substitution

$$\mathfrak{s}: (x, t) \mapsto (-x, -t)$$

maps the perturbed variety for t onto that for $-t$; it is the identity on P_3 and translates P_1 and P_2 by $(\gamma, \theta) \mapsto (\gamma, \theta + \pi)$. For odd q , composing $\mathfrak{s}$ with the symmetry $b \mapsto -b$ of Smith’s equations gives a homeomorphism of $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ onto $\mathcal{R}_{-t}(Q_{1/3} + Q_{1/q})$ which acts on P_3 by the translation $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$. This translation permutes the corners, and pairings with a fixed curve can change under it. Remark 6.2 gives the verification. *Throughout, $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ denotes Smith’s perturbed curve for small $t < 0$, and every pairing computed in Sections 6 and 7 is a pairing with this curve.* Remark 6.17 records what changes for $t > 0$.

2.6. Twisted complexes over the Kotelskiy–Watson–Zibrowius algebra. Since tangle character varieties miss one corner, Cazassus, Herald, Kirk and Kotelskiy study them in the wrapped Fukaya category of the pillowcase with that corner removed, and generate the relevant subcategory by two arcs $a^\circ, a^\bullet$ ending at the other corners [CHKK22, §11.5]. The endomorphism algebra of $a^\circ \oplus a^\bullet$ is modelled by the associative two-idempotent algebra $\mathcal{B}$ of Kotelskiy, Watson and Zibrowius [KWZ19], whose idempotents $\iota_\circ$ and $\iota_\bullet$ correspond to a° and $a^\bullet$.

To encode curves we use, following [KWZ19], the dual pair of arcs $s^\circ, s^\bullet$ in the chart obtained by sending a distinguished corner to infinity (Figure 1). Both are properly embedded with both ends at the puncture at infinity, and we call their union the *arc system*. Its complement has three

faces, each containing one of the remaining corners, and we write L, M, R for the length-one paths that cross them. A positive path is a word X^k with $X \in \{L, M, R\}$ and $k \geq 1$, and we write $i \xrightarrow{X^k} j$ for a matrix entry from generator i to generator j labelled by X^k . Same-face words compose by $X^m X^n = X^{m+n}$, products of words with two different face labels vanish, and the idempotents are units; the relations $DS = SD = 0$ of the quiver presentation of $\mathcal{B}$ are built into this notation. The words L^k and R^k join generators of equal idempotent, and M^k joins generators of different idempotents exactly when k is odd. The algebra $\mathcal{B}$ is bigraded [KWZ19]. We ignore the gradings, as in the comparison in [CHKK22, §11.5], except in Section 7, where the δ -grading enters through a cited pairing result.

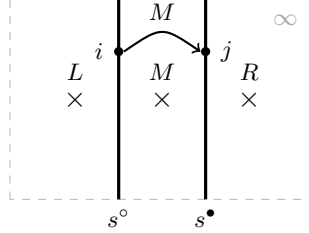


FIGURE 1. The chart for the algebra $\mathcal{B}$. A distinguished corner of the pillowcase is sent to infinity, leaving the other three corners ($\times$) in the plane. The arcs $s^\circ, s^\bullet$ of the arc system cut the chart into three faces, one for each remaining corner, labelled L, M and R . A generator is an intersection of a curve with s° or $s^\bullet$ and carries the idempotent $\iota_\circ$ or $\iota_\bullet$; a segment running from i to j through the face X contributes an arrow $i \xrightarrow{X^k} j$, where k counts the crossings of the arcs. Each of the faces L and R is bounded by a single arc of the system, so their words join generators of equal idempotent; the face M is bounded by both arcs.

An immersed curve in the chart that misses the corners determines an object of the category $\text{Tw}(\mathcal{B})$ of twisted complexes in the standard way [KWZ19]: its intersections with the arc system are the generators, each carrying $\iota_\circ$ or $\iota_\bullet$ according to the arc it meets, and a segment of the curve running from one intersection to the next through the face X contributes the arrow labelled by the corresponding power of X . Each pillowcase intersection has two lifts to the torus double cover, interchanged by the elliptic involution, so intersection counts on the torus are twice those on the pillowcase.

A finite twisted complex is a finite idempotent module V with a matrix δ over $\mathcal{B}$ satisfying $\delta^2 = 0$; it is a type D structure over $\mathcal{B}$ in the sense of [KWZ19], and we use the two descriptions interchangeably. An entry of δ labelled by an idempotent is a unit, and Gaussian elimination along it deletes its source and its target without changing the homotopy type; we call this a *unit cancellation*. A twisted complex is *reduced* if every entry of δ is a sum of positive words. For finite twisted complexes $\mathcal{E} = (V_{\mathcal{E}}, \delta_{\mathcal{E}})$ and $N = (V_N, \delta_N)$, the morphism space $\text{Mor}(\mathcal{E}, N)$ consists of the matrices f over $\mathcal{B}$ from $V_{\mathcal{E}}$ to V_N , with differential $f \mapsto \delta_N f + f \delta_{\mathcal{E}}$. We call $\dim_{\mathbb{F}_2} H \text{Mor}(\mathcal{E}, N)$, when it is finite, the *pairing*, or *wrapped dimension*, of $\mathcal{E}$ with N . For curves in pairing position, Kotelskiy, Watson and Zibrowius identify $\text{Mor}(\mathcal{E}, N)$ with the Lagrangian Floer complex of the two curves [KWZ19, Thm. 5.22], so the pairing is a Floer homology dimension; we write $HF(\mathcal{E}, N)$ for $H \text{Mor}(\mathcal{E}, N)$ in this case. The hypotheses under which this identification is used are stated in Section 7.

The endomorphism differential of (V, δ) is $D_\delta(b) = \delta b + b \delta$. A matrix b over $\mathcal{B}$ is a *Maurer–Cartan element* for (V, δ) if

$$(\delta + b)^2 = D_\delta(b) + b^2 = 0, \quad (2.7)$$

and then $(V, \delta + b)$ is again a twisted complex. There is no infinite series in (2.7): for a suitable choice of Hamiltonian perturbations of the generating arcs, μ^1 and all μ^k with $k \geq 3$ vanish and μ^2 is concatenation of paths [CHKK22, §11.5]. Cazassus, Herald, Kirk and Kotelskiy deduce this from Bocklandt's description of the wrapped Fukaya category of a punctured surface [Boc16, Thm. 7.6] and the constructions of [HKK17, §3]; the vanishing of the higher operations for a formal arc system is in [HKK17, §3.4]. The Maurer–Cartan elements used in Section 6 are supported near self-intersection points of a curve, where they encode smoothings of the curve.

Kotelskiy, Watson and Zibrowius associate to a four-ended tangle T in a ball type D structures $\widetilde{\text{BN}}(T)$ and $\widetilde{Kh}(T)$ over $\mathcal{B}$, invariants of T up to homotopy equivalence, with $\widetilde{Kh}(T)$ the mapping cone of $H \cdot \text{id}$ on $\widetilde{\text{BN}}(T)$ [KWZ19, Def. 6.5]; both are represented by immersed curves, possibly with local systems, in the four-punctured sphere [KWZ19]. Their geometric pairing theorem [KWZ19, Thm. 1.9] computes reduced Khovanov homology of a link $L = T_1 \cup T_2$ obtained by gluing two such tangles as

$$\widetilde{Kh}(L; \mathbb{F}_2) \cong HF(\widetilde{Kh}(mT_1), \widetilde{\text{BN}}(T_2)),$$

where m denotes the mirror. For rational tangles these curves are the perturbed traceless character varieties, $\widetilde{\text{BN}}$ corresponding to the tangle and $\widetilde{Kh}$ to the tangle with an earring [KWZ19, §§1.8, 8.2], up to the identification of charts made in Section 7. Section 7 compares Smith's curve, with a Maurer–Cartan element, to $\widetilde{\text{BN}}(Q_{1/3} + Q_{1/q})$.

2.7. The knots. We use the convention that positive knots have negative signature, as do [DS24a, DS24b, Wue26c] and the tables of [HHK18]; for instance $\sigma(T(3, 5)) = -8$ [HHK18, p. 60]. Braid generators are written σ_1, σ_2 and are positive crossings, so that the closure of $(\sigma_1 \sigma_2)^n$ is the positive torus knot $T(3, n)$.

For odd $q \geq 3$ the pretzel knot $P(-2, 3, q)$ is the numerator closure

$$P(-2, 3, q) = \text{num}(Q_{-1/2} + Q_{1/3} + Q_{1/q}) \quad (2.8)$$

of a sum of rational tangles, in the convention of §2.5. Its first two members are the torus knots $P(-2, 3, 3) = T(3, 4)$ and $P(-2, 3, 5) = T(3, 5)$; for $q \geq 7$ it is hyperbolic (Section 4). Section 6 uses the decomposition (2.8) with the earring on $Q_{-1/2}$; Section 4 uses a different decomposition, obtained from Hedden, Herald and Kirk's decomposition of $T(3, 5)$ by regluing.

For $n \geq 7$ with $n \equiv 1, 2 \pmod{6}$ and $m \geq 1$, the twisted torus knot $T(3, n; 2, m)$ is the closure of the three-strand braid

$$(\sigma_1 \sigma_2)^n \sigma_1^{2m},$$

that is, $T(3, n)$ with m positive full twists added on two adjacent strands. It is a positive braid closure. Section 5 treats these knots with decompositions obtained from those of $T(3, n)$ in the same way.

3. THE INSTANTON SIDE

Throughout this section $K_q = P(-2, 3, q)$ with $q \geq 3$ odd. For a knot K let $\ell(K) = \sum_i |a_i(\Delta_K)|$ be the coefficient norm of its Alexander polynomial. Kronheimer and Mrowka's integral spectral sequence from reduced Khovanov homology, their rational identification of $I^\natural$ with KHI , and the Alexander-graded Euler characteristic of KHI give

$$\ell(K) \leq \dim_{\mathbb{Q}} I^\natural(K; \mathbb{Q}) \leq \dim_{\mathbb{Q}} \widetilde{Kh}(K; \mathbb{Q}) \quad (3.1)$$

[KM11b, Props. 1.2 and 1.4], [KM10, Thm. 1.1 and Cor. 1.2]. We prove Theorem 1.1 by showing that both bounds equal $q + 2$ and that the spectral sequence collapses over $\mathbb{Z}$. We then record the signature of K_q and the $\mathbb{Z}/4$ -graded dimensions of $I^\natural(K_q)$ that Section 4 compares with the pillowcase.

3.1. The Alexander polynomial. For $m \geq 1$ write $[m]_x = 1 + x + \cdots + x^{m-1}$.

Lemma 3.1. *For every odd $q \geq 3$ put*

$$Q_q(x) = (x-2)[2]_x[3]_x[q]_x + [2]_x[3]_x + [2]_x[q]_x + [3]_x[q]_x.$$

Then

$$\Delta_{K_q}(t) \doteq Q_q(-t) = 1 - t + \sum_{j=3}^q (-1)^{j+1} t^j - t^{q+2} + t^{q+3}, \quad (3.2)$$

where $\doteq$ denotes equality up to a unit $\pm t^k$. In particular $\ell(K_q) = q+2$ and $\det K_q = |\Delta_{K_q}(-1)| = |q-6|$.

Proof. Hironaka proves that the Alexander polynomial of the $(2, 3, q, -1)$ -pretzel knot is $Q_q(-t)$, with Q_q the first displayed expression [Hir01, Thm. 1.2]; the erratum [Hir02] corrects the statement for pretzel links with more than three positive entries and leaves the three-entry case used here unchanged. The $(2, 3, q, -1)$ -pretzel knot is K_q , because the rational tangles of fractions $1/2$ and -1 sum to the rational tangle of fraction $-1/2$; Hironaka records this identification for $q = 7$ [Hir01, p. 442]. Expanding the geometric sums gives $Q_q(x) = 1 + x - \sum_{j=3}^q x^j + x^{q+2} + x^{q+3}$, and substituting $x = -t$ gives the right-hand side of (3.2), since q is odd. It has four coefficients of absolute value 1 outside the sum and $q-2$ inside, so $\ell(K_q) = q+2$. At $t = -1$ the terms outside the sum contribute $2+2$ and each of the $q-2$ terms inside contributes -1 , so $\Delta_{K_q}(-1) = 6-q$. $\square$

Remark 3.2. At $q = 7$, $Q_7(x) = 1 + x - x^3 - x^4 - x^5 - x^6 - x^7 + x^9 + x^{10}$ is Lehmer's polynomial. The substitution $t \mapsto -t$ in (3.2) is part of Hironaka's formula, not a normalization ambiguity of the Alexander polynomial. The same polynomial is obtained from the reduced Burau representation of the braid $(\sigma_1 \sigma_2)^5 \sigma_1^{q-5}$, whose closure is K_q for $q \geq 5$ (Section 4).

The following facts are used for the hyperbolicity and novelty statements of Section 4.

Lemma 3.3. *For every odd $q \geq 3$ the knot K_q is neither quasi-alternating nor alternating, and hence it is not two-bridge. For $q \geq 7$ it is not a torus knot.*

Proof. For a quasi-alternating knot K the rank of $I^\natural(K)$ equals $|\det K|$ [KM11b, Cor. 1.6]. By (3.1) and Lemma 3.1 the rank of $I^\natural(K_q)$ is at least $q+2$, whereas $|\det K_q| = |q-6| < q+2$. For an alternating knot the coefficients of the Alexander polynomial alternate in sign [Cro59, Mur58], so that $|\Delta_K(-1)| = \ell(K)$; for K_q the two sides are $|q-6|$ and $q+2$. Two-bridge knots are alternating.

For the last assertion, recall that for coprime $1 < a < b$ the Alexander polynomial $\Delta_{T(a,b)} = (1-t)(1-t^{ab})/((1-t^a)(1-t^b))$ begins $1-t+t^a-\cdots$ and has degree $(a-1)(b-1)$; it is unchanged under mirroring. In (3.2) the coefficient of t^2 is 0 and that of t^3 is 1, so a torus knot with the same Alexander polynomial would have $a = 3$, and comparing degrees gives $2(b-1) = q+3$, so $b = (q+5)/2 \geq 6$. For $b \geq 6$ the coefficient of t^5 in $\Delta_{T(3,b)} = (1-t)(1+t^3+t^6+\cdots)(1+t^b+\cdots)(1-t^{3b})$ is 0, while the coefficient of t^5 in (3.2) is 1. $\square$

3.2. Reduced Khovanov homology and the integral spectral sequence. Manion computes the reduced Khovanov homology over $\mathbb{Z}$ of the pretzel knots $P(-p, r, s)$ with $p, r, s > 0$ and $2 \leq p \leq r, s$ that are not quasi-alternating, and proves that it is free [Man18, Thm. 1.1 and Thm. 3.3]. For p even it is $\mathbb{Z}^{p^2} \oplus \mathbb{Z}^{(r-p)(s-p)}$, supported in two adjacent δ -gradings. By Lemma 3.3, K_q satisfies these hypotheses for every odd $q \geq 3$, and substituting $(p, r, s) = (2, 3, q)$ gives

$$\widetilde{Kh}(K_q; \mathbb{Z}) \text{ is free of rank } 4 + (3-2)(q-2) = q+2. \quad (3.3)$$

The reduced complex of the mirror is the integral dual complex, so by the universal coefficient theorem the reduced Khovanov homology of the mirror is also free of rank $q+2$.

Proof of Theorem 1.1. Kronheimer and Mrowka construct a spectral sequence of abelian groups whose E_2 page is the reduced Khovanov homology of the mirror and whose abutment is $I^\natural(K_q; \mathbb{Z})$ [KM11b, Prop. 1.2 and Thm. 8.12]; the underlying filtration, indexed by the vertices of a cube of resolutions, has finite length. After tensoring with $\mathbb{Q}$, (3.3) gives $\dim_{\mathbb{Q}} I^\natural(K_q; \mathbb{Q}) \leq q + 2$. Kronheimer and Mrowka also identify $I^\natural(K_q; \mathbb{Q})$ with $KHI(K_q; \mathbb{Q})$ [KM11b, Prop. 1.4]; the Alexander-graded Euler characteristic of KHI is the Alexander polynomial up to sign, so the dimension of KHI is at least $\ell(K_q)$ [KM10, Thm. 1.1 and Cor. 1.2]. These statements are made over $\mathbb{C}$ and pass to $\mathbb{Q}$ by flat base change. By Lemma 3.1, $\dim_{\mathbb{Q}} I^\natural(K_q; \mathbb{Q}) \geq \ell(K_q) = q + 2$, so the two rational bounds agree. Total dimension cannot increase from one page to the next in a spectral sequence of finite-dimensional vector spaces, so every differential vanishes after tensoring with $\mathbb{Q}$.

The E_2 page of the integral spectral sequence is free of rank $q + 2$. Inductively each page is free, and a homomorphism between free abelian groups whose rationalization vanishes is zero; hence every integral differential vanishes. The associated graded group of $I^\natural(K_q; \mathbb{Z})$ is therefore free of rank $q + 2$, and its finite filtration splits because extensions of free abelian groups split. This proves $I^\natural(K_q; \mathbb{Z}) \cong \mathbb{Z}^{q+2}$, and the assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change and the universal coefficient theorem. $\square$

Remark 3.4. The rational squeeze is Lobb and Zentner’s proof that the Kronheimer–Mrowka spectral sequence of K_q is trivial [LZ20, Prop. 4.7]; their $I^\natural$ is Kronheimer and Mrowka’s $I^\natural$ of the mirror, and the rank is unchanged by mirroring. Smith uses the same argument for $q = 5$ [Smi24, proof of Thm. 5.9]. The rational rank also follows from the work of Li and Ye: K_q is an instanton L-space knot [LY21, Prop. 7.1(2)], so each Alexander-graded summand of $KHI(K_q)$ has dimension at most one [LY21, Thm. 1.9], and the dimension of KHI is then the number of nonzero coefficients of the Alexander polynomial [KM10, Thm. 1.1]. The freeness in Theorem 1.1 additionally uses Manion’s integral calculation. For $q \geq 7$ it also follows from the work of Daemi and Scaduto (Proposition 3.6 below). For the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$ it can be read off from the integral S-complexes that Daemi and Scaduto write down [DS24a, §§9.3–9.4]: there the differential has rank one, because the rank of $I^\natural$ is known, and one of its components counts a single instanton, so the homology is torsion free. Finally, $\det K_q = |q - 6|$ never vanishes and equals 1 exactly for $q = 5, 7$, the two members whose double branched cover is an integral homology sphere.

3.3. The signature. We use the convention that positive knots have negative signature, so that $\sigma(T(3, 4)) = -6$ and $\sigma(T(3, 5)) = -8$; this is the convention of [HHK18] and [Vue26c].

Proposition 3.5. *For odd $q \geq 1$,*

$$\sigma(K_q) = \begin{cases} -(q + 3), & q \in \{1, 3, 5\}, \\ -(q + 1), & q \geq 7. \end{cases}$$

Proof. This is [DS24b, Lemma 8.29], stated there in the same convention; Daemi and Scaduto argue by induction from $K_1 = T(2, 5)$, using the congruence $|\sigma(K)| + 1 \equiv \det K \pmod{4}$ and the change of the signature under a crossing change between links of nonzero determinant. An independent check is the formula of Gordon and Litherland [GoLi78, Thm. 6], applied to the checkerboard surface of the standard diagram: two discs joined by three twisted bands with -2 , 3 and q half-twists. With respect to the two white regions between the bands, its Goeritz form is $\eta \begin{pmatrix} 1 & -3 \\ -3 & q+3 \end{pmatrix}$ for a sign η fixed by the choice of colouring, with determinant $q - 6$. The strands of K_q run antiparallel through the band with -2 half-twists and parallel through the other two, so the Gordon–Litherland correction term is $\mu = \eta(q + 3)$, one contribution from each of the $3 + q$ crossings of the two odd bands. The form has signature 2η for $q \geq 7$ and 0 for $q \leq 5$, so

$\sigma(K_q) = -\eta(q+1)$ for $q \geq 7$ and $-\eta(q+3)$ for $q \leq 5$. The value $\sigma(K_5) = \sigma(T(3,5)) = -8$ fixes $\eta = 1$. $\square$

3.4. The $\mathbb{Z}/4$ -graded instanton homology. Daemi and Scaduto associate to a knot K an S-complex

$$\tilde{C}(K) = C(K) \oplus C(K)[-1] \oplus \mathbb{Z}_{(0)}, \quad C[i]_j = C_{i+j}, \quad (3.4)$$

a decomposition of $\mathbb{Z}/4$ -graded modules in which $C(K)$ is generated by the irreducible critical points of the singular Chern–Simons functional and the summand $\mathbb{Z}_{(0)}$, in degree 0, by the reducible one. The differential has the shape

$$\tilde{d} = \begin{pmatrix} d & 0 & 0 \\ v & -d & \delta_2 \\ \delta_1 & 0 & 0 \end{pmatrix},$$

where δ_1 and δ_2 count instantons to and from the reducible [DS24b, Def. 2.1 and (1.9)], [DS24a, Def. 3.21]. Its homology is $I^\natural(K)$: there is a chain homotopy equivalence between Kronheimer and Mrowka’s complex and $\tilde{C}(K)$, homogeneous for the $\mathbb{Z}/4$ -gradings, and for knots in S^3 the resulting isomorphism on homology has degree $\sigma(K)$ modulo 4 [DS24a, Thm. 8.9]. We write $I(K)$ for the homology of $(C(K), d)$, the irreducible instanton homology.

Proposition 3.6. *Let $q \geq 7$ be odd and put $N_1 = \lceil \frac{q+1}{4} \rceil$ and $N_3 = \lfloor \frac{q+1}{4} \rfloor$. Then $I^\natural(K_q; \mathbb{Z})$ is free of rank $q+2$, and in Kronheimer and Mrowka’s absolute grading its graded ranks are*

$$(1 + N_3, N_1, N_1, N_3) \quad \text{in degrees } \text{gr} - \sigma(K_q) = 0, 1, 2, 3. \quad (3.5)$$

The same holds with coefficients in $\mathbb{Q}$ and in $\mathbb{F}_2$.

Proof. For $q \geq 7$, Daemi and Scaduto prove $I(K_q) \cong \mathbb{Z}_{(1)}^{N_1} \oplus \mathbb{Z}_{(3)}^{N_3}$, with the subscripts denoting $\mathbb{Z}/4$ degrees in the grading of (3.4) [DS24b, Prop. 8.30]; the degrees come from [DS24b, Prop. 8.28] and the freeness from [DS24b, Rem. 8.27]. The filtration $C[-1] \subset C[-1] \oplus \mathbb{Z}_{(0)} \subset \tilde{C}$ induces a spectral sequence

$$I(K_q) \oplus I(K_q)[-1] \oplus \mathbb{Z}_{(0)} \implies I^\natural(K_q; \mathbb{Z}) \quad (3.6)$$

[DS24b, (8.5)]. Daemi and Scaduto state the homology groups of their §8 as $\mathbb{Z}/2$ -graded, but the spectral sequence respects the $\mathbb{Z}/4$ -grading: by the shape of $\tilde{d}$, the summands $C[-1]$ and $C[-1] \oplus \mathbb{Z}_{(0)}$ are subcomplexes, and (3.4) is a decomposition of $\mathbb{Z}/4$ -graded modules, so (3.6) is a spectral sequence of $\mathbb{Z}/4$ -graded groups. Since a class of degree j in I sits in degree $j+1$ in $I[-1]$, the initial page has ranks $1 + N_3, N_1, N_1$ and N_3 in degrees 0, 1, 2 and 3, for a total of $1 + 2(N_1 + N_3) = q + 2$.

By Theorem 1.1, the rank of the abutment is also $q + 2$. Hence every differential vanishes after tensoring with $\mathbb{Q}$, and, as in the proof of Theorem 1.1, every integral differential vanishes and the filtration of $I^\natural(K_q; \mathbb{Z})$ splits. So $I^\natural(K_q; \mathbb{Z})$ is free with the graded ranks of the initial page in the grading of (3.4). By [DS24a, Thm. 8.9], Kronheimer and Mrowka’s absolute grading is this grading shifted by $\sigma(K_q)$; since $\sigma(K_q)$ is even, the sign of the shift does not matter modulo 4. The assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change. $\square$

For $q = 7, 9, 11$ and 13 the vector (3.5) is $(3, 2, 2, 2)$, $(3, 3, 3, 2)$, $(4, 3, 3, 3)$ and $(4, 4, 4, 3)$. For $q = 3, 5$ the same spectral sequence has initial rank $q + 4$ and a differential of rank one. The graded ranks of $I^\natural$ of the torus knots $K_3 = T(3, 4)$ and $K_5 = T(3, 5)$ are $(1, 1, 2, 1)$ and $(2, 1, 2, 2)$ in degrees $\text{gr} - \sigma = 0, 1, 2, 3$ [Wue26c, (2.5) and Table 2], and these are the vectors with which the pillowcase homology of the two torus knots is compared in [Wue26c]. They are instances of the graded formula of [Wue26b], which determines $I^\natural(K; \mathbb{Q})$ for every torus knot K , and the S-complexes of the two knots are described in [Wue26a].

4. PILLOWCASE HOMOLOGY OF THE PRETZEL KNOTS

In this section we prove Theorem 1.2. The decomposition is the one used by Hedden, Herald and Kirk for $T(3, 5)$ [HHK18, §10], reglued along the Conway sphere by $m = (q - 5)/2$ Dehn twists. The proof has four steps.

- (1) The reglued knot is $K_q = P(-2, 3, q)$ (§4.1). The argument uses only the fundamental group and the signature, and no planar diagram of the decomposition.
- (2) On the pillowcase the regluing replaces L_1 by its image under a linear shear (§4.2).
- (3) The unperturbed variety of the $T(3, 5)$ tangle maps onto two straight polygons (§4.3). This rests on a word in $\pi_1(Y \setminus T)$ that is trivial on the whole central circle of the variety.
- (4) The perturbed curves are those of [Wue26c], sheared (§4.4). From them we compute the generators, the gradings and the bigons by hand (§§4.5–4.7), for both ways in which the perturbation can resolve the double points.

No invariance theorem of [HHK18] is used. As in [Wue26c], results of [HHK18] are cited by their numbering in arXiv:1501.00028v1.

Throughout, $\varepsilon = (\varepsilon_A, \varepsilon_B)$ denotes the parameters of the holonomy perturbation on the tangle side, and $\epsilon > 0$ the earring parameter of §2.2. As in [Wue26c, §6], ε is fixed first and ϵ is then taken sufficiently small, and all statements about intersection points with L_0 hold for every sufficiently small ϵ .

4.1. The decomposition.

The tangle. Let (Y, T) be the tangle of [HHK18, §10] for $T(3, 5)$, with the integers $(r, s) = (2, -1)$ satisfying $3r + 5s = 1$; this is the tangle of [Wue26c, §3] for $n = 5$. The group $\pi_1(Y \setminus T)$ is free on two elements A, B (the elements A_1, B_1 of [HHK14, §11]). At $\varepsilon = 0$ the meridians of the four boundary punctures are

$$a = A^2 B^3, \quad b = B^{-2} A^{-1}, \quad c = B^{-2} a B^2, \quad (4.1)$$

by [HHK14, (42)] with $(s + p, q - r, r, s) = (2, 3, 2, -1)$ (see [Wue26c, (3.2)]), and the fourth meridian d satisfies $ba = cd$ [HHK18, (31)]. More generally, for $n = 3\nu + 2$ with $\nu \geq 1$, the tangle of [HHK18, §10] for $T(3, n)$ with $(r, s) = (\nu + 1, -1)$ has

$$a = A^2 B^{2\nu+1}, \quad b = B^{-(\nu+1)} A^{-1}, \quad c = B^{-(\nu+1)} a B^{\nu+1}. \quad (4.2)$$

The trivial tangle and the twist. Let (D, U) be the trivial tangle of [HHK18, (17)]: D is the unit ball, and U consists of the two vertical arcs $\{(\pm 1/\sqrt{2}, 0, t) : |t| \leq 1/\sqrt{2}\}$, one joining the punctures a, d and the other joining b, c . The group $\pi_1(D \setminus U)$ is free, and its boundary meridians satisfy $a = d$ and $b = c$ [HHK14, Prop. 6.1], [FKP17, p. 4]. Let $e \subset \partial D$ be the equator $\{z = 0\}$; it separates $\{a, b\}$ from $\{c, d\}$ and bounds the horizontal disc $D_e = D \cap \{z = 0\}$, which meets each arc of U once. Let t_e be the Dehn twist along e that acts on $\pi_1(S^2 \setminus \{a, b, c, d\}) = \langle a, b, c, d \mid ba = cd \rangle$, with base point on the side of c, d , by

$$t_e(a) = (ba) a (ba)^{-1}, \quad t_e(b) = (ba) b (ba)^{-1}, \quad t_e(c) = c, \quad t_e(d) = d. \quad (4.3)$$

It fixes $ba = cd$. Hedden, Herald and Kirk use the half twist g in the hemisphere containing c and d that exchanges c and d , and note that g^2 is the Dehn twist about e [HHK18, proof of Thm. 9.2, p. 49]. With base point near a , g_* fixes a, b and sends $c \mapsto d$, $d \mapsto d^{-1}cd$. So g_*^2 conjugates c and d by $(cd)^{-1} = (ba)^{-1}$ and fixes a, b , and composing $(g_*^2)^{-1}$ with conjugation by $(ba)^{-1}$ gives t_e^{-1} . Thus t_e and g^2 are the same mapping class.

Recovering a knot from (Y, T) requires a choice of identification of ∂D with ∂Y [HHK18, after (17)]. For $h \in \mathbb{Z}$ let

$$K^{(h)} = (Y, T) \cup_{t_e^h} (D, U)$$

be the knot obtained by precomposing the identification of [HHK18, (17)], on the side of Y , with t_e^h : a boundary loop x of D is identified with the loop $t_e^h(x)$ of Y .

Lemma 4.1. *For every $h \in \mathbb{Z}$, $K^{(h)}$ is a knot in S^3 and*

$$\pi_1(S^3 \setminus K^{(h)}) \cong \langle A, B \mid t_e^h(b) = c \rangle, \quad t_e^h(b) = (ba)^h b (ba)^{-h},$$

with a, b, c the words of the tangle. The knots $K^{(h)}$ and $K^{(h-1)}$ differ by one full twist of the two arcs of U about each other along D_e ; in particular they are related by one crossing change.

Proof. Gluing two balls along their boundary gives S^3 , and t_e fixes each puncture, so the strands are connected as in $K^{(0)}$, which is a knot. By van Kampen's theorem, $\pi_1(S^3 \setminus K^{(h)})$ is the amalgam of $\pi_1(Y \setminus T) = F(A, B)$ and $\pi_1(D \setminus U)$, free on $x = a = d$ and $y = b = c$, over the four-punctured sphere. Eliminating x and y leaves the relations $t_e^h(b) = t_e^h(c) = c$ and $t_e^h(a) = t_e^h(d) = d$. The second follows from the first, because t_e^h fixes $ba = cd$: $t_e^h(a) = t_e^h(b)^{-1}ba = c^{-1}ba = d$. Regluing by t_e^{-1} amounts to replacing U by its image under a twist map supported in a collar of ∂D , which is U with one full twist of its two arcs along D_e . Changing one of the two crossings of a full twist of two strands undoes it. $\square$

At $h = 0$ the relation $b = c$ reads $B^{-2}A^{-1} = B^{-2}A^2B^5$, that is $A^3B^5 = 1$, the group of $T(3, 5)$, as it must be.

Clay and Watson define the twisted torus knots $T_{3,n}^m$ by adding m positive full twists on two of the three strands of $T(3, n)$ [CW13, §3.2]; $T_{3,n}^m$ is the closure of the braid $(\sigma_1\sigma_2)^n\sigma_1^{2m}$ (§2.7). For $n = 3\nu + 2$ they give the presentation [CW13, Prop. 22]

$$\pi_1(S^3 \setminus T_{3,n}^m) \cong \langle \alpha, \beta \mid \alpha^2(\beta^{-\nu}\alpha)^m\alpha = \beta^{2\nu+1}(\beta^{-\nu}\alpha)^m\beta^{\nu+1} \rangle, \quad m \geq 0. \quad (4.4)$$

Lemma 4.2. *Let $n = 3\nu + 2$ with $\nu \geq 1$, let (Y, T) be the tangle with the words (4.2), and put $C = AB^\nu$. For every $m \geq 0$ the relation $t_e^{-m}(b) = c$ is equivalent to the vanishing of the cyclic word*

$$R_{n,m} = B^{\nu+1}A C^m A^2 B^{2\nu+1} C^{-m},$$

and under $\alpha = A$, $\beta = B^{-1}$ the relator of (4.4) is a cyclic permutation of $R_{n,m}$. In particular $\pi_1(S^3 \setminus K^{(-m)}) \cong \pi_1(S^3 \setminus T_{3,n}^m)$.

Proof. Since $ba = B^{-(\nu+1)}AB^{2\nu+1} = B^{-(\nu+1)}CB^{\nu+1}$, we have $(ba)^j = B^{-(\nu+1)}C^jB^{\nu+1}$. Hence $t_e^{-m}(b) = B^{-(\nu+1)}C^{-m}A^{-1}B^{-(\nu+1)}C^mB^{\nu+1}$, and $t_e^{-m}(b) = c$ becomes $C^{-m}A^{-1}B^{-(\nu+1)}C^m = A^2B^{2\nu+1}$. The inverse of the corresponding relator is $A^2B^{2\nu+1}C^{-m}B^{\nu+1}AC^m$, a cyclic permutation of $R_{n,m}$. On the other side, substitute $\alpha = A$ and $\beta = B^{-1}$ in the relator of (4.4):

$$\alpha^2(\beta^{-\nu}\alpha)^m\alpha\beta^{-(\nu+1)}(\beta^{-\nu}\alpha)^{-m}\beta^{-(2\nu+1)} \longmapsto A^2(B^\nu A)^m A B^{\nu+1}(B^\nu A)^{-m} B^{2\nu+1}.$$

Using $A(B^\nu A)^m = C^m A$ and $(B^\nu A)^{-m} = B^\nu C^{-m} B^{-\nu}$, this is $AC^m A^2 B^{2\nu+1} C^{-m} B^{\nu+1}$, again a cyclic permutation of $R_{n,m}$. $\square$

We have also checked the identity of Lemma 4.2 as an identity of reduced free-group words for $n \in \{5, 8, \dots, 29\}$ and $m \leq 15$; with t_e^{+m} in place of t_e^{-m} the relator of (4.4) is never obtained.

The group of a knot does not distinguish it from its mirror image, so the chirality of the base case enters as a premise.

Remark 4.3 (Base case). We take $K^{(0)}$ to be the positive torus knot $T(3, 5)$, of signature -8 . This is the standing convention of the torus-knot computations of [HHK18, §§10–11] and [Wue26c], where the decomposition [HHK18, (17)] of $T(3, 5)$ is compared with Kronheimer and Mrowka's instanton homology using $\sigma = -8$. It is consistent with a computer check: from a planar diagram of [HHK18, Fig. 18], SnapPy [CDGW] finds an orientation-preserving isometry, compatible with the meridians, between the complement of the link $K^{(0)} \cup e$ and the complement of the closure of

the positive braid $(\sigma_1\sigma_2)^5$ together with an unknotted circle encircling two of its strands. That check is not used in the proofs. The group cannot detect this premise; the identification of the knots below and the graded comparison in Theorem 1.2 both depend on it.

Theorem 4.4. *For every $m \geq 1$, the knot $K^{(-m)}$ is the pretzel knot $K_{5+2m} = P(-2, 3, 5 + 2m)$, the closure of the positive braid $(\sigma_1\sigma_2)^5\sigma_1^{2m}$, and $\sigma(K^{(-m)}) = -(2m + 6)$.*

Proof. By Lemma 4.2 with $\nu = 1$, the groups of $K^{(-m)}$ and of $T_{3,5}^m$ are isomorphic. Clay and Watson state that $T_{3,5}^m$ is $P(-2, 3, 5 + 2m)$ [CW13, §3.2]; for $m = 1$ this is also stated in [DS24b, p. 164]. We use this only up to mirror image. In particular both knots are nontrivial, since the Alexander polynomial of their common group is that of K_{5+2m} , which is not constant (Lemma 3.1). Both groups are generated by two elements, so both knots are prime by Norwood's theorem [Nor82]. Prime knots with isomorphic groups are equivalent, by a homeomorphism of S^3 that may reverse orientation [GoLu89, Cor. 2.1]. Hence $K^{(-m)}$ is K_{5+2m} or its mirror image.

To fix the chirality, recall $\sigma(K_q) = -(q + 1)$ for $q \geq 7$ and $\sigma(K_5) = \sigma(T(3, 5)) = -8$ (Proposition 3.5); the mirror image has the opposite signature. By Lemma 4.1, $K^{(-m-1)}$ and $K^{(-m)}$ are related by one crossing change, so their signatures differ by at most 2 [Mur65, proof of Thm. 10.1, (10.2)]. Suppose $\sigma(K^{(-m)}) = -s_m$ with $s_0 = 8$ and $s_m = 2m + 6$ for $m \geq 1$; for $m = 0$ this is Remark 4.3. The two candidates for $K^{(-m-1)}$ have signature $\pm(2m + 8)$, and only $-(2m + 8)$ lies within 2 of $-s_m$. Hence $K^{(-m-1)} = K_{2m+7}$, and induction on m completes the proof. $\square$

The same argument applies to every $n \equiv 2 \pmod{3}$ once the signature of $T(3, n; 2, m)$ is known; this is used in Section 5.

Proposition 4.5. *For odd $q \geq 7$ the knot K_q is hyperbolic and is not a two-bridge knot.*

Proof. By Lemma 3.3, K_q is neither two-bridge nor a torus knot. It is the Montesinos knot with rational tangles $-1/2, 1/3, 1/q$, all of denominator at least 2. By Oertel's theorem [Oer84, Cor. 5], the complement of such a Montesinos link carries a complete hyperbolic structure unless the link is a torus link or one of the links $K(\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})$, $K(\frac{2}{3}, -\frac{1}{3}, -\frac{1}{3})$, $K(\frac{1}{2}, -\frac{1}{4}, -\frac{1}{4})$, $K(\frac{1}{2}, -\frac{1}{3}, -\frac{1}{6})$ or a mirror image of one of these. For each of the four listed links the sum of the fractions is 0, whereas for a knot it is nonzero [Oer84, Thm. 1]; so none of them, and none of their mirror images, is a knot. Conventions for the sign of a rational tangle differ between authors, which may replace K_q by its mirror image; the list of exceptions is closed under mirror image. Hence K_q is hyperbolic. $\square$

Oertel's corollary does not list the torus links, which is why Lemma 3.3 is needed. Explicitly, apart from two-bridge knots the only non-hyperbolic Montesinos knots are $P(-2, 3, 3) = T(3, 4)$ and $P(-2, 3, 5) = T(3, 5)$ and their mirror images [IJ09, Rem. 1], [Mei14, p. 440]. Futer and Guéritaud give a published proof of the classification of non-hyperbolic arborescent links, which they attribute to Bonahon and Siebenmann [FG09, Thm. 1.5].

4.2. The twist acts by a shear. For $h \in \mathbb{Z}$ let $S_h: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the linear map $S_h(\gamma, \theta) = (\gamma, \theta - 2h\gamma)$.

Lemma 4.6. *Let (Y, T) be a 2-tangle as in [HHK18, (17)], with a holonomy perturbation supported in the interior of Y , and let $h \in \mathbb{Z}$. For the decomposition $(Y, T) \cup_{t_h} (D, U)$:*

- (a) $R_\pi(Y, T)$ is unchanged, its restriction map is $S_h \circ L_1$, where L_1 is the restriction map for the identification of [HHK18, (17)], and L_0 is unchanged;
- (b) S_h descends to a homeomorphism of P fixing each corner and to an orientation-preserving diffeomorphism of P^* ; it preserves γ , hence the open strip $\{0 < \gamma < \pi\}$, the two edges, and the vertical degree of closed curves that miss the edges;

- (c) S_h maps proper immersions of intervals to proper immersions, adding $-2h$ to both limiting slopes, and preserves embeddedness of lifts and the absence of fishtails;
- (d) for closed curves S_h preserves z and $\mu(\cdot, \ell_1)$; hence it maps restricted immersed 1-manifolds to restricted immersed 1-manifolds;
- (e) the corner images of the two abelian representations are unchanged;
- (f) for every immersed path α , $\mu(S_h \circ \alpha, \ell_1) = \mu(\alpha, S_h^{-1} \ell_1)$, where $S_h^{-1} \ell_1$ is the constant line field spanned by $(1, 1 + 2h)$.

Proof. (a) The perturbation is supported in the interior of Y , so $R_\pi(Y, T)$ does not change. Let a representation restrict to $a \mapsto \mathbf{i}$, $b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}$, $c \mapsto e^{\theta \mathbf{k}} \mathbf{i}$ on ∂Y . Then $\rho(ba) = -e^{\gamma \mathbf{k}}$, so by (4.3), $\rho(t_e^h a) = e^{2h\gamma \mathbf{k}} \mathbf{i}$, $\rho(t_e^h b) = e^{(2h+1)\gamma \mathbf{k}} \mathbf{i}$, and c is fixed. Conjugating by $e^{-h\gamma \mathbf{k}}$ gives $(\mathbf{i}, e^{\gamma \mathbf{k}} \mathbf{i}, e^{(\theta-2h\gamma) \mathbf{k}} \mathbf{i})$, the point $S_h(\gamma, \theta)$, with $\gamma \in [0, \pi]$ unchanged. The trivial tangle keeps the identification of [HHK18, (17)], so L_0 is unchanged.

(b) S_h is linear of determinant 1 and commutes with -1 ; it sends $(2\pi, 0)$ to $(2\pi, -4\pi h)$ and fixes $(0, 2\pi)$, so it normalises G . It maps $(\pi a, \pi b)$ to $(\pi a, \pi(b - 2ha)) \equiv (\pi a, \pi b)$ modulo $2\pi\mathbb{Z}^2$.

(c) Embeddedness of lifts and the absence of fishtails are invariant under the homeomorphism S_h of P^* .

(d) S_h fixes the corners and preserves orientation, so it preserves z . For a closed curve, $\mu(\cdot, \ell)$ is the total rotation of the tangent line divided by π , for every constant line field ℓ , because the deck transformations have derivative $\pm I$. Along the path $s \mapsto S_{sh}$, $0 \leq s \leq 1$, this total rotation is a continuous integer, hence constant. The rest follows from (b), (c) and Definition 2.1.

(e) This follows from (b).

(f) The tangent line of $S_h \circ \alpha$ equals ℓ_1 exactly when the tangent line of α equals $S_h^{-1} \ell_1$. Since S_h preserves orientation, the senses of passage agree, and the endpoint conventions correspond. $\square$

The computation in (a) is that of [HHK18, (28)]: there the half twist g acts on the pillowcase by $(\gamma, \theta) \mapsto (\gamma, \theta - \gamma)$, and [HHK18, (29)–(30)] obtain the action of $\mathrm{PSL}_2(\mathbb{Z})$ from g and a 3-cycle. Since $t_e = g^2$, t_e^h acts by S_h . Our additions are the observation that regluing by t_e^h is a choice of identification permitted by [HHK18, (17)], and properties (b)–(f). The action of the mapping class group on the pillowcase is also [CHK22, Thm. B], and its counterpart for immersed curves in Khovanov homology is [KWZ19, Thm. 8.1].

The pretzel case. From now on $q \geq 7$ is odd and $m = (q - 5)/2$, so that

$$S_{-m}(\gamma, \theta) = (\gamma, \theta + (q - 5)\gamma).$$

By Theorem 4.4 and Lemma 4.6, the decomposition $K_q = (Y, T) \cup_{t_e^{-m}} (D, U)$ has restriction map $S_{-m} \circ L_1$, where L_1 is the restriction map of [Wue26c] for $T(3, 5)$. On $\mathbb{R}^2$ we keep $\varphi = \theta - \gamma$; then

$$\varphi \circ S_{-m} = \lambda, \quad \lambda(\gamma, \theta) = \theta + (q - 6)\gamma,$$

and $S_{-m}^{-1} \ell_1$ is the line field spanned by $(1, 6 - q)$.

4.3. The unperturbed curve. We use the following facts about the tangle (Y, T) for $n = 5$, in the notation of [Wue26c, §§3–5]. A perturbed traceless representation has $A \mapsto e^{uQ_A}$ and $B \mapsto e^{vQ_B}$ with $Q_A = \mathbf{i}$ and $Q_B = \tau \mathbf{i} + \sqrt{1 - \tau^2} \mathbf{j}$; $W_\varepsilon \subset [0, \pi]^2 \times [-1, 1]$ is the variety of such representations, $W = W_0$, and $\Pi: W_\varepsilon \rightarrow P$ is the restriction map, normalised by $a \mapsto \mathbf{i}$, $b \mapsto e^{\gamma \mathbf{k}} \mathbf{i}$, $c \mapsto e^{\theta \mathbf{k}} \mathbf{i}$ with $\gamma \in [0, \pi]$.

- (V1) $W = B \cup C_0$, where $B = \{(\frac{\pi}{2}, \frac{\pi}{2})\} \times [-1, 1]$ is the binary dihedral arc and C_0 is the central circle; $C_0 \cap B = \{c_+, c_-\}$ with $c_\pm = (\frac{\pi}{2}, \frac{\pi}{2}, \pm \tau_1)$, $\tau_1 = \sqrt{3}/2$; the halves $C_0^L = C_0 \cap \{v \leq \frac{\pi}{2}\}$ and $C_0^R = C_0 \cap \{v \geq \frac{\pi}{2}\}$ meet $\{v = \frac{\pi}{2}\}$ only at $c_\pm$ [Wue26c, Prop. 3.6, Prop. 3.10(2)].

- (V2) $\Pi(B)$ lifts to the segment $t \mapsto (t, 0)$, $t \in [0, \pi]$, with $\gamma = \arccos \tau$ on B . The endpoints $r_{\pm} = (\frac{\pi}{2}, \frac{\pi}{2}, \pm 1)$ map to $(0, 0)$ and $(\pi, 0)$, and $\Pi(c_+) = (\frac{\pi}{6}, 0)$, $\Pi(c_-) = (\frac{5\pi}{6}, 0)$ [Wue26c, Prop. 3.9]. We write B_0, B_1, B_2 for the pieces of B with $\tau \in [\tau_1, 1], [-\tau_1, \tau_1], [-1, -\tau_1]$.
- (V3) Along each half of C_0 , from c_+ to c_- , φ is strictly monotone with $d\varphi \neq 0$ at interior points and increases by $10\pi/3$; at interior points $\frac{\pi}{6} < \gamma < \frac{5\pi}{6}$ [Wue26c, Thm. 4.10, Props. 4.11–4.12].
- (V4) v is a regular parameter of C_0 at $c_{\pm}$, and on a lift $\Pi = \Pi(c_{\pm}) \pm Vw^2 + O(w^3)$ with $w = |v - \frac{\pi}{2}|$ and $V = (2/\sqrt{3}, 12/\sqrt{3})$, of slope 6 [Wue26c, Lemma 4.13].
- (V5) $c_{\pm}$ are nondegenerate double points of W . With respect to the equations (E1), (E2) of [Wue26c, Lemma 3.1], the kernel of $d\Psi(c_{\pm})$ is spanned by $T_B = \partial_{\tau}$ and $X = -2\tau\partial_u + \partial_v$, the covector $\ell = (-2\tau, -1)$ spans the left kernel, $\ell \cdot D^2\Psi(T_B, X) = -8\tau$ and $\ell \cdot D^2\Psi(T_B, T_B) = 0$ [Wue26c, Prop. 3.11(2)].

Lemma 4.7. *Let $\varepsilon = 0$ and put $w = (ba)^{-3}c(ba)^3a \in \pi_1(Y \setminus T)$. Then $\rho(w) = 1$ for every $\rho \in C_0$. Consequently $\theta \equiv 6\gamma - \pi \pmod{2\pi}$ on $\Pi(C_0)$, and for $l \in \mathbb{Z}$ the lift of each half of C_0 that starts at the point $(\frac{\pi}{6}, 2\pi l)$ over $\Pi(c_+)$ is the segment*

$$\{(\gamma, 6\gamma - \pi + 2\pi l) : \frac{\pi}{6} \leq \gamma \leq \frac{5\pi}{6}\},$$

traversed with γ strictly increasing from c_+ to c_- and with $d\gamma \neq 0$ at interior points.

Proof. We write quaternions as $p = p_0 + \mathbf{p}$ with $p_0 = \operatorname{Re} p$, so that $\mathbf{p}\mathbf{q} = -\mathbf{p} \cdot \mathbf{q} + \mathbf{p} \times \mathbf{q}$ for pure quaternions.

Step 1: the word. By (4.1), $ba = B^{-2}(AB)B^2$, so $(ba)^3 = B^{-2}g$ with $g = (AB)^3B^2$. Since $c = B^{-2}aB^2$, we get $w = g^{-1}B^2 \cdot B^{-2}aB^2 \cdot B^{-2}g \cdot a = g^{-1}aga$. If $\rho(a)$ and $\rho(g)$ are orthogonal pure unit quaternions, they anticommute, so $\rho(g)^{-1}\rho(a)\rho(g) = -\rho(a)$ and $\rho(w) = -\rho(a)^2 = 1$. It remains to prove that orthogonality.

Step 2: notation. Let $\rho \in W$. Then $\beta = \rho(b)$ is a pure unit quaternion. Write $N = \rho(B) = \cos v + \sin v Q$ with $Q = Q_B$, and put

$$\zeta = \beta \cdot Q, \quad U = \cos v \beta + \sin v (\beta \times Q), \quad E = 4 \cos^2 v + 4\zeta^2 \sin^2 v - 3.$$

From $b = B^{-2}A^{-1}$ we get $\rho(A) = -\beta N^{-2}$.

Step 3: $\rho(a)$. Conjugation by β is the rotation by π about β , so $\beta Q \beta^{-1} = Q'$ with $Q' = 2\zeta\beta - Q$, and $\beta N^{-2}\beta = -e^{-2vQ'}$. Hence $\rho(a) = \rho(A)^2 N^3 = -e^{-2vQ'} N$. Using $\operatorname{Re}(e^{sP} e^{tR}) = \cos s \cos t - (P \cdot R) \sin s \sin t$ and $Q' \cdot Q = 2\zeta^2 - 1$,

$$\operatorname{Re} \rho(a) = -\cos v \cdot E, \quad \mathbf{Vec} \rho(a) = -\sin 3v Q + 2\zeta \sin 2v U.$$

Step 4: $C_0 \subset \{E = 0\}$. On W the element $\rho(a)$ is traceless, so $\cos v \cdot E = 0$. On $C_0 \setminus \{c_{\pm}\}$ we have $v \neq \frac{\pi}{2}$ by (V1), so $E = 0$ there, and by continuity on all of C_0 .

Step 5: $\rho(g)$. Put $\xi = \beta N^{-1}$, so that $\rho(AB) = -\xi$ and $\operatorname{Re} \xi = \zeta \sin v$. The Cayley–Hamilton identity $\xi^2 = 2(\operatorname{Re} \xi)\xi - 1$ gives $\xi^3 = (4\zeta^2 \sin^2 v - 1)\xi - 2\zeta \sin v$, and $\xi N^2 = \beta N$, so

$$\rho(g) = -\xi^3 N^2 = -(4\zeta^2 \sin^2 v - 1) \beta N + 2\zeta \sin v N^2, \quad \operatorname{Re} \rho(g) = \zeta \sin v \cdot E.$$

On $\{E = 0\}$ we have $4\zeta^2 \sin^2 v - 1 = -2 \cos 2v$, so $\rho(g)$ is pure, with $\mathbf{Vec} \rho(g) = 2 \cos 2v U + 2\zeta \sin v \sin 2v Q$.

Step 6: orthogonality. With $U \cdot U = 1 - \zeta^2 \sin^2 v$, $U \cdot Q = \zeta \cos v$ and $\sin 3v = \sin v (4 \cos^2 v - 1)$, a direct expansion gives

$$\mathbf{Vec} \rho(g) \cdot \mathbf{Vec} \rho(a) = 2\zeta (2\zeta^2 \sin^2 v \sin 2v + 2 \cos 2v \sin 2v - \sin 3v \cos v) = \zeta \sin 2v \cdot E.$$

So on C_0 the quaternions $\rho(a)$ and $\rho(g)$ are orthogonal pure units, and $\rho(w) = 1$ by Step 1.

Step 7: the pillowcase. In the normalisation of Π , $\rho(ba) = -e^{\gamma \mathbf{k}}$ and $\mathbf{i} e^{t \mathbf{k}} = e^{-t \mathbf{k}} \mathbf{i}$, so $\rho((ba)^j c (ba)^{-j} a) = -e^{(\theta + 2j\gamma) \mathbf{k}}$. For $j = -3$, $\rho(w) = 1$ gives $\theta - 6\gamma \in \pi + 2\pi \mathbb{Z}$.

Step 8: the lifts. By (V3) the lift of a closed half starting at a lift $(\frac{\pi}{6}, 2\pi l)$ of $\Pi(c_+)$ stays in the open strip except at its ends. On it $\theta - 6\gamma + \pi$ is continuous with values in $2\pi\mathbb{Z}$, hence equal to its value $2\pi l$ at the start, so $\varphi = 5\gamma - \pi + 2\pi l$. By (V3), φ is strictly monotone with $d\varphi \neq 0$ at interior points and increases by $10\pi/3$; hence $\gamma = (\varphi + \pi - 2\pi l)/5$ increases strictly from $\frac{\pi}{6}$ to $\frac{5\pi}{6}$, with $d\gamma \neq 0$ at interior points. $\square$

Each displayed identity of Steps 1–7 has also been verified symbolically for indeterminate (v, ζ) (see Appendix A). Hedden, Herald and Kirk observed, by a computer algebra computation, that the two halves of the central circle for $T(3, 5)$ map linearly with slope 6 [HHK18, §11.7, p. 59]; Lemma 4.7 proves this. For their own decomposition of the pretzel knots $P(p, q, r)$ with p even, Fukumoto, Kirk and Pinzón-Caicedo prove the analogous straightness [FKP17, Thm. 5.1], and they describe the unperturbed curve as the segments $\theta = (r - 5)\gamma$ and $\theta = (r + 1)\gamma + \pi$ [FKP17, §5, p. 28]. These are the images under S_{-m} of $\Pi(B)$ and of the line of Lemma 4.7, with $r = q$. We do not identify their decomposition with ours, and we do not use their result; we regard it as the antecedent of Lemma 4.7.

By Proposition 5.7 of [Wue26c], after a small perturbation the arc of L_1 is built from B_0 , one half of C_0 and B_2 , possibly through B_1 and the other half. Write H for the half joined to B_0 and H' for the other half, and orient both from c_+ to c_- . Let $O = (0, 0)$, and

$$J_1 = (\frac{\pi}{6}, 0), \quad J_2 = (\frac{5\pi}{6}, 4\pi), \quad J_3 = (\frac{\pi}{6}, 4\pi), \quad J_4 = (\frac{5\pi}{6}, 8\pi), \quad E = (\pi, 8\pi), \quad E' = (\pi, 4\pi).$$

Corollary 4.8. *At $\varepsilon = 0$ the lifted pieces are straight segments, traversed monotonically and regular in their interiors: $B_0: O \rightarrow J_1$ along $(1, 0)$; $H: J_1 \rightarrow J_2$ along $(1, 6)$; $B_1^{-1}: J_2 \rightarrow J_3$ along $(-1, 0)$; $H': J_3 \rightarrow J_4$ along $(1, 6)$; and $B_2: J_4 \rightarrow E$, or $B_2: J_2 \rightarrow E'$ when B_2 follows H , along $(1, 0)$. The lift of the closed curve $B_1^{-1} \cdot H'$ is the $(0, 4\pi)$ -periodic polygon with vertices $(\frac{5\pi}{6}, 4\pi j)$ and $(\frac{\pi}{6}, 4\pi j)$, $j \in \mathbb{Z}$.*

Proof. The pieces of B are described by (V2): $\gamma = \arccos \tau$ is a diffeomorphism from $(-1, 1)$ onto $(0, \pi)$. The halves are described by Lemma 4.7, applied to the lifts of [Wue26c, Lemma 6.1], whose endpoints are the listed vertices. $\square$

4.4. The perturbed curves. We recall the local description of [Wue26c, §5.2]. Near a double point c the variety W_ε is the zero set of a reduced function $g(s, t; \varepsilon)$, with Hessian $\ell \cdot D^2\Psi(c)$ restricted to $\ker d\Psi(c)$. It has a unique critical point near c , with critical value $\kappa_c(\varepsilon)$, and in Morse coordinates $g = \kappa_c(\varepsilon) + s't'$, where $\{t' = 0\} = B$ and $\{s' = 0\} = C_0$ at $\varepsilon = 0$. Moreover $\kappa_c(0) = 0$ and $d\kappa_c(0) = \ell \cdot \partial_\varepsilon \Psi(0; c)$. For $a > 0$ the cone $\mathcal{C}_a$ is the set of ε whose linear parts $d\kappa_{c_\pm}(0)\varepsilon$ have absolute value at least $a|\varepsilon|$ for both double points [Wue26c, Def. 5.5]. Let P_B and V be the vectors with $\Pi(s', 0) = \Pi(c) + s'P_B + O(s'^2)$ and $\Pi(0, t') = \Pi(c) + t'V + O(t'^3)$ at $\varepsilon = 0$; by (V2) and (V4), $P_B \parallel (1, 0)$ and $V \parallel \pm(1, 6)$.

For $\varepsilon \in \mathcal{C}_a$ small, each of $c_\pm$ is resolved: the two arcs of W_ε near c each join a half-branch of B to a half-branch of C_0 . Following [Wue26c, §6.1], there are two combinatorial types:

- the *joined resolution*, in which $R_\pi(Y, T)$ is a single arc $B_0 \cdot H \cdot B_1^{-1} \cdot H' \cdot B_2$;
- the *split resolution*, in which $R_\pi(Y, T)$ is the arc $B_0 \cdot H \cdot B_2$ together with the *remnant circle* $B_1^{-1} \cdot H'$.

Here each product denotes the resolved curve, C^1 -close to the listed pieces away from $c_\pm$ [Wue26c, Prop. 5.7]. The analysis of [Wue26c] does not need to know which type occurs. After the shear it matters, and we determine it.

Lemma 4.9. *Let $c \in \{c_+, c_-\}$, $a > 0$, and let λ' be a linear function on $\mathbb{R}^2$ with $\lambda'(P_B) \neq 0$ and $\lambda'(V) \neq 0$. Let U_c , of radius r_0 , and ε_0 be as in [Wue26c, Lemma 5.6]. After shrinking r_0 and*

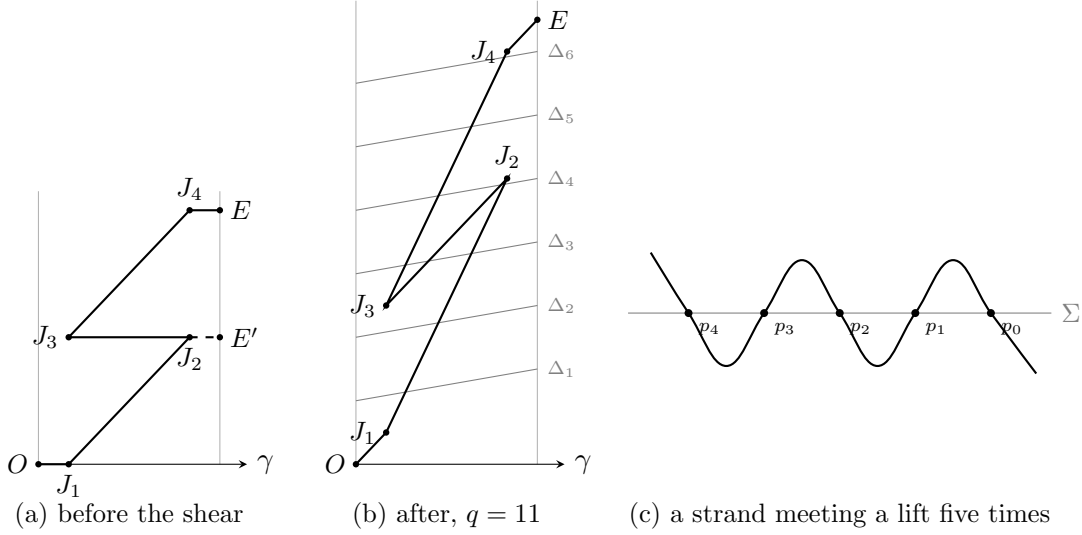


FIGURE 2. (a) The lift of the unperturbed curve for $T(3, 5)$, at $\varepsilon = 0$ (Corollary 4.8); the dashed segment is B_2 when it follows H . Vertical units are multiples of π . (b) Its image under S_{-m} for $q = 11$, with the lines Δ_k ; the lines Δ_3 and Δ_4 are crossed three times. (c) The configuration of Lemma 4.16: the points $p_0, \dots, p_4$ are ordered by decreasing γ , and the lobes alternate above and below the strand Σ .

ε_0 , the following holds for every $\varepsilon \in \mathcal{C}_a$ with $0 < |\varepsilon| < \varepsilon_0$ and each of the two arcs of $W_\varepsilon \cap U_c$, traversed from its end on B to its end on C_0 or conversely.

- (1) Call arrival direction the direction of traversal along the first half-branch towards c , and departure direction that along the second half-branch away from c ; each is a multiple of $\pm P_B$ or $\pm V$. For any prescribed $\vartheta > 0$, the velocity of Π along the arc stays within angle ϑ of the closed convex cone spanned by the arrival and departure directions, which has opening less than π . Consequently, if ℓ' is a constant line field not parallel to the arrival or departure direction, and ϑ is small, the net number of passages of the tangent line through ℓ' is ± 1 if one of the two directions of ℓ' lies inside that cone, the sign being that of the rotation from the arrival to the departure direction, and 0 otherwise.
- (2) If the two joined half-branches move λ' in opposite senses away from c , then $\lambda' \circ \Pi$ is strictly monotone along the arc.
- (3) Along the arc, $|\lambda' \circ \Pi - \lambda'(\Pi(c))| \leq C_{\lambda'}(r_0 + |\varepsilon|)$, with $C_{\lambda'}$ independent of ε .

Proof. In the proof of [Wue26c, Lemma 5.6] the arc is parametrised by t' , with $s' = -\kappa/t'$, $\kappa = \kappa_c(\varepsilon)$ and $|\kappa|/r_0 \leq |t'| \leq r_0$, and its velocity is $(\kappa/t'^2)P_B + 2t'V + e(t')$ with $|e| \leq C(r_0 + |\varepsilon|^{2/3}a^{-1/3})A(t')$, where $A(t') = |\kappa|/t'^2 + |t'|$. Along the arc the signs of κ and t' are fixed, so the main term is a positive combination of $\text{sign}(\kappa)P_B$ and $\text{sign}(t')V$, of length at least $c_0A(t')$. At the end where $|t'| \approx r_0$ the term $2t'V$ dominates, and at the end where $|t'| \approx |\kappa|/r_0$ the term $(\kappa/t'^2)P_B$ does; these are the directions of traversal along the two half-branches. Shrinking r_0 and ε_0 makes $|e|/A$ as small as we wish, which gives the first assertion of (1). The tangent direction moves continuously in a sector of opening less than π , which gives the count of passages. For (2), λ' applied to the main term is $\kappa\lambda'(P_B)/t'^2 + 2\lambda'(V)t'$. In the case of opposite senses both summands have the same sign, so its absolute value is at least a fixed multiple of A , and the error does not change the sign. For (3), Π_ε is uniformly Lipschitz on U_c and $\Pi_\varepsilon(c) = \Pi(c) + O(|\varepsilon|)$. $\square$

The lemma avoids one step of the proof of [Wue26c, Lemma 5.6], the uniqueness of the critical point of φ in the case of equal senses, which requires C^1 control of e/A ; the arguments below use only (1)–(3). For $\lambda' = \lambda$ we have $\lambda(1, 0) = q - 6 \neq 0$ and $\lambda(1, 6) = q \neq 0$, so the lemma applies at c_+ and c_- .

Proposition 4.10. *Let $a > 0$ and $\varepsilon \in \mathcal{C}_a$ with $|\varepsilon|$ small. The critical values at $c_\pm$ are*

$$\kappa_\pm(\varepsilon) = \mp \frac{\sqrt{3}}{2} \varepsilon_A - \frac{1}{2} \varepsilon_B + O(|\varepsilon|^2),$$

for the covector $\ell = (-2\tau, -1)$ of (V5). The resolution is

$$\text{joined} \iff |\varepsilon_B| < \sqrt{3} |\varepsilon_A|, \quad \text{split} \iff |\varepsilon_B| > \sqrt{3} |\varepsilon_A|.$$

For small a both types occur in $\mathcal{C}_a$: the joined sectors contain the ε_A -axis, and the split sectors the ε_B -axis. The two lines excluded from $\mathcal{C}_a$ are $\{\varepsilon_B = \mp \sqrt{3} \varepsilon_A\}$, the boundaries between the sectors.

Proof. At $c = (\frac{\pi}{2}, \frac{\pi}{2}, \tau)$, $\tau = \pm \tau_1$, put $\Lambda = \ell \cdot D^2\Psi(T_B, X) = -8\tau$ (V5). The tangent T_C of C_0 at c is the second null direction of the reduced Hessian; it has the form $X + \alpha T_B$, with ∂_v -component 1 because v is a regular parameter of C_0 at c (V4). Since $\ell \cdot D^2\Psi(T_B, T_B) = 0$, we have $\ell \cdot D^2\Psi(T_B, T_C) = \Lambda$. So in coordinates (s, t) along (T_B, T_C) the reduced function is $g = \kappa_c(\varepsilon) + \Lambda st + (\text{higher order})$, and the arc of $\{g = 0\}$ in the quadrant with $\text{sign}(s) \text{sign}(t) = -\text{sign}(\kappa_c \Lambda)$ joins the half-branch of B in direction $\text{sign}(s)T_B$ to the half-branch of C_0 in direction $\text{sign}(t)T_C$. Now $+T_C$ points into C_0^R at both double points, and $+T_B = \partial_\tau$ points into B_0 at c_+ and into B_1 at c_- . At c_+ , $\Lambda < 0$, so B_0 is joined to the half with $\text{sign}(t) = \text{sign}(\kappa_+)$; at c_- , $\Lambda > 0$, so B_2 (where $s < 0$) is joined to the half with $\text{sign}(t) = \text{sign}(\kappa_-)$. The resolution is split exactly when B_0 and B_2 are joined to the same half, that is, when $\kappa_+ \kappa_- > 0$.

For the critical values, the perturbed phases for $n = 5$ are $\Phi_1 = 2u + 3\varepsilon_A \sin u$, $\Theta_1 = 3v - 2\varepsilon_B \sin v$, $\Phi_2 = -u - 2\varepsilon_A \sin u$ and $\Theta_2 = -v + \varepsilon_B \sin v$, and $\Psi = (E2, E1)$ at $\varepsilon = 0$ [Wue26c, Lemma 3.1]. At c this gives $\partial_\varepsilon \Psi_1 = (-3\tau, 2)$ and $\partial_\varepsilon \Psi_2 = (2, -\tau)$, so $d\kappa_c(0) = \ell \cdot \partial_\varepsilon \Psi = -2\tau(2, -\tau) - (-3\tau, 2) = (-\tau, 2\tau^2 - 2) = (-\tau, -\frac{1}{2})$. Hence $\kappa_+ \kappa_- = \frac{1}{4} \varepsilon_B^2 - \frac{3}{4} \varepsilon_A^2 + O(|\varepsilon|^3)$. On $\mathcal{C}_a$ both linear parts are at least $a|\varepsilon|$ in absolute value, so for $|\varepsilon|$ small the signs of $\kappa_\pm$ are those of their linear parts. $\square$

In particular the proposition decides which alternative of [Wue26c, Prop. 5.7(4)] occurs for $n = 5$; the numerical observation of [Wue26c, Rem. 5.8] for $n = 5$ concerns perturbations in the joined sectors.

Let $\tilde{A}$ be the lift of the arc of [Wue26c, §6.1], starting at O , and $\hat{A} = S_{-m}(\tilde{A})$. In the split case let $\tilde{C}$ be the lift of the remnant circle through the lifts of Corollary 4.8, and $\hat{C} = S_{-m}(\tilde{C})$. The values of φ at the sheared vertices are

$$\begin{aligned} \varphi(S_{-m}O) &= 0, & \varphi(S_{-m}J_1) &= \frac{(q-6)\pi}{6}, & \varphi(S_{-m}J_2) &= 4\pi + \frac{5(q-6)\pi}{6}, & \varphi(S_{-m}J_3) &= 4\pi + \frac{(q-6)\pi}{6}, \\ \varphi(S_{-m}J_4) &= 8\pi + \frac{5(q-6)\pi}{6}, & \varphi(S_{-m}E) &= (q+2)\pi, & \varphi(S_{-m}E') &= (q-2)\pi. \end{aligned} \tag{4.5}$$

We write $\varphi_i = \varphi(S_{-m}J_i)$. Since $q - 6$ is odd, each φ_i differs from $4\pi\mathbb{Z}$ by an odd multiple of $\pi/6$, and so lies at distance at least $\pi/6$ from $2\pi\mathbb{Z}$.

Proposition 4.11. *Let $a > 0$. There are neighbourhoods U_{J_i} of the junctions $S_{-m}J_i$ (the images of $U_{c_\pm}$) and of the corners such that for every $\varepsilon \in \mathcal{C}_a$ with $0 < |\varepsilon|$ sufficiently small, the threshold depending on q :*

- (1) $\hat{A}$ is an embedded proper arc from $(0, 0)$ to $(\pi, (q+3)\pi)$ in the joined case and to $(\pi, (q-1)\pi)$ in the split case, with interior in the open strip. It leaves $(0, 0)$ with limiting slope $q - 5 + O(|\varepsilon|)$, and $\varphi > 0$ on its initial segment.

- (2) Outside $\bigcup_i U_{J_i}$ and the corner neighbourhoods, $\hat{A}$ is C^1 -close to the image under S_{-m} of the polygon of Corollary 4.8, and φ is strictly monotone with $|d\varphi|$ bounded below: increasing near $S_{-m}(B_0)$, $S_{-m}(H)$, $S_{-m}(H')$, $S_{-m}(B_2)$, and decreasing near $S_{-m}(B_1^{-1})$. In U_{J_1} , in U_{J_4} , and at the junction of H with B_2 in the split case, φ is strictly monotone. In U_{J_2} and U_{J_3} , $|\varphi - \varphi_i| < \pi/12$. Near the corners $|d\varphi|$ is bounded below.
- (3) For a sub-arc α of $\hat{A}$ with endpoints outside $\bigcup_i U_{J_i}$, $\mu(\alpha, \ell_1)$ is the number of neighbourhoods U_{J_2} traversed by α minus the number of U_{J_3} traversed. All critical points of φ on $\hat{A}$ lie in $U_{J_2} \cup U_{J_3}$.

In the split case $\hat{C}$ is a restricted immersed circle that misses the edges and has vertical degree 2. Its lift is $(0, 4\pi)$ -periodic and alternates between pieces C^1 -close to $S_{-m}(B_1^{-1}) + (0, 4\pi j)$, on which φ decreases from $\varphi_2 + 4\pi j$ to $\varphi_3 + 4\pi j$, and pieces C^1 -close to $S_{-m}(H') + (0, 4\pi j)$, on which φ increases from $\varphi_3 + 4\pi j$ to $\varphi_2 + 4\pi(j+1)$; the junctions are of the types J_2 and J_3 , and (2), (3) hold on it verbatim.

Proof. (1) By [Wue26c, Lemma 6.1], $\tilde{A}$ is embedded with interior in the strip and ends at E or E' , and S preserves γ and maps $(\pi, 2\pi j)$ to $(\pi, 2\pi j + (q-5)\pi)$. The limiting slope is that of $\Pi(B)$, which is 0, plus $q-5$, up to $O(|\varepsilon|)$, by [Wue26c, Cor. 5.3] and Lemma 4.6(c). In the corner chart of [Wue26c, Lemma 5.1] the lift is $y(c_1(\varepsilon), c_2(\varepsilon)) + O(y^3)$ with $(c_1, c_2)(0)$ a positive multiple of $(1, 0)$, so on $\hat{A}$ we get $\varphi = y(c_2 + (q-6)c_1) + O(y^3)$, which is positive for small $y > 0$.

(2) Away from the junctions and corners, $\tilde{A}$ is C^1 -close to the polygon of Corollary 4.8 [Wue26c, Prop. 5.7(2)]. The image under S_{-m} of the polygon has velocities that are positive multiples of $(1, q-5)$ on $S_{-m}(B_0)$ and $S_{-m}(B_2)$, of $(1, q+1)$ on $S_{-m}(H)$ and $S_{-m}(H')$, and of $(-1, 5-q)$ on $S_{-m}(B_1^{-1})$, along which $d\varphi$ is $q-6$, q and $-(q-6)$. At the junctions we apply Lemma 4.9 with $\lambda' = \lambda$:

junction	first half-branch, away from c	second half-branch	conclusion
J_1 at c_+ , $B_0 \rightarrow H$	B_0 decreases λ	H increases λ	(2): monotone
J_2 at c_- , $H \rightarrow B_1^{-1}$	H decreases λ	B_1^{-1} decreases λ	(3)
J_3 at c_+ , $B_1^{-1} \rightarrow H'$	B_1 increases λ	H' increases λ	(3)
J_4 at c_- , $H' \rightarrow B_2$	H' decreases λ	B_2 increases λ	(2): monotone
split, at c_- , $H \rightarrow B_2$	H decreases λ	B_2 increases λ	(2): monotone

Choose r_0 with $C_\lambda r_0 < \pi/24$ and then $|\varepsilon|$ with $C_\lambda |\varepsilon| < \pi/24$. Near the corners we use [Wue26c, Cor. 5.3] and Lemma 4.6(c); its proof uses only that the limiting slope differs from 1, and here it is $q-5 \geq 2$.

(3) By Lemma 4.6(f), $\mu(S_{-m} \circ \tilde{\alpha}, \ell_1) = \mu(\tilde{\alpha}, \ell')$, where ℓ' is spanned by $(1, 6-q)$, at angle $-\arctan(q-6)$, equivalently $\pi - \arctan(q-6) \in (\frac{\pi}{2}, \frac{3\pi}{4}]$. We compute on the unshaped curve. On the C^1 -close pieces the directions are near $(\pm 1, 0)$ or $(1, 6)$, and near the corners the slope is $O(|\varepsilon|)$; none of these is parallel to $(1, 6-q)$, so there are no passages there. In U_{J_2} the velocity moves, by Lemma 4.9(1), from near $(1, 6)$ to near $(-1, 0)$ inside the cone of angles $[\arctan 6, \pi]$. This cone contains the direction at angle $\pi - \arctan(q-6)$ and not its opposite, and the rotation is counterclockwise; so the count is $+1$. In U_{J_3} the motion is reversed and the count is -1 . In U_{J_1} , U_{J_4} and the split junction the cone is $[0, \arctan 6]$, which contains neither direction. A critical point of φ on $\hat{A}$ is a point where the tangent of $\hat{A}$ is parallel to $(1, 1)$, that is, where the tangent of $\tilde{A}$ is parallel to $(1, 6-q)$; by the above this happens only in $U_{J_2} \cup U_{J_3}$.

For the remnant circle, the properties of being restricted, avoiding the edges and having vertical degree 2 follow from [Wue26c, Lemma 6.2] and Lemma 4.6(b)–(d). Its pieces are those of Corollary 4.8, with junctions of type J_2 (from H' to B_1^{-1} , at c_-) and J_3 (from B_1^{-1} to H' , at c_+), and the estimates above apply to it by compactness and periodicity. $\square$

Remark 4.12. The smallness threshold for ε in Proposition 4.11 depends on q through the constant C_λ , which grows linearly in q . The margins that matter in the next subsection are the distances from φ_3 to the nearest value in $2\pi\mathbb{Z}$ above it and from φ_2 to the nearest value in $2\pi\mathbb{Z}$ below it. Both are at least $\pi/6$; the first equals $\pi/6$ exactly when $q \equiv 5 \pmod{12}$, and the second exactly when $q \equiv 11 \pmod{12}$. Above the threshold a cancelling pair of generators can be missing (a line Δ_k crossed once instead of three times); the homology is the same.

4.5. Generators. Put

$$N_q = \#\left\{j \in \mathbb{Z} : \frac{q-6}{12} < j < \frac{5(q-6)}{12}\right\} = \left\lfloor \frac{5(q-6)}{12} \right\rfloor - \left\lfloor \frac{q-6}{12} \right\rfloor;$$

both bounds are non-integers because $q-6$ is odd. Thus $N_7 = 0$, $N_9 = 1$, $N_{11} = N_{13} = 2$, $N_{15} = 3$ and $N_{17} = N_{19} = 4$. We call a transverse crossing of a lift of L_1 with a line Δ_k an *up-crossing* or a *down-crossing* according as φ increases or decreases through $2\pi k$ there.

Proposition 4.13. *For $\varepsilon \in \mathcal{C}_a \setminus \{0\}$ sufficiently small, and then ϵ sufficiently small:*

- (1) L_0 and L_1 meet transversally. The arc carries r_+ , which lies on S_+^0 , and every component carries a pair $x_P^\pm \in S_\pm^k$ over each transverse crossing P of (one period of) its lift with a line Δ_k , with $k \geq 1$ for the arc. No generator lies over the double point of L_0 .
- (2) In the joined case, $\hat{A}$ crosses Δ_k three times (up, down, up) if $\varphi_3 < 2\pi k < \varphi_2$, that is for $k = 2 + j$ with $\frac{q-6}{12} < j < \frac{5(q-6)}{12}$, and once (up) for every other $k \in \{1, \dots, \frac{q+1}{2}\}$; there are no other crossings. The arc carries $q + 2 + 4N_q$ generators.
- (3) In the split case, $\hat{A}$ crosses $\Delta_1, \dots, \Delta_{(q-3)/2}$ once each (up) and carries $q - 2$ generators; one period of $\hat{C}$ crosses N_q lines downward and $N_q + 2$ upward, and $\hat{C}$ carries $4 + 4N_q$ generators.

In both cases the complex has $q + 2 + 4N_q$ generators.

Proof. The argument of [Wue26c, Prop. 6.3] applies. By Proposition 4.11(2), outside small neighbourhoods of its transverse crossings with the lines Δ_k and of the corners, each curve stays a positive distance from $\bigcup_k \Delta_k$: in U_{J_2} and U_{J_3} the values of φ stay within $\pi/12$ of φ_i , and the same holds in U_{J_1} , U_{J_4} by Lemma 4.9(3). Near a crossing, $|d\varphi|$ is bounded below and the slope is bounded away from 1, so the curve meets each of S_+^k and S_-^k once, transversally. The corner $(0,0)$ contributes exactly r_+ [HHK18, Lemma 5.1], [Wue26c, Cor. 5.3(4)]. Since $\hat{A}$ leaves $(0,0)$ with slope $q - 5 > 1$, it meets the strand S_+^0 , which passes $(0,0)$ through $(-\epsilon, \epsilon)$, near $\gamma = 2\epsilon/(q-6) > 0$, and it does not meet S_-^0 near the corner; so $r_+ \in S_+^0$. The far endpoint lies on $\{\varphi \in (2\mathbb{Z} + 1)\pi\}$, away from every Δ_k .

In the joined case φ is monotone along the pieces over $(0, \varphi_2)$ (near $S_{-m}(B_0)$ and $S_{-m}(H)$), over (φ_3, φ_2) (near $S_{-m}(B_1^{-1})$) and over $(\varphi_3, (q+2)\pi)$ (near $S_{-m}(H')$ and $S_{-m}(B_2)$). Here $\varphi_3 > 0$ and $(q+2)\pi - \varphi_2 = (q+18)\pi/6 > 0$. The lines with $2\pi k \in (\varphi_3, \varphi_2)$ are those with $k = 2 + j$ and $\frac{q-6}{12} < j < \frac{5(q-6)}{12}$, and the lines with $0 < 2\pi k < (q+2)\pi$ are $k = 1, \dots, \frac{q+1}{2}$. This gives $1 + 2(\frac{q+1}{2} + 2N_q)$ generators. In the split case φ increases monotonically along the arc over $(0, (q-2)\pi)$; on one period of the circle it decreases over (φ_3, φ_2) , crossing N_q lines, and then increases over $(\varphi_3, \varphi_2 + 4\pi)$, crossing $N_q + 2$ lines.

At $\varepsilon = 0$ every crossing of the sheared polygons with a line Δ_k lies on a piece of B , where $\varphi = (q-6)\gamma + 4\pi j$, or on a half of C_0 , where $\varphi = q\gamma + (2j+1)\pi$; so its abscissa is $2\pi i/(q-6)$ or $(2i+1)\pi/q$ with $i \in \mathbb{Z}$. Neither equals $\pi/2$, because $q-6$ and q are odd. Hence for small perturbations no crossing lies near the double point $(\frac{\pi}{2}, \frac{\pi}{2})$ of L_0 . $\square$

For $q = 7$ there are $9 = 1 + |\sigma(K_7)|$ generators. In general the count exceeds $1 + |\sigma(K_q)| = q + 2$ by $4N_q$; we compare it with the traceless characters in Remark 4.19. The combinatorics of the

sheared polygons used in this proof (the crossing lists, N_q , and the abscissae below) have also been checked in exact rational arithmetic for all odd $q \leq 401$ (Appendix A).

4.6. Gradings.

Lemma 4.14. *Let p, p' be generators on strands of the same sign $\sigma \in \{+, -\}$, with lifts $\tilde{p} \in S_\sigma^k$ and $\tilde{p}' \in S_\sigma^{k'}$ joined by a sub-path $\tilde{\alpha}$ of a lift of a component of L_1 that lies in the open strip and has slope bounded away from 1 at $\tilde{p}$ and $\tilde{p}'$. Then*

$$\text{gr}(p, p') \equiv \mu(\alpha_1, \ell_1) + 2(k - k') \pmod{4},$$

where α_1 is the projection of $\tilde{\alpha}$, run from p' to p .

Proof. This is the proof of [Wue26c, Lemma 6.8], without its hypothesis that $\tilde{\alpha}$ contain no critical point of φ . Apply (2.3) with α_0 the branch of L_0 on the front face from p to the point over p' ; its lift runs along S_σ^k from $\tilde{p}$ to $\tilde{p}' - (0, 2\pi(k' - k))$ inside the open strip. Then $\mu(\alpha_0, \ell_1) = 0$, because the slope of the strand stays on one side of 1. The triple indices at p and p' agree: both are 0 for $\sigma = +$ and 1 for $\sigma = -$, since the tangent line of L_1 is bounded away from slope 1. Finally the lift of $\alpha_0 * \alpha_1$ runs in the open strip from $\tilde{p}$ to $\tilde{p} + (0, 2\pi(k - k'))$, so $\alpha_0 * \alpha_1$ is homotopic in P^* to $k - k'$ times the core of the annulus obtained from the strip, which encircles two corners; hence $z(\alpha_0 * \alpha_1) \equiv 2(k - k')$. $\square$

Proposition 4.15. *Put $\deg = \text{gr} - \sigma(K_q)$, normalised by $\text{gr}(r_+) = \sigma(K_q)$ as in (2.5). For a crossing P of $\hat{A}$ with Δ_k ,*

$$\deg x_P^+ = \begin{cases} 2k, & P \text{ an up-crossing,} \\ 2k + 1, & P \text{ a down-crossing,} \end{cases} \quad \deg x_P^- = \deg x_P^+ - 1.$$

On the remnant circle the same formula holds for the relative grading: for crossings P, P' of $\hat{C}$ with $\Delta_k, \Delta_{k'}$, $\deg x_P^+ - \deg x_{P'}^+ \equiv 2(k - k') + \delta_P - \delta_{P'}$, with $\delta_P = 1$ for down-crossings and 0 for up-crossings.

Proof. On the arc, apply Lemma 4.14 to $p = x_P^+ \in S_+^k$ and $p' = r_+ \in S_+^0$ along $\hat{A}$. By Proposition 4.11(3), μ counts +1 for each U_{J_2} and -1 for each U_{J_3} passed. Up-crossings lie before U_{J_2} or after U_{J_3} , so $\mu = 0$; down-crossings lie between them, so $\mu = +1$. Then $\text{gr}(x_P^+, x_P^-) = 1$ [HHK18, §5.1]. On the circle the same argument applies along its lift: between two up-crossings the neighbourhoods U_{J_2} and U_{J_3} alternate, so their net count is 0, and from an up-crossing to the next down-crossing it is +1. Over one period $\mu = 0$, $z \equiv 0$ and k changes by 2, which is consistent. $\square$

4.7. Bigons. A bigon lifts to an immersed disc in $\mathbb{R}^2 \setminus (\pi\mathbb{Z})^2$ whose boundary is a sub-path of one lift of a component of L_1 together with an arc of one strand [Wue26c, Lemma 6.4]. The following lemma counts such discs in the configuration of Figure 2(c). Each strand S_σ^k is a graph over γ and leaves the open strip at both ends; a transverse crossing with it is an up- or down-crossing according as the curve passes from below the strand to above it or conversely.

Lemma 4.16. *Let $\tilde{L}$ be an embedded lift of a component of L_1 that lies in the open strip $\{0 < \gamma < \pi\}$, apart from endpoints at lattice points, and is either an arc or a properly embedded line. Let $\Sigma = S_\sigma^k$, and suppose that $\tilde{L} \cap \Sigma = \{p_0, p_1, \dots, p_{2d}\}$, all transverse and in this order along $\tilde{L}$, with p_{2i} up-crossings and p_{2i+1} down-crossings, and with $\gamma(p_0) > \gamma(p_1) > \dots > \gamma(p_{2d})$. Then the bigons whose boundary lifts to a sub-path of $\tilde{L}$ and an arc of Σ are exactly $2d$ bigons, one from p_{2i-1} to p_{2i-2} and one from p_{2i-1} to p_{2i} for each $1 \leq i \leq d$; each of them lifts to an embedded disc in $\mathbb{R}^2$.*

Proof. For $0 \leq s < 2d$ let Γ_s be the union of the sub-path $\tilde{L}[p_s, p_{s+1}]$ and the arc $\Sigma[p_{s+1}, p_s]$. The first meets Σ only at its ends, because the points are consecutive along $\tilde{L}$, and the second contains no point of $\tilde{L}$ in its interior, because they are consecutive in the order of γ . So Γ_s is a Jordan curve; let D_s be its bounded complementary component. The two closed half-planes complementary to the strip are connected, unbounded and disjoint from Γ_s , so D_s lies in the strip and contains no lattice point. Between consecutive crossings the curve stays on one side of Σ , so D_s lies above Σ if p_s is an up-crossing and below Σ if it is a down-crossing: the domains alternate. They are pairwise disjoint, since their boundaries meet only at shared endpoints and $\overline{D_s} \cap \Sigma = \Sigma[p_{s+1}, p_s]$.

Existence. At the corners of D_s the four continuations leave D_s without entering it: the rays of $\tilde{L}$ beyond p_s and p_{s+1} meet Γ_s nowhere else, by embeddedness and the order of γ , and run to lattice points outside the strip or to infinity; the rays of Σ leave the strip. So both corners are convex, and D_s projects to a bigon. If D_s lies above Σ , its counterclockwise boundary runs along Σ in the direction of increasing γ , from p_{s+1} to p_s ; if below, from p_s to p_{s+1} . In both cases the bigon runs from the down-crossing to the up-crossing, in agreement with Proposition 4.15 and [HHK18, Prop. 3.8].

Uniqueness. The L_1 -side of a bigon as in the statement is $\tilde{L}[p_i, p_j]$ for some $i \neq j$, and its L_0 -side is $\Sigma[p_j, p_i]$, since Σ is an embedded line. If $|i - j| \geq 2$, its boundary is, as a 1-cycle, the sum of the $|i - j|$ consecutive curves Γ_s between p_i and p_j , each traversed with its sub-path of $\tilde{L}$ in the same direction. Since the domains alternate above and below Σ , consecutive curves Γ_s are traversed with opposite orientations, so the boundary winds -1 times around the points of one of the domains. This is impossible for the boundary of an orientation-preserving immersed disc, whose winding number around a point is its number of preimages. If $|i - j| = 1$ the boundary is Γ_s , and only the direction from the down-crossing is counterclockwise. There is at most one bigon from a given generator to another [HHK18, Cor. 3.13]. $\square$

Proposition 4.17. (1) *In the joined case the complex is the direct sum, over the strands S_σ^k , of subcomplexes $C_{k,\sigma}$. If Δ_k is crossed once, $C_{k,\sigma}$ has one generator. If Δ_k is crossed three times, at P_1 (up), P_2 (down) and P_3 (up), then $\partial x_{P_2}^\sigma = x_{P_1}^\sigma + x_{P_3}^\sigma$, and $H(C_{k,\sigma}) = \mathbb{F}_2$, generated by $[x_{P_1}^\sigma]$ in degree $2k$ ($\sigma = +$) or $2k - 1$ ($\sigma = -$). There are no other bigons; in particular none involves r_+ . The differential has rank $2N_q$.*

(2) *In the split case the arc carries no bigon. The complex of the remnant circle is the direct sum of four chains, one for each pair $(\sigma, k \bmod 2)$; each has generators $p_0, \dots, p_{2d}$ with $\partial p_{2i-1} = p_{2i-2} + p_{2i}$, and homology $\mathbb{F}_2$ in the degree of its up-crossings. The circle contributes rank 1 in each of the four degrees, for every choice of absolute grading on it; its differential has rank $2N_q$.*

Proof. By [Wue26c, Lemma 6.4] the boundary of a bigon lifts to a sub-path of one lift of a component of L_1 and an arc of one strand. For the arc there is one lift through each generator, $\hat{A}$. The only elements of G preserving the open strip are the vertical translations, so for the circle a sub-path of a lift is unique up to the translation $T = (0, 4\pi)$, which carries S_σ^k to S_σ^{k+2} .

(1) On a strand met three times, the order along $\hat{A}$ is P_1 (near $S_{-m}(H)$), P_2 (near $S_{-m}(B_1^{-1})$), P_3 (near $S_{-m}(H')$), because $\varphi_1 < 2\pi k < \varphi_2 < \varphi_4$. At $\varepsilon = 0$ the abscissae of the three crossings are

$$\gamma_1 = \frac{(2k+1)\pi}{q}, \quad \gamma_2 = \frac{2\pi(k-2)}{q-6}, \quad \gamma_3 = \frac{(2k-3)\pi}{q},$$

from $q\gamma - \pi = 2\pi k$ on $S_{-m}(H)$, $4\pi + (q-6)\gamma = 2\pi k$ on $S_{-m}(B_1^{-1})$ and $q\gamma + 3\pi = 2\pi k$ on $S_{-m}(H')$. With $x = 2\pi(k-2)$,

$$\gamma_2 - \gamma_3 = \frac{6\left(x - \frac{(q-6)\pi}{6}\right)}{q(q-6)}, \quad \gamma_1 - \gamma_2 = \frac{6\left(\frac{5(q-6)\pi}{6} - x\right)}{q(q-6)}.$$

Since $\varphi_3 < 2\pi k < \varphi_2$ and both $\varphi_2 - 2\pi k$ and $2\pi k - \varphi_3$ are at least $\pi/6$, both differences are at least $\pi/(q(q-6))$. So for small perturbations $\gamma(P_1) > \gamma(P_2) > \gamma(P_3)$, and Lemma 4.16 applies with $d = 1$. A strand met once carries one generator, and S_+^0 carries only r_+ , since $\varphi > 0$ after the corner. The generators on S_σ^k have degrees $(u, u+1, u)$ with $u = 2k$ or $2k-1$ (Proposition 4.15). There are N_q such lines, each with two strands, so the differential has rank $2N_q$.

(2) The arc is φ -monotone, so it carries no bigon [Wue26c, Lemma 6.4]. Along the lift of the circle the pieces alternate, dec_j near $S_{-m}(B_1^{-1}) + (0, 4\pi j)$ and inc_j near $S_{-m}(H') + (0, 4\pi j)$, and each crosses a given line Δ_k at most once. By Proposition 4.11, the range of φ on dec_j is the intersection of its ranges on inc_{j-1} and inc_j . Hence the crossings of Δ_k along the lift alternate between up and down, beginning and ending with an up-crossing (as $\varphi \rightarrow \pm\infty$ along the lift), and a down-crossing on dec_j has the up-crossings on inc_{j-1} and inc_j as neighbours. The triple $(\text{inc}_{j-1}, \text{dec}_j, \text{inc}_j)$ is the triple $(S_{-m}(H), S_{-m}(B_1^{-1}), S_{-m}(H'))$ of the arc translated by $(0, 4\pi(j-1))$, so the abscissa computation above shows that γ decreases strictly along the sequence, and Lemma 4.16 applies to every strand. The complex of the circle in P^* is the quotient by $\langle T \rangle$: its generators are the T -orbits of crossing points, represented exactly once on the four strands $S_\sigma^{k_0}$ with $k_0 \in \{0, 1\}$, and its bigons are the T -orbits of the domains of Lemma 4.16. Each chain has homology $\mathbb{F}_2$ in the degree of its up-crossings. By Proposition 4.15 these degrees are $2k_0$ and $2k_0 - 1$ up to a common shift, that is, 0, 2, 3, 1. Finally $4 + 4N_q = \sum (2d+1)$ over the four chains, so $\sum d = 2N_q$ is the rank of the differential. $\square$

4.8. Proof of Theorem 1.2.

Theorem 4.18. *Let $q \geq 7$ be odd and $a > 0$. For the decomposition $K_q = (Y, T) \cup_{t_e^{-(q-5)/2}} (D, U)$, every sufficiently small $\varepsilon \in \mathcal{C}_a \setminus \{0\}$, the threshold depending on q , and every sufficiently small earring parameter:*

- (1) L_1 is a restricted immersed 1-manifold, (L_0, L_1) is admissible, and $C(L_0, L_1)$ has $q+2+4N_q$ generators;
- (2) the differential has rank $2N_q$ and is described by Proposition 4.17; it is nonzero for $q \geq 9$, and no bigon involves r_+ ;
- (3) with $\text{gr}(r_+) = \sigma(K_q)$, $H^\natural(Y, T, \pi)$ has rank $q+2$ and graded dimensions $(1 + N_3, N_1, N_1, N_3)$ in degrees $\text{gr} - \sigma(K_q) = 0, 1, 2, 3$, $N_1 = \lceil \frac{q+1}{4} \rceil$, $N_3 = \lfloor \frac{q+1}{4} \rfloor$.

This holds in both resolutions, and for every choice of absolute grading on the remnant circle.

Proof. (1) By Proposition 4.11(1) and Lemma 4.6(c), the arc is a proper immersion with embedded lift, so it has no fishtail and is a restricted immersed arc; the remnant circle is restricted by Proposition 4.11. Transversality is Proposition 4.13. Admissibility follows as in [Wue26c, Lemma 6.6]: loops on L_0 have classes in the span of $e_{(0,0)}$ and $e_{(\pi,\pi)}$ in $H_1(P^*)$, while a loop running j times around the remnant circle, which is homotopic in the strip to twice the core, has class $\pm 2j(e_{(0,0)} + e_{(0,\pi)})$, which lies in that span only if $j = 0$. The generator count is Proposition 4.13.

(2) This is Proposition 4.17, with $N_q \geq 1$ for $q \geq 9$.

(3) In the joined case, sum Proposition 4.17(1) over $k = 1, \dots, K$ with $K = \frac{q+1}{2}$, and add r_+ in degree 0. The strand S_+^k contributes degree $2k$ and S_-^k degree $2k-1$; modulo 4, odd k gives degrees 2 and 1, and even k gives 0 and 3. So the homology is $(1 + \lfloor K/2 \rfloor, \lceil K/2 \rceil, \lceil K/2 \rceil, \lfloor K/2 \rfloor)$,

and $\lceil K/2 \rceil = N_1$, $\lfloor K/2 \rfloor = N_3$. In the split case the arc crosses Δ_k once for $k = 1, \dots, K - 2$, which gives $(\lfloor K/2 \rfloor, \lceil K/2 \rceil - 1, \lceil K/2 \rceil - 1, \lfloor K/2 \rfloor - 1)$, and the circle adds $(1, 1, 1, 1)$ by Proposition 4.17(2). The sum is the same vector. $\square$

Proof of Theorem 1.2. By Theorem 4.4, $K_q = K^{(-m)}$ with $m = (q - 5)/2$, so K_q is the union of the tangle of [HHK18, §10] for $T(3, 5)$ with $(r, s) = (2, -1)$ and the trivial tangle, glued by t_e^{-m} ; this is a decomposition as in [HHK18, (17)], and by Lemma 4.6 its restriction map is $S_{-m} \circ L_1$. Theorem 4.18 gives restrictedness, admissibility, the generators and the differential, and the graded dimensions of $H^\natural(Y, T, \pi)$, with $\text{gr}(r_+) = \sigma(K_q) = -(q + 1)$ (Proposition 3.5). By Proposition 3.6, $I^\natural(K_q; \mathbb{F}_2)$ has the same graded dimensions in Kronheimer and Mrowka's grading, so $H^\natural(Y, T, \pi) \cong I^\natural(K_q; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces. This holds for every sufficiently small perturbation in the open cone $\mathcal{C}_a$, in either of its kinds of sector (Proposition 4.10), with zero bounding cochain; in particular Conjecture 2.5 holds for K_q . $\square$

For $q = 3$ and $q = 5$ the knots are $T(3, 4)$ and $T(3, 5)$, and the corresponding statement over $\mathbb{Q}$ is [Wue26c, Thm. 1.1, Cor. 1.2]. The knots K_q with $q \geq 7$ are hyperbolic and not two-bridge (Proposition 4.5).

Remark 4.19. At $\varepsilon = 0$ the intersection points of $S_{-m}(\Pi(W))$ with Δ are the restrictions of the traceless characters of K_q , and the generator count of Proposition 4.13 is one more than twice the number of irreducible traceless characters. It can be written $q + 2 + 4N_q = q - 6 + 4(\lfloor \frac{5q+6}{12} \rfloor - \lfloor \frac{q+6}{12} \rfloor)$, which is the count of [FKP17, §5, p. 29] read consistently: the general formulas printed there give 13 generators at $q = 7$, whereas the explicit count on p. 28 is 9. For $q \geq 9$ the traceless character variety therefore has more points than are needed to generate $I^\natural(K_q)$, and the complex is not minimal. This agrees with the remark of Daemi and Scaduto that a direct computation from the character variety shows that $P(-2, 3, 7)$ is I -basic, while such a computation does not work for $P(-2, 3, q)$ with $q > 7$ [DS24b, p. 161].

Remark 4.20. Put $\chi(x_P^-) = x_P^+$ and $\chi(x_P^+) = \chi(r_+) = 0$. By Proposition 4.17 every bigon pairs $x_{P_2}^\sigma$ with $x_{P_1}^\sigma$ and $x_{P_3}^\sigma$ for both signs σ , so the differential commutes with χ . The pillowcase complex is thus an S-complex over $\mathbb{F}_2$, in the sense of (3.4), with $\delta_1 = \delta_2 = 0$ and $v = 0$, and its irreducible part has a differential of rank N_q . The homology of the span of the generators x_P^- has one class for each line Δ_k , $1 \leq k \leq \frac{q+1}{2}$, in degree $2k - 1$ (in the split case, with the remnant circle graded so that its generators x^- have odd degree, as in [Wue26c, Def. 6.13]): it has rank N_1 in degree 1 and N_3 in degree 3, the ranks of the irreducible instanton homology $I(K_q)$ of [DS24b, Prop. 8.30]. A chain-level comparison with $\tilde{C}(K_q; \mathbb{F}_2)$ is open. For torus knots the corresponding conjecture is formulated in [Wue26b, Conj. 1.5] and proved in part for $T(3, n)$ in [Wue26c, Thm. 1.3]; there the irreducible part has zero differential and the pillowcase differential is a component of type δ_1 , whereas for K_q with $q \geq 9$ it lies in the irreducible part.

Remark 4.21. The proofs above are by hand, but we have compared them with numerical computations of the perturbed curves, using the continuation code of [Wue26c] for W_ε and independent code for the shear, the generators, the degrees from (2.3) and the bigons whose lifts are embedded discs (Appendix A).

- *Joined sectors.* For ten perturbations with $|\varepsilon_A|$ between 0.002 and 0.3 and $|\varepsilon_B|$ between 0.0003 and 0.05, two earring parameters, and $q \in \{7, 9, \dots, 25, 29, 33\}$, the homology agrees with Theorem 4.18 in all 240 cases. Every bigon found runs from a down-crossing to an up-crossing on one strand, the number of bigons is twice the rank of the differential, and the degrees agree with Proposition 4.15. The generator count $q + 2 + 4N_q$ is attained in 178 cases; in the other 62, at the larger perturbations, cancelling pairs are missing, as in Remark 4.12.

- *Split sectors.* For five perturbations with $|\varepsilon_B| > \sqrt{3}|\varepsilon_A|$ and $|\varepsilon| \leq 0.031$, and $q = 7, 9, \dots, 21$, all 40 cases agree: the arc is φ -monotone with $q - 2$ generators, and the remnant circle has homology of rank 1 in each degree. The circle carries $4 + 4N_q$ generators in 37 cases; in the other three ($q = 17$, at the three larger perturbations) a cancelling pair is missing. A strand meets the lift of the circle up to seven times ($q = 21$).
- *Resolution type.* At $|\varepsilon| = 0.03$, in 24 equally spaced directions of which four lie on the excluded lines, the observed type agrees with Proposition 4.10 in all 20 remaining directions; the split ones are those at $75^\circ, 90^\circ, 105^\circ, 255^\circ, 270^\circ$ and 285° .

5. TWISTED TORUS KNOTS

For $n \equiv 1, 2 \pmod{6}$ the traceless character variety of the tangle of [HHK18, §10] for $T(3, n)$ is, before and after a small holonomy perturbation, a disjoint union of an arc and $2\lfloor n/6 \rfloor$ circles, and the image of the unperturbed arc in the pillowcase is a straight segment [Wue26c, Props. 3.6, 3.9 and 5.7]. There is no central circle, so the argument of Section 4 goes through without an analogue of Lemma 4.7. Regluing the trivial tangle by powers of the twist t_e produces the twisted torus knots $T(3, n; 2, m)$, and Theorem 5.1 below computes their pillowcase homology, identifies the knots, and compares the result with their instanton homology. On the pillowcase side, the one input beyond [Wue26c] and Section 4 is Hedden, Herald and Kirk's computation of the Floer homology of a circle that misses the edges [HHK18, Thm. 5.4], which we use for the closed components.

5.1. Statement. For $n \geq 1$ coprime to 3 and $m \geq 0$, the *twisted torus knot* $T(3, n; 2, m)$ is the closure of the positive braid

$$\beta_{n,m} = (\sigma_1 \sigma_2)^n \sigma_1^{2m} \in B_3,$$

that is, $T(3, n)$ with m positive full twists added on two adjacent strands; $T(3, n; 2, 0) = T(3, n)$. Clay and Watson write $T_{3,n}^m$ for this knot [CW13, §3.2], and Daemi and Scaduto write $T(3, n; 2, m)$ [DS24b, §8.3]. The knots of Section 4 are the case $n = 5$: $T(3, 5; 2, m) = P(-2, 3, 5 + 2m)$. Throughout we put

$$\nu = \lfloor n/3 \rfloor, \quad M = \begin{cases} m, & n \equiv 1 \pmod{3}, \\ m + 1, & n \equiv 2 \pmod{3}, \end{cases}$$

and for $n \equiv 1, 2 \pmod{6}$ we write $n = 6j + 1$ or $n = 6j + 2$, so that $\nu = 2j$.

For $n \geq 4$ coprime to 3 let (Y_n, T_n) be the tangle of [HHK18, §10] for $T(3, n)$ with the parameters $(r, s) = ((2n+1)/3, -2)$ if $n \equiv 1$ and $(r, s) = ((n+1)/3, -1)$ if $n \equiv 2 \pmod{3}$, as in [Wue26c], and let $K_h(3, n) = (Y_n, T_n) \cup_{t_e^h} (D, U)$ be the regluing of Section 4. Lemma 4.1 and its proof apply to every n , with the boundary words a, b, c of [Wue26c, (3.2)]: $\pi_1(S^3 \setminus K_h(3, n)) = \langle A, B \mid t_e^h(b) = c \rangle$ with $t_e^h(b) = (ba)^h b (ba)^{-h}$, and $K_{h-1}(3, n)$ differs from $K_h(3, n)$ by one full twist of two strands, hence by one crossing change. As in Section 4 we use the convention, shared with [HHK18] and [Wue26c], that $K_0(3, n)$ is the positive torus knot $T(3, n)$.

We write $\varepsilon = (\varepsilon_A, \varepsilon_B)$ for the parameters of the holonomy perturbation π of [Wue26c] and L_1^Y for the resulting curve of (Y_n, T_n) . By Lemma 4.6, the curve of (Y_n, T_n) in the decomposition of $K_{-m}(3, n)$ is $L_1 = S_{2m} \circ L_1^Y$, where $S_{2m}(\gamma, \theta) = (\gamma, \theta + 2m\gamma)$. For the trivial side we use the earring curve $L_0 = L_0^{\varepsilon, g}$ of [HHK18, Def. 3.4] with $\varepsilon > 0$ and $g = \delta \sin$; in [Wue26c], $\delta = 0$. Recall that $\ell(K)$ denotes the sum of the absolute values of the coefficients of Δ_K .

Theorem 5.1. *Let $n \geq 7$ with $n \equiv 1$ or $2 \pmod{6}$, let $m \geq 1$, define j, ν, M as above, and put $K = K_{-m}(3, n)$ and $(a, b) = (\lfloor M/2 \rfloor, \lceil M/2 \rceil)$.*

- (i) $K = T(3, n; 2, m)$.
- (ii) $\sigma(K) = \sigma(T(3, n)) - 2m = -(4\nu + 2M)$, and $\ell(K) = |\sigma(K)| + 1$.

- (iii) For every sufficiently small $\varepsilon \in \mathbb{R}^2 \setminus \{0\}$, L_1 is a restricted immersed 1-manifold consisting of an arc and $2j$ circles, and the circles miss the left and right edges and have vertical degree 2. There is $\epsilon_0 > 0$, depending on ε , such that for $0 < \epsilon < \epsilon_0$ and $|\delta| < \epsilon^2/2$ the arc meets L_0 transversely in $1 + 2M$ points and carries no bigon. For each such ϵ and almost every such δ the pair (L_0, L_1) is admissible, and with $\text{gr}(r_+) = \sigma(K)$ the graded dimensions of $H^\natural(Y_n, T_n, \pi)$ are

$$(1 + 2j + a, 2j + b, 2j + b, 2j + a) \quad \text{in degrees } \text{gr} - \sigma(K) = 0, 1, 2, 3,$$

whatever absolute gradings are chosen on the circles.

- (iv) $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, and $H^\natural(Y_n, T_n, \pi) \cong I^\natural(K; \mathbb{F}_2)$ as $\mathbb{Z}/4$ -graded vector spaces.

In particular, this decomposition of $T(3, n; 2, m)$ satisfies the conclusion of [HHK18, Conj. 6.5].

Remark 5.2. By Lemma 5.3 below, $T(3, 6j + 2; 2, m) = T(3, 6j + 1; 2, m + 1)$. The two families in Theorem 5.1 therefore consist of the knots $T(3, 6j + 1; 2, M)$ with $M \geq 1$, and the vector in (iii) depends only on (j, M) ; for $M \geq 2$ the knot arises from both tangles (Y_{6j+1}, T_{6j+1}) and (Y_{6j+2}, T_{6j+2}) . For $M = 1$ the knot $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$ is a torus knot, and Theorem 5.1 gives a second decomposition of it, with the homology found in [Wue26c]. For $M \geq 2$ the knot is neither a torus knot, nor alternating, nor a pretzel knot $P(-2, 3, q)$ (Lemma 5.6).

5.2. Alexander polynomials. Let ψ be the reduced Burau representation of B_3 ,

$$\psi(\sigma_1) = \begin{pmatrix} -t & 1 \\ 0 & 1 \end{pmatrix}, \quad \psi(\sigma_2) = \begin{pmatrix} 1 & 0 \\ t & -t \end{pmatrix}.$$

For $\beta \in B_3$ whose closure $\hat{\beta}$ is a knot, $\Delta_{\hat{\beta}}(t) \doteq \frac{1-t}{1-t^3} \det(I - \psi(\beta))$ by Burau's formula [Bir74, Thm. 3.11]; here $\doteq$ denotes equality up to a unit $\pm t^k$. Daemi and Scaduto record a formula for the Alexander polynomials of all twisted torus links [DS24b, Prop. 8.32]; we need the coefficient norm, which the closed form below displays directly.

Lemma 5.3. For $\nu, m \geq 0$, $\sigma_1^{-1}(\sigma_1\sigma_2)^{3\nu+2}\sigma_1^{2m}\sigma_1 = (\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2m+2}$ in B_3 . Hence $T(3, 3\nu + 2; 2, m) = T(3, 3\nu + 1; 2, m + 1)$ as oriented knots, and for every n coprime to 3 the knot $T(3, n; 2, m)$ is the closure of $(\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M}$.

Proof. The full twist $\Delta^2 = (\sigma_1\sigma_2)^3$ is central, and $(\sigma_1\sigma_2)^2 = \sigma_1(\sigma_2\sigma_1\sigma_2) = \sigma_1^2\sigma_2\sigma_1$ by the braid relation. Hence $(\sigma_1\sigma_2)^{3\nu+2}\sigma_1^{2m} = \Delta^{2\nu}\sigma_1^2\sigma_2\sigma_1^{2m+1}$, and conjugation by σ_1 gives $\Delta^{2\nu}\sigma_1\sigma_2\sigma_1^{2m+2} = (\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2m+2}$. Conjugate braids have isotopic closures. $\square$

For example, $P(-2, 3, 5 + 2m) = T(3, 5; 2, m) = T(3, 4; 2, m + 1)$ and $T(3, 6j + 1; 2, 1) = T(3, 6j + 2)$.

Proposition 5.4. Let $\nu, M \geq 0$ and let $K_{\nu, M}$ be the closure of $(\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M}$. Then

$$\Delta_{K_{\nu, M}}(t) \doteq \underbrace{\sum_{i=0}^{\nu-1} (t^{3i} - t^{3i+1})}_{L(t)} + \underbrace{\sum_{i=0}^{2M} (-1)^i t^{3\nu+i}}_{L_{\text{mid}}(t)} + \underbrace{\sum_{i=0}^{\nu-1} (t^{6\nu+2M-3i} - t^{6\nu+2M-3i-1})}_{L^*(t)}.$$

The three sums have disjoint exponent ranges, every coefficient is 0 or ± 1 , the nonzero coefficients alternate in sign, and $\ell(K_{\nu, M}) = 4\nu + 2M + 1$.

Proof. Put $P = \psi(\sigma_1\sigma_2) = \begin{pmatrix} 0 & -t \\ t & -t \end{pmatrix}$ and $T = t^{2M}$. Since $\text{tr } P = -t$ and $\det P = t^2$, the Cayley–

Hamilton theorem gives $P^3 = t^3 I$, and induction on M gives $\psi(\sigma_1)^{2M} = \begin{pmatrix} T & (1 - T)/(1 + t) \\ 0 & 1 \end{pmatrix}$.

The matrix $X = t^{3\nu} P\psi(\sigma_1)^{2M}$ has $\text{tr } X = -t^{3\nu+1}(t+T)/(1+t)$ and $\det X = t^{6\nu+2}T$, so $\det(I-X) = 1 - \text{tr } X + \det X$ is

$$F_{\nu,M}(t) := \det(I - \psi((\sigma_1\sigma_2)^{3\nu+1}\sigma_1^{2M})) = 1 + t^{3\nu+2} \frac{1+t^{2M-1}}{1+t} + t^{6\nu+2M+2} \quad (5.1)$$

(for $M=0$ the middle term is $t^{3\nu+1}$). Since $(1-t)/(1-t^3) = 1/(1+t+t^2)$, it suffices to show that $(1+t+t^2)(L+L_{\text{mid}}+L^*) = F_{\nu,M}$. From $(1+t+t^2)(1-t) = 1-t^3$ and $L^*(t) = t^{6\nu+2M}L(t^{-1})$,

$$(1+t+t^2)L = 1-t^{3\nu}, \quad (1+t+t^2)L^* = t^{6\nu+2M+2} - t^{3\nu+2M+2}.$$

Next, $L_{\text{mid}} = t^{3\nu}(1+t^{2M+1})/(1+t)$ and $(1+t+t^2)(1+t^{2M+1}) - (1+t)(1+t^{2M+2}) = t^2 + t^{2M+1}$, so

$$(1+t+t^2)L_{\text{mid}} = t^{3\nu} + t^{3\nu+2M+2} + t^{3\nu+2} \frac{1+t^{2M-1}}{1+t}.$$

The three products add up to $F_{\nu,M}$. The sums L , L_{mid} and L^* occupy the exponents $0, \dots, 3\nu-2$, then $3\nu, \dots, 3\nu+2M$, then $3\nu+2M+2, \dots, 6\nu+2M$, so no cancellation occurs, and they contribute 2ν , $2M+1$ and 2ν nonzero coefficients. Inside each sum the signs alternate; the last coefficient of L and the first of L^* are -1 , and L_{mid} begins and ends with $+1$. $\square$

The degree $6\nu+2M = 2(n+m-1)$ is twice the genus of the positive braid closure. For $T(3, 7; 2, 2)$, where $\nu = M = 2$, the coefficients of $t^0, \dots, t^{16}$ are

$$1, -1, 0, 1, -1, 0, 1, -1, 1, -1, 1, 0, -1, 1, 0, -1, 1,$$

and $\ell = 13 = |\sigma| + 1$ (Proposition 5.5).

5.3. Signature.

Proposition 5.5. *Let $n \equiv 1, 2 \pmod{6}$ and $m \geq 0$. Then $\det T(3, n; 2, m) = 2m+1$ for $n \equiv 1$ and $2m+3$ for $n \equiv 2 \pmod{6}$, and*

$$\sigma(T(3, n; 2, m)) = \sigma(T(3, n)) - 2m = -(4\nu + 2M).$$

Consequently $\ell(T(3, n; 2, m)) = |\sigma(T(3, n; 2, m))| + 1$.

Proof. At $t = -1$, $\psi(\sigma_1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\psi(\sigma_2) = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$, so $Q = \psi(\sigma_1\sigma_2)|_{t=-1} = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}$, which has order 6. Since $1+t+t^2 = 1$ at $t = -1$, $\det T(3, n; 2, m) = |\det(I - Q^n\psi(\sigma_1)^{2m})|_{t=-1}|$. For $n \equiv 1 \pmod{6}$ the matrix $Q^n\psi(\sigma_1)^{2m}|_{t=-1}$ is $\begin{pmatrix} 0 & 1 \\ -1 & 1-2m \end{pmatrix}$, and $\det \begin{pmatrix} 1 & -1 \\ 1 & 2m \end{pmatrix} = 2m+1$; for $n \equiv 2$ it is $\begin{pmatrix} -1 & 1-2m \\ -1 & -2m \end{pmatrix}$, and $\det \begin{pmatrix} 2 & 2m-1 \\ 1 & 2m+1 \end{pmatrix} = 2m+3$.

By Murasugi's theorem [Mur65, Thm. 5.6], if $|\sigma(K)| \equiv 2\mu \pmod{4}$ then $|\Delta_K(-1)| \equiv (-1)^\mu \pmod{4}$. As m increases by one, the determinant alternates between 1 and 3 modulo 4, so the parity of $\sigma/2$ changes and $\sigma(T(3, n; 2, m+1)) \neq \sigma(T(3, n; 2, m))$. Changing one crossing of σ_1^{2m+2} in $\beta_{n,m+1}$ from positive to negative gives $\beta_{n,m}$. For knots $K_\pm$ related by such a change, $\sigma(K_+) - \sigma(K_-) \in \{0, -2\}$ in our convention (the standard Seifert-matrix computation; see [DS24b, p. 163]). Hence $\sigma(T(3, n; 2, m+1)) = \sigma(T(3, n; 2, m)) - 2$. Finally $|\sigma(T(3, n))| = 2(n-1) - 4\lfloor n/6 \rfloor$ [Lit79] (see [Wue26c, proof of Prop. 6.3]), which is 4ν for $n = 6j+1$ and $4\nu+2$ for $n = 6j+2$. The last assertion follows from Proposition 5.4 and Lemma 5.3. $\square$

For $n \equiv 2 \pmod{6}$ this agrees with the formula $\sigma(T(3, 3n'+2; 2, k)) = -4n' - 2 - 2k$, $k > 0$, of [DS24b, p. 163]. The argument needs the determinant to alternate modulo 4; for $n = 5$ it does not at the first step, and indeed $\sigma(P(-2, 3, 5)) = \sigma(P(-2, 3, 7)) = -8$.

Lemma 5.6. *Let $\nu \geq 2$ and $M \geq 2$. Then $K_{\nu,M}$ is not a torus knot, not alternating, and not a pretzel knot $P(-2, 3, q)$.*

Proof. By Proposition 5.4, $\Delta_{K_{\nu,M}} = 1 - t + t^3 - \dots$, with zero coefficient of t^2 . As in the proof of Lemma 3.3, a torus knot with this Alexander polynomial would be $T(3, b)$ with $2(b-1) = 6\nu + 2M$. By Lemma 5.3, $T(3, b)$ is $K_{\nu',0}$ or $K_{\nu',1}$ for some ν' , so by Proposition 5.4 its Alexander polynomial has no more than three consecutive nonzero coefficients, whereas $\Delta_{K_{\nu,M}}$ has $2M + 1 \geq 5$. Next, $\det K_{\nu,M} = |F_{\nu,M}(-1)| = |2 + (-1)^\nu(2M-1)| < 4\nu + 2M + 1 = \ell(K_{\nu,M})$, while for an alternating knot the determinant equals ℓ (see the proof of Lemma 3.3). Finally, the coefficient of t^5 in $\Delta_{K_{\nu,M}}$ is 0, because $5 = 3 \cdot 1 + 2$ is not an exponent of L and L_{mid} begins at $3\nu \geq 6$, whereas in $\Delta_{P(-2,3,q)}$ it is 1 for $q \geq 5$ (Lemma 3.1); and $P(-2, 3, q)$ is a torus knot for $q \leq 5$. $\square$

5.4. Identification of the knots. Clay and Watson prove that for $\nu \geq 0$ and $m \geq 0$ the group of $T(3, 3\nu + 2; 2, m) = T_{3,3\nu+2}^m$ is

$$\pi_{3\nu+2}^m = \langle \alpha, \beta \mid \alpha^2(\beta^{-\nu}\alpha)^m\alpha = \beta^{2\nu+1}(\beta^{-\nu}\alpha)^m\beta^{\nu+1} \rangle$$

[CW13, §3.2 and Prop. 22].

Lemma 5.7. (a) *If $n = 3\nu + 2 \geq 5$ and $m \geq 0$, then $\alpha \mapsto A$, $\beta \mapsto B^{-1}$ induces an isomorphism $\pi_{3\nu+2}^m \cong \pi_1(S^3 \setminus K_{-m}(3, n))$.*

(b) *If $n = 3\nu + 1 \geq 4$ and $m \geq 1$, then $\alpha \mapsto AB^{2\nu+1}$, $\beta \mapsto B$ induces an isomorphism $\pi_{3\nu+2}^{m-1} \cong \pi_1(S^3 \setminus K_{-m}(3, n))$.*

Proof. Both maps are automorphisms of the free group on two letters. The group of a one-relator presentation is unchanged, up to isomorphism, if the relator is replaced by its image under an automorphism of the free group, by a cyclic permutation, or by its inverse. So it suffices to show that each map carries Clay and Watson's relator $\alpha^2 Y^m \alpha \beta^{-(\nu+1)} Y^{-m} \beta^{-(2\nu+1)}$, where $Y = \beta^{-\nu} \alpha$ (with $m-1$ in place of m in (b)), to a cyclic permutation of the relator, or of its inverse, of the presentation $\langle A, B \mid (ba)^{-m} b (ba)^m = c \rangle$.

(a) Here $a = A^2 B^{2\nu+1}$, $b = B^{-(\nu+1)} A^{-1}$ and $c = B^{-(\nu+1)} a B^{\nu+1}$ [Wue26c, (3.2)]. With $C = AB^\nu$ we have $ba = B^{-(\nu+1)} C B^{\nu+1}$, so the relation reads $C^{-m} A^{-1} B^{-(\nu+1)} C^m = A^2 B^{2\nu+1}$; a cyclic permutation of the inverse of the corresponding relator is $R = AC^m A^2 B^{2\nu+1} C^{-m} B^{\nu+1}$. Under $\alpha \mapsto A$, $\beta \mapsto B^{-1}$ we have $Y \mapsto B^\nu A$, $A(B^\nu A)^m = C^m A$ and $(B^\nu A)^{-m} = B^\nu C^{-m} B^{-\nu}$, so Clay and Watson's relator goes to $AC^m A^2 \cdot B^{\nu+1} \cdot B^\nu C^{-m} B^{-\nu} \cdot B^{2\nu+1} = R$.

(b) Here $a = AB^\nu$, $b = B^{-(2\nu+1)} A^{-2}$ and $c = B^{-(2\nu+1)} AB^{3\nu+1}$ [Wue26c, (3.2)]. With $Z = B^{\nu+1} A$ we have $ba = B^{-(2\nu+1)} Z^{-1} B^{2\nu+1}$, so the relation reads $Z^m A^{-2} B^{-(2\nu+1)} Z^{-m} = AB^\nu$, whose relator has inverse $R_H = AB^\nu Z^m B^{2\nu+1} A^2 Z^{-m}$. Under $\alpha \mapsto AB^{2\nu+1}$, $\beta \mapsto B$ we have $Y \mapsto B^{-\nu} AB^{2\nu+1} = B^{-(2\nu+1)} Z B^{2\nu+1}$. Using $B^{2\nu+1} A = B^\nu Z$, the left side $\alpha^2 Y^{m-1} \alpha$ goes to $AB^\nu Z^m B^\nu Z B^{2\nu+1}$ and the right side $\beta^{2\nu+1} Y^{m-1} \beta^{\nu+1}$ to $Z^{m-1} B^{3\nu+2}$. The relator therefore goes to $AB^\nu Z^m B^\nu Z B^{-(\nu+1)} Z^{1-m}$, which equals R_H because $B^\nu Z = B^{2\nu+1} A$ and $AB^{-(\nu+1)} Z = A^2$. $\square$

At $m = 0$ in (a), and at $m = 1$ in (b), Clay and Watson's relation is $\alpha^3 = \beta^{3\nu+2}$, the group of the torus knot $T(3, 3\nu + 2)$; part (a) at $\nu = 1$ is Lemma 4.2.

Proof of Theorem 5.1 (i) and (ii). Let $K = K_{-m}(3, n)$. For $n = 3\nu + 2$, Lemma 5.7(a) identifies $\pi_1(S^3 \setminus K)$ with the group of $T(3, n; 2, m)$. For $n = 3\nu + 1$, part (b) identifies it with the group of $T(3, 3\nu + 2; 2, m-1)$, which is $T(3, n; 2, m)$ by Lemma 5.3. Both groups are generated by two elements, and both knots are nontrivial by Proposition 5.4, so both knots are prime by Norwood's theorem [Nor82]. Prime knots with isomorphic groups are equivalent [GoLu89, Cor. 2.1], by a homeomorphism of S^3 that may reverse orientation. Hence K is $T(3, n; 2, m)$ or its mirror image.

The chirality follows by induction on m , as in the proof of Theorem 4.4. By our convention $K_0(3, n) = T(3, n)$. Suppose that $\sigma(K_{-m}(3, n)) = \sigma(T(3, n)) - 2m$ for some $m \geq 0$. The knot $K_{-(m+1)}(3, n)$ differs from $K_{-m}(3, n)$ by one crossing change, so their signatures differ by at most 2 [Mur65, (10.2)]. By Proposition 5.5, $T(3, n; 2, m+1)$ and its mirror image have signatures $\mp s$ with $s = |\sigma(T(3, n))| + 2m + 2$, and only $-s$ lies within 2 of $\sigma(T(3, n)) - 2m = -(s - 2)$, since $s + (s - 2) > 2$. Hence $K_{-(m+1)}(3, n) = T(3, n; 2, m+1)$. This proves (i), and (ii) is Proposition 5.5. $\square$

5.5. The pillowcase homology.

Proof of Theorem 5.1 (iii). Let $s = 2$ for $n \equiv 1$ and $s = 4$ for $n \equiv 2 \pmod{6}$, so that $s + 2m - 1 = 2M + 1$.

The arc. By [Wue26c, Prop. 5.7(3) and Lemma 6.1], for small $\varepsilon \neq 0$ the lift $\tilde{A}^V$ of the arc of L_1^V starting at $(0, 0)$ is, away from its ends, C^1 -close to the segment from $(0, 0)$ to $(\pi, s\pi)$, along which $\theta = s\gamma$ [Wue26c, Prop. 3.9]. Near $(0, 0)$ it has the form $yc(\varepsilon) + y^3R(y, \varepsilon)$, where c and R are real-analytic and $c(0)$ is a positive multiple of $(1, s)$ [Wue26c, Lemma 5.1], and the same holds at its other end. The shear S_{2m} is linear. Hence the lift $\tilde{A} = S_{2m}\tilde{A}^V$ is C^1 -close, away from its ends, to the segment from $(0, 0)$ to $(\pi, (s + 2m)\pi)$, on which $\varphi = (2M + 1)\gamma$; near $(0, 0)$ it equals $yS_{2m}c(\varepsilon) + y^3S_{2m}R(y, \varepsilon)$, and the φ -component of $S_{2m}c(0)$ is $(2M + 1)c_1(0) > 0$. So, as in [Wue26c, Cor. 5.3], φ is strictly monotone along $\tilde{A}$ with $|d\varphi|$ bounded below in arclength, and the limiting slopes $s + 2m + O(|\varepsilon|)$ differ from 1. The interior of $\tilde{A}$ lies in the open strip $0 < \gamma < \pi$, and $\tilde{A}$ ends on the line $\varphi = (2M + 1)\pi$. It therefore crosses exactly the lines $\Delta_1, \dots, \Delta_M$, each once and transversally, and meets Δ_0 only at its initial point. Since $\tilde{A}$ is embedded, the arc has no fishtail and is a restricted immersed arc.

The circles. By [Wue26c, Prop. 5.7(3)], the other components of L_1^V are circles C^1 -close to the $2\lfloor \nu/2 \rfloor = 2j$ non-central circles [Wue26c, Prop. 3.6]. They are restricted, miss the left and right edges, and have vertical degree 2 [Wue26c, Cor. 4.8 and Lemma 6.6]. By Lemma 4.6 their images under S_{2m} have the same three properties. Along these images φ need not be monotone: S_{2m} adds $2m\gamma$ to φ , and γ is not monotone along a closed curve.

The earring curve. The strands $S_{\pm}^k$ of $L_0 = L_0^{\varepsilon, g}$ are the translates of $\pm \tilde{L}_0^{\varepsilon, g}$ by $(0, 2\pi k)$, where $\tilde{L}_0^{\varepsilon, g}$ is given by [HHK18, (13)]. At the point with parameter t the slope of either is $(1 - \varepsilon \cos t)/(1 + \varepsilon \cos t) + \delta \cos \gamma$, and over the front strip $0 < \gamma < \pi$ one has $\cos t > \varepsilon/2$ on S_+^k and $\cos t < -\varepsilon/2$ on S_-^k , for small ε . Hence for $|\delta| < \varepsilon^2/2$ the strands S_+^k have slope less than 1 and the strands S_-^k slope greater than 1 over the front strip, as for $\delta = 0$ [Wue26c, §2.2], and they lie within $\sqrt{2}\varepsilon + |\delta|$ of Δ_k . The proofs of [Wue26c, Cor. 5.3(4), Prop. 6.3 and Lemma 6.8] use only these properties of the strands and the properties of the arc established above. Therefore, for small ε and $|\delta| < \varepsilon^2/2$, the arc meets L_0 transversely in exactly $1 + 2M$ points: the generator r_+ of [HHK18, Lemma 5.1], which lies on S_+^0 because $\tilde{A}$ leaves $(0, 0)$ above Δ_0 , and a pair $x_k^{\pm} \in S_{\pm}^k$ near the crossing of $\tilde{A}$ with Δ_k for $k = 1, \dots, M$.

Transversality on the circles. Fix ε and let C be a circle of L_1 . Since $L_0^{\varepsilon, \delta \sin} = c_{\delta \sin} \circ L_0^{\varepsilon, 0}$, where $c_g(\gamma, \theta) = (\gamma, \theta + g(\gamma))$ [HHK18, (11), p. 18], $L_0^{\varepsilon, \delta \sin}$ is transverse to C if and only if $L_0^{\varepsilon, 0}$ is transverse to $c_{-\delta \sin}(C)$. Consider $\Phi(t, x, \delta) = \tilde{L}_0(t) - c_{-\delta \sin}(\tilde{C}(x)) \in \mathbb{R}^2$, where $\tilde{L}_0$ is a lift of $L_0^{\varepsilon, 0}$ and $\tilde{C}$ a lift of C . Its derivative in t is the tangent vector of $\tilde{L}_0$, which is never vertical, and its derivative in δ is $(0, \sin \gamma(\tilde{C}(x)))$, which is vertical and nonzero because C misses the edges. So Φ is a submersion, $\Phi^{-1}(0)$ is a 1-manifold, and δ is a regular value of the projection $\Phi^{-1}(0) \rightarrow \mathbb{R}$ exactly when 0 is a regular value of $\Phi(\cdot, \cdot, \delta)$. By Sard's theorem, applied to the countably many pairs of lifts, $L_0^{\varepsilon, \delta \sin}$ is transverse to every circle of L_1 for almost every δ . For such δ the pair (L_0, L_1) is admissible: L_0 is a circle, the intersections are transverse, and the

argument of [Wue26c, Lemma 6.6] applies, since L_0 lies near Δ and each circle of L_1 lies in the open strip with vertical degree 2.

Bigons and gradings on the arc. By [Wue26c, Lemma 6.4], the boundary of a bigon from p to q lifts to a closed curve made of a sub-path of one lift of a component of L_1 and an arc of one strand, which contains lifts $\tilde{p}$, $\tilde{q}$ of both corners. Every lift of the arc is a translate of $\tilde{A}$ under the deck group G , which permutes the strands, and $\tilde{A}$ meets each strand at most once. So a bigon between generators on the arc would have $\tilde{p} = \tilde{q}$, hence $p = q$, which is impossible since $\text{gr}(p, q) = 1$ [HHK18, Prop. 3.8]. Thus the arc carries no bigon. For the degrees, $\deg = \text{gr} - \sigma(K)$ satisfies $\deg r_+ = 0$ by [HHK18, (19)] and $\deg x_k^- = \deg x_k^+ - 1$ [HHK18, §5.1]. Since φ has no critical point on $\tilde{A}$, [Wue26c, Lemma 6.8] applies to $r_+ \in S_+^0$ and $x_1^+ \in S_+^1$, and to x_{k-1}^+ and x_k^+ ; each step changes the degree by 2. Hence $\deg x_k^+ = 2k$ and $\deg x_k^- = 2k - 1$ modulo 4, and the homology of the arc, which equals its complex, has dimensions $(1 + a, b, b, a)$, since a and b count the even and the odd $k \in \{1, \dots, M\}$.

The total. By [HHK18, Thm. 5.4], each circle contributes $(1, 1, 1, 1)$, whatever its absolute grading. There is no differential between different components [HHK18, p. 37], and $H^\natural(Y_n, T_n, \pi)$ is the direct sum of the contributions of the components [HHK18, p. 34]. Adding $2j$ copies of $(1, 1, 1, 1)$ to $(1 + a, b, b, a)$ gives the stated vector. $\square$

Remark 5.8. We have computed the sheared curves and their complexes numerically, by the method of [Wue26c, Rem. 6.12] with $\delta = 0$, for $n = 7, 8, 13$, $m = 1, 2, 3$, and the two perturbations $(n\varepsilon_A, \varepsilon_B, \epsilon) = (0.3, 0.05, 0.01)$ and $(-0.3, -0.03, 0.02)$. In all 18 cases the arc carries $1 + 2M$ generators and no bigon, the homology of each circle, computed from its complex rather than from [HHK18, Thm. 5.4], is $(1, 1, 1, 1)$, and the total is the vector of Theorem 5.1(iii). These computations are not used in the proofs.

5.6. Comparison with instanton homology.

Proposition 5.9. *Let $n \geq 7$ with $n \equiv 1, 2 \pmod{6}$, let $m \geq 0$, and let $K = T(3, n; 2, m)$. Then $I^\natural(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, and in Kronheimer and Mrowka's absolute grading its graded ranks are*

$$(1 + \lfloor |\sigma(K)|/4 \rfloor, \lceil |\sigma(K)|/4 \rceil, \lceil |\sigma(K)|/4 \rceil, \lfloor |\sigma(K)|/4 \rfloor) \quad \text{in degrees } \text{gr} - \sigma(K) = 0, 1, 2, 3.$$

The same holds with coefficients in $\mathbb{Q}$ and in $\mathbb{F}_2$.

Proof. Since $n = 3(2j + 1) - 2$ or $n = 3 \cdot 2j + 2$, and $\det K \neq 0$ by Proposition 5.5, K is I -basic by [DS24b, Prop. 8.31], applied with $p = 3$ and with m full twists: its irreducible instanton homology $I(K)$ has rank $|\sigma(K)|/2$. The proof of [DS24b, Prop. 8.31] applies [DS24b, Lemma 8.23] starting from $T(3, n)$, whose irreducible complex has rank $|\sigma(T(3, n))|/2$ equal to the absolute value of its Euler characteristic [DS24b, Ex. 8.11], so that $I(T(3, n))$ is free abelian; by [DS24b, Rem. 8.27], $I(K)$ is free abelian too. By [DS24b, Prop. 8.28], $I(K) \cong \mathbb{Z}_{(1)}^{\lceil |\sigma|/4 \rceil} \oplus \mathbb{Z}_{(3)}^{\lfloor |\sigma|/4 \rfloor}$ in the grading of (3.4). For $n \equiv 2 \pmod{6}$ and $m \geq 1$ this is [DS24b, Cor. 8.36], with their n equal to $2j$ and their k equal to m .

The rest is the proof of Proposition 3.6. The initial page of the spectral sequence (3.6) is free, with ranks $1 + \lfloor |\sigma|/4 \rfloor$, $\lceil |\sigma|/4 \rceil$, $\lceil |\sigma|/4 \rceil$ and $\lfloor |\sigma|/4 \rfloor$ in degrees 0, 1, 2 and 3, for a total of $1 + |\sigma(K)|$. By (3.1) and Proposition 5.5, $\dim_{\mathbb{Q}} I^\natural(K; \mathbb{Q}) \geq \ell(K) = 1 + |\sigma(K)|$. Hence every differential vanishes rationally, hence integrally, and $I^\natural(K; \mathbb{Z})$ is free with these graded ranks in the grading of (3.4). Kronheimer and Mrowka's grading is this grading shifted by $\sigma(K)$ [DS24a, Thm. 8.9]. The assertions over $\mathbb{Q}$ and $\mathbb{F}_2$ follow by base change and the universal coefficient theorem. $\square$

Proof of Theorem 5.1 (iv). By (i), (ii) and Proposition 5.9, $I^{\natural}(K; \mathbb{Z})$ is free of rank $|\sigma(K)| + 1$, with $|\sigma(K)|/4 = 2j + M/2$. Hence $\lfloor |\sigma(K)|/4 \rfloor = 2j + a$ and $\lceil |\sigma(K)|/4 \rceil = 2j + b$, and the graded dimensions of $I^{\natural}(K; \mathbb{F}_2)$ are the vector of (iii). $\square$

The restriction to $n \equiv 1, 2 \pmod{6}$ comes from the pillowcase side. For $n \equiv 4, 5 \pmod{6}$ the curve of [Wue26c] contains the central circle, and the monotonicity of φ along its halves after the shear requires control of γ along them; Section 4 obtains it for $n = 5$ from the straightness of the image of the central circle (Lemma 4.7). We have not treated $n = 4$, whose twisted knots are again the pretzel knots by Lemma 5.3, nor $n \equiv 4, 5 \pmod{6}$ with $n \geq 10$.

6. SMITH'S DECOMPOSITION

Smith's decomposition of $K_q = P(-2, 3, q)$ is (2.8) with the earring on the -2 band, $T_1 = \widehat{Q_{-1/2}}$ and $T_2 = Q_{1/3} + Q_{1/q}$ [Smi24]. The red curve $\mathcal{R}^{\natural}(\widehat{Q_{-1/2}})$ does not depend on q ; the blue curve is Smith's holonomy-perturbed curve $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$. In contrast with the decomposition of Section 4, the pairing of these two curves with zero bounding cochain does not have the rank of $I^{\natural}(K_q)$: at $q = 7$ it is 7 rather than 9 (Proposition 6.7), and the finite statistics of §6.6 differ from $q + 2$ by ± 2 for $q = 5, 7, 11, 13$. Conditional on the conjecture of Cazassus, Herald, Kirk and Kotelskiy [CHKK22, Conj. D], Smith deduces that T_2 must carry a nonzero bounding cochain when $q = 5$ [Smi24, proof of Thm. 5.9]. At $q = 7$ we work in the two-arc algebra of Section 2: local smoothings of the blue curve give Maurer–Cartan elements to all orders, and a second closure selects one homotopy class among them, conditionally on a localization hypothesis. Section 7 relates the selected class to the Khovanov invariant of T_2 . Statements labelled *Computation* are outputs of the finite procedures of §6.6 and are recorded as data; the statement labelled *Hypothesis* is assumed, not proved.

6.1. Piecewise-linear models and Smith's cross-resolution. We work in Smith's coordinates (2.6), with the corner rule for rational lines and the fiber-product description of Conway sums recalled in §2.5. As Smith does, we call the images of the lines $\gamma = 0$ and $\gamma = \pi$ the *edges* of the pillowcase, and the circles of representations that a Conway sum acquires over them *edge circles* [Smi24, Thms. 3.15 and 3.26]. Smith's perturbation resolves each internal edge circle by a diagonal cut-and-paste [Smi24, Thm. 4.22], which replaces it by two arcs crossing once; we call these arcs *connectors*, and the remaining arcs of the resolved curve, which come from the fiber product, *inherited strands*. The earring curve E_r of $\widehat{Q_r}$ is a Herald–Kirk figure-eight, the doubled arc closing through the corner folds with one pinch [HK24, Def. 6.7]; this specifies a finite curve, not a bounding cochain. The closing tangle enters the pairing through its mirror image, so E_r surrounds the line of Q_{-r} (§7.1); for instance the red curve $E_{-1/2} = \mathcal{R}^{\natural}(\widehat{Q_{-1/2}})$ lies along the line of slope $1/2$.

Write $L_{q, \text{PL}}$ for the piecewise-linear curve obtained from the lines of slopes $1/3$ and $1/q$ through $(0, 0)$ by the fiber product and the crossed local pairing at every edge circle. These lines are the translates by $(0, \pi)$ of the lines of $Q_{1/3}$ and $Q_{1/q}$, and since the fiber product adds the θ -coordinates the two shifts cancel: both fiber products have the same image and the same edge circles. The model represents Smith's perturbed curve for negative values of his perturbation parameter (Remark 6.2). The red curve $\mathcal{R}^{\natural}(\widehat{Q_{-1/2}})$ is one immersed circle with one pinch; the blue curve has $q - 1$ resolved edge circles and is a single proper immersed arc. The model $L_{q, \text{PL}}$ has 40, 82 and 240 transverse self-intersections at $q = 5, 7, 11$. These are counts for the model: the number is not invariant under regular homotopy, and a numerical trace of Smith's perturbed curve at $q = 7$ has 50 (Remark 6.15). At $q = 7$ the model also has twelve tangential self-contacts on the edges, where connectors meet inherited strands without crossing; we remove them by a small perturbation that creates no crossings. We require $\gcd(q, 3) = 1$, since otherwise corner

circles appear; this excludes $q = 3, 9, 15, \dots$ from the members computed here, but not from Theorem 1.1.

Proposition 6.1 (Smith’s cross-resolution and the model). *Let q be odd and $\gcd(q, 3) = 1$.*

- (a) *The fiber product of the rational lines of slopes $1/3$ and $1/q$ has $q - 1$ edge circles, all internal, and no corner circles.*
- (b) *For all sufficiently small nonzero values t of Smith’s C_3 -perturbation parameter, after an arbitrarily small additional perturbation where Smith’s construction requires one, $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ contains a subspace homeomorphic to the curve obtained from the fiber product by replacing each internal circle with two intervals joined to its ends crosswise; this is the pairing used to define $L_{q, \text{PL}}$. As $t \rightarrow 0$ the image of each new interval approaches the image of the corresponding circle. The remaining components are circles, the auxiliary components.*
- (c) *For a fixed complementary tangle, there is an arbitrarily small perturbation of it for which the Floer pairing with $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ equals the pairing with its main components. This remains valid when either tangle carries an earring.*

Proof. Smith identifies the two rational character varieties with straight intervals of slopes $1/3$ and $1/q$ [Smi24, Prop. 2.27], and rational tangles of finite slope form a good pair without preliminary perturbations [Smi24, example following Thm. 3.26]. Because the denominators 3 and q are coprime, the sum has no corner circles [Smi24, discussion following Prop. 5.8], so every edge circle is internal. This is (a). For $t > 0$, part (b) is Smith’s cross-resolution theorem [Smi24, Thm. 4.22], together with his remark preceding [Smi24, Thm. 4.20] that an arbitrarily small additional perturbation makes the relevant spaces manifolds when transversality fails. For $t < 0$ it follows from the case of $-t$: the homeomorphism $\hat{s}$ of Remark 6.2 maps $\mathcal{R}_{-t}(Q_{1/3} + Q_{1/q})$ onto $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ and the unperturbed fiber product onto itself, and it acts on the pillowcase of the sum by the translation $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$. Part (c) is Smith’s auxiliary-component result [Smi24, Prop. 5.7], which includes the earring case; its proof uses only that the auxiliary components converge to finitely many points of P^* as $t \rightarrow 0$, and by the same homeomorphism this also holds for $t < 0$. $\square$

Smith states informally that [Smi24, Thm. 4.22] determines the image of the main component up to regular homotopy ([Smi24, caption of Fig. 3] and the outline of the proof in his introduction); the formal statement gives the homeomorphism type and the convergence in (b). We use $L_{q, \text{PL}}$ as a representative of that regular-homotopy class, but no proof below depends on this identification; at $q = 7$ the type-D encoding of a traced copy of Smith’s curve is strictly isomorphic to that of $L_{7, \text{PL}}$ (Remark 6.15).

Remark 6.2 (The sign of the perturbation). In Smith’s configuration C_3 the summands are glued along two inner boundary spheres, with pillowcases P_1 and P_2 , and the sum is read off on the outer sphere, with pillowcase P_3 ; write $C_{3,t}$ for the perturbed character variety of the configuration. Its representations are parametrized by the images of the meridians a, b, c of the outer sphere and of a meridian x of the first inner sphere: the perturbation element is $p = \exp(t \operatorname{Im}(ac^{-1}x))$, the remaining meridians are $d = ca^{-1}b$ and $y = xp^{-1}a^{-1}pb$, and the perturbed variety is cut out by $\operatorname{Re}(y) = 0$ [Smi24, Prop. 4.1 and (4.1.1)]. The substitution $\mathfrak{s}: (x, t) \mapsto (-x, -t)$ leaves p and d unchanged and replaces y by $-y$, so it maps the perturbed variety for t onto that for $-t$. It is the identity on P_3 , and a direct computation with Smith’s boundary words [Smi24, Fig. 20] shows that it acts on P_1 and on P_2 by the translation $(\gamma, \theta) \mapsto (\gamma, \theta + \pi)$. For odd q a line of slope $1/q$ is invariant under translation by (π, π) , so translation by $(0, \pi)$ carries the line of slope $1/q$ through $(0, 0)$ to the line through $(\pi, 0)$ and $(0, \pi)$. Consequently $\mathfrak{s}$ carries the composition of $C_{3,t}$ with the lines of slopes $1/3$ and $1/q$ through $(0, 0)$, for $t > 0$, onto the composition of $C_{3,-t}$

with the lines of $Q_{1/3}$ and $Q_{1/q}$, without moving the image in P_3 . Hence the piecewise-linear model, which is built from the lines through $(0, 0)$ and compared with their composition for $t > 0$, represents the main component of $\mathcal{R}_t(Q_{1/3} + Q_{1/q})$ in Smith's coordinates for small $t < 0$, which is the convention of §2.5.

The substitution $b \mapsto -b$ is also a symmetry of Smith's equations for each t , since p does not involve b and y changes sign; it translates each of P_1, P_2, P_3 by $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$. The composite $\hat{s}: (b, x, t) \mapsto (-b, -x, -t)$ therefore translates P_1 and P_2 by (π, π) , which preserves the lines of $Q_{1/3}$ and $Q_{1/q}$, and P_3 by $(\pi, 0)$, so the curve for $t > 0$ is the translate by $(\pi, 0)$ of the curve for $-t$. This translation is a symplectomorphism of P^* that permutes the corners, but the translated curve is a different representative, and pairings of its smoothings with the earrings can differ from those for $t < 0$ (Remark 6.17).

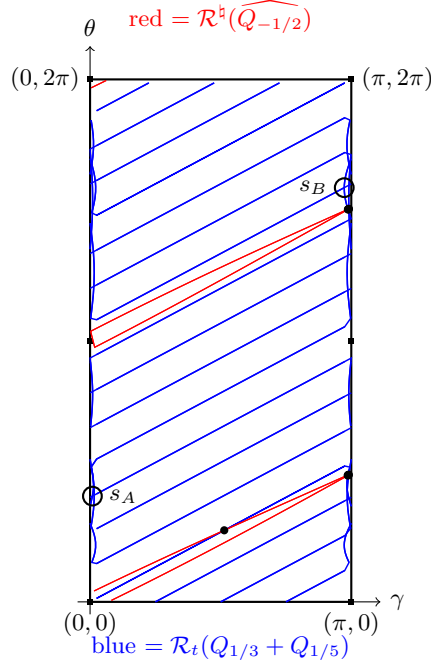


FIGURE 3. The two piecewise-linear model curves at $q = 5$ on the fundamental domain $[0, \pi] \times [0, 2\pi)$ of the pillowcase, in Smith's coordinates (compare [Smi24, Fig. 31]). Blue models Smith's perturbed curve $\mathcal{R}_t(Q_{1/3} + Q_{1/5})$, a single immersed arc with 40 self-intersections clustered along the edges; the small arcs hugging the edges $\gamma \in \{0, \pi\}$ are the connectors resolving the four edge circles, drawn at amplitude $\varepsilon \approx 0.05$ – 0.07 , and the V-shaped turnbacks at the border are strands reflecting off the edges in the quotient. Red is the earring figure-eight $\mathcal{R}^h(\widehat{Q_{-1/2}})$. Across the interior the blue strands run nearly parallel to red (slopes $8/15$ and $1/2$, crossing angle ≈ 0.026), which is why shallow-angle corner tests are needed. Solid dots are the nine generators: one interior, and two clusters of four near $\gamma = \pi$ that merge at this scale. The circled self-intersections s_A, s_B are the support of Computation 6.20(i). Curves appear as several arcs because the fundamental domain folds the double cover.

6.2. The $Q_{1/3}$ correspondence is not embedded. Write P_i^* for the smooth locus of P_i , P^- for a pillowcase with the sign of its symplectic form reversed, and $C_{3,t}^{\text{reg}} \looparrowright P_1^- \times P_2^- \times P_3^-$ for the

regular locus of Smith's perturbed three-sphere correspondence, his $\mathcal{R}^*$ [Smi24, Def. 4.9], which maps to the product of pillowcases by a Lagrangian immersion [Smi24, Cor. 4.11].

Lemma 6.3 (Reduction to a correspondence between surfaces). *Let $A_{1/3} \looparrowright P_1$ be the rational line of $Q_{1/3}$. Wherever the fiber product is transverse,*

$$F_{1/3,t} := A_{1/3} \times_{P_1} C_{3,t}^{\text{reg}} \looparrowright P_2^- \times P_3$$

is an immersed Lagrangian surface. On a torus-cover chart of the unperturbed main stratum with $\gamma \notin \{0, \pi\}$, a branch of $A_{1/3}$ has $\theta_1 = \gamma/3 + c$, and $F_{1/3,0}$ is the graph of the symplectic shear $(\gamma, \eta) \mapsto (\gamma, \eta + \gamma/3 + c)$; on compact subsets of such a chart the projection of $F_{1/3,t}$ to P_2 remains a local diffeomorphism for small $|t|$. After composition with the rational line of $Q_{1/7}$, the complement of these graph charts lies in neighborhoods of the six internal edge circles of Proposition 6.1 and of the two corner endpoints of the composed arc. If the fiber product is transverse along the entire regular locus, the target projection $F_{1/3,t} \rightarrow P_3^$ is proper.*

Proof. On the unperturbed main stratum Smith's boundary-coordinate formulas give $p_1 = (\gamma, \beta)$, $p_2 = (\gamma, \theta - \beta)$ and $p_3 = (\gamma, \theta)$. Imposing $\beta = \gamma/3 + c$ and writing $\eta = \theta - \beta$ gives the displayed graph, which preserves $d\gamma \wedge d\theta$, so the transverse composition is Lagrangian; the local-diffeomorphism property persists on compact subsets by the implicit function theorem. For $Q_{1/3}$ and $Q_{1/7}$ there are six internal edge circles, and [Smi24, Thm. 4.22] replaces precisely their neighborhoods by the crossed pieces used to construct $L_{7,\text{PL}}$. For properness, let $\bar{A}_{1/3}$ be the compact rational arc and $\bar{F}_{1/3,t} := \bar{A}_{1/3} \times_{P_1} \mathcal{R}_{D_t}(C_3)$, a compact space. The boundary restriction of $\mathcal{R}_{D_t}(C_3)$ lies either in $P_1^* \times P_2^* \times P_3^*$ or in the product of the three corner strata [Smi24, Prop. 4.3], and its regular Lagrangian locus is the former [Smi24, Cor. 4.11]. Hence $\bar{F}_{1/3,t} \setminus F_{1/3,t}$ maps to the corners of P_3 , and the inverse image of a compact subset of P_3^* is compact and lies in $F_{1/3,t}$. $\square$

Proposition 6.4 (A triple-point obstruction). *For every sufficiently small $t < 0$, the partial fiber product of Lemma 6.3 has three distinct transverse points whose images in $P_2^* \times P_3^*$ are all $((0, \pi/2), (0, \pi/2))$. Consequently $F_{1/3,t} \rightarrow P_2^- \times P_3$ is not embedded.*

Proof. Put $v(\varphi) = \cos \varphi \mathbf{i} + \sin \varphi \mathbf{k}$ and fix $\gamma = 0$, $\theta = \pi/2$ in Smith's parametrization, so that $a = b = \mathbf{i}$ and $c = d = \mathbf{j}$. For $\beta = 0$ or π write $(\varphi, u) = (\pi/2 - \alpha, t \sin \alpha)$ if $\beta = 0$ and $(\varphi, u) = (\pi/2 + \alpha, -t \sin \alpha)$ if $\beta = \pi$. Then $x = v(\varphi)$, the perturbation element is $p = e^{-u\mathbf{j}}$, and direct quaternion multiplication shows that $y = xp^{-1}a^{-1}pb$ is purely imaginary, so $\text{Re}(y) = 0$ holds identically along these two families. On the torus cover the first boundary value is $(\gamma_1, \theta_1) = (2u, \varphi + 2u)$.

Take first $t > 0$ and the line $A_{1/3}^0 = \{3\theta_1 - \gamma_1 \in 2\pi\mathbb{Z}\}$ through $(0, 0)$, the translate of $A_{1/3}$ by $(0, \pi)$. Its equation $3\varphi + 4u \in 2\pi\mathbb{Z}$ gives

$$\begin{aligned} 3(\pi/2 - \alpha) + 4t \sin \alpha &= 0, & \alpha &\rightarrow \pi/2, & \beta &= 0, \\ 3(\pi/2 + \alpha) - 4t \sin \alpha &= 2\pi, & \alpha &\rightarrow \pi/6, & \beta &= \pi, \\ 3(\pi/2 + \alpha) - 4t \sin \alpha &= 4\pi, & \alpha &\rightarrow 5\pi/6, & \beta &= \pi, \end{aligned}$$

with α -derivatives $-3, 3, 3$ at $t = 0$, so the implicit function theorem gives three solutions for small $t > 0$. They are distinct, since their limiting x -values are, and none lies over a corner of P_1 , since $|\gamma_1| = 2t \sin \alpha > 0$. Using $xp = p^{-1}x$, the second-boundary words become $(v(\varphi + 2u), v(\varphi + 2u), \mathbf{j}, \mathbf{j})$, so $p_2 = (0, \pi/2)$ at all three points, and $(a, b, c, d) = (\mathbf{i}, \mathbf{i}, \mathbf{j}, \mathbf{j})$ gives $p_3 = (0, \pi/2)$. Since $\partial_\beta \text{Re}(y) = -2t \sin^2 \alpha + O(t^2)$ and $\partial_\alpha(3\theta_1 - \gamma_1) = \pm 3 + O(t)$, the fiber product is transverse at all three points. Finally, the symmetry $\mathfrak{s}$ of Remark 6.2 maps the fiber product of $A_{1/3}^0$ with $C_{3,t}$ onto that of $A_{1/3}$ with $C_{3,-t}$, preserving distinctness, transversality

and γ_1 , fixing p_3 and translating p_2 by $(0, \pi)$. Since $(0, 3\pi/2) = \iota(0, \pi/2)$, the three image points again map to $((0, \pi/2), (0, \pi/2))$; replacing t by $-t$ gives the statement. $\square$

Gao's wrapped representability theorem [Gao17, Thm. 7.24 and Cor. 7.25] applies to *admissible* correspondences, which are properly embedded, exact and cylindrical, with vanishing relative first Chern class [Gao17, Def. 6.3, Def. 7.2 and the standing convention of §7.1], and whose projection to the target is proper; only the geometric composition may be immersed [Gao17, Assumption 7.10], and the unobstructedness input [Gao17, Thm. 7.12] has the same hypotheses. By Proposition 6.4, Gao's theorem cannot be invoked for $F_{1/3,t}$, whose exactness, cylindricity and Chern class have not been checked either. The triple point is not generic, since the three sheets are pairwise transverse, but splitting it into double points leaves $F_{1/3,t}$ non-embedded. Gao expects the results of his §6 to extend to immersed Lagrangians [Gao17, Rem. 6.1], but no immersed version of [Gao17, Thm. 7.24] is stated. Fukaya's compact theorem [Fuk25] does not apply verbatim to P^* , and the torus double cover does not help by itself: Smith's perturbed C_3 variety has singular points with neighborhoods that are cones on tori [Smi24, Thm. 4.16], and the required equivariant correspondence functor has not been constructed [CHKK22, §11.5].

6.3. The pairing with zero bounding cochain at $q = 7$. We use the two-arc algebra $\mathcal{B}$, the arc system $s^\circ, s^\bullet$ with its faces L, M, R , the twisted complexes of curves, unit cancellations, morphism complexes and pairings as in §2.6, ignoring gradings, and we omit exponents equal to one in arrows $i \xrightarrow{X^k} j$. A perturbation b of a twisted complex is Maurer–Cartan exactly when (2.7) holds, a finite identity. A recorded degree is the relative quantum grading gr_q of [KWZ19]; it plays no part in the Maurer–Cartan equation or in any pairing.

Finiteness of the morphism complex. Since $\mathcal{B}$ is infinite dimensional, so is the morphism complex $\text{Mor}(\mathcal{E}, N)$ of two finite twisted complexes, and its homology need not be finite. The *identity part* $\text{Mor}^\times$ of $\text{Mor}(\mathcal{E}, N)$ is spanned by the entries labelled by the two vertex idempotents, and for $X \in \{L, M, R\}$ its X -part Mor^X by the entries labelled by positive powers of X . Put $U_L = L$, $U_R = R$ and $U_M = M^2$.

Lemma 6.5 (Mapping-cone reduction of the morphism complex). *Let $\mathcal{E}$ and N be reduced finite twisted complexes over $\mathcal{B}$.*

- (a) *The differential maps $\text{Mor}^\times$ into $\text{Mor}^+ := \text{Mor}^L \oplus \text{Mor}^M \oplus \text{Mor}^R$ and preserves each Mor^X . Hence $\text{Mor}(\mathcal{E}, N)$ is the mapping cone of a chain map $\beta: \text{Mor}^\times \rightarrow \text{Mor}^+$, where $\text{Mor}^\times$ is finite dimensional with zero differential.*
- (b) *Each Mor^X is a free $\mathbb{F}_2[U_X]$ -module of finite rank, with U_X acting by multiplication, and its differential is $\mathbb{F}_2[U_X]$ -linear. Consequently $H(\text{Mor}^X) \cong \mathbb{F}_2[U_X]^{f_X} \oplus \bigoplus_i \mathbb{F}_2[U_X]/(p_{X,i})$ for some $f_X \geq 0$ and nonconstant polynomials $p_{X,i}$.*
- (c) *$H\text{Mor}(\mathcal{E}, N)$ is finite dimensional if and only if $f_L = f_M = f_R = 0$, that is, if and only if each $\text{Mor}^X \otimes_{\mathbb{F}_2[U_X]} \mathbb{F}_2(U_X)$ is acyclic. In that case, with $e = \dim \text{Mor}^\times$, $\tau = \dim_{\mathbb{F}_2} H(\text{Mor}^+) = \sum_{X,i} \deg p_{X,i}$, and r the rank of $\beta_*: \text{Mor}^\times \rightarrow H(\text{Mor}^+)$,*

$$\dim_{\mathbb{F}_2} H\text{Mor}(\mathcal{E}, N) = e + \tau - 2r. \quad (6.1)$$

Proof. Composing an idempotent entry with the positive words of $\delta_{\mathcal{E}}$ and δ_N gives positive words, and a positive power of X composed with a word in a different face is zero; this is (a). For (b), the entries of Mor^X joining a fixed pair of generators are the powers X^k , $k \geq k_0$, of a fixed parity when $X = M$; multiplication by U_X shifts k by 1 for $X = L, R$ and by 2 for $X = M$, so Mor^X is free on the entries of lowest length, and the differential commutes with U_X because powers of X commute. The decomposition is the structure theorem over the principal ideal domain $\mathbb{F}_2[U_X]$. For (c), the cone gives the exact sequence $0 \rightarrow \text{coker } \beta_* \rightarrow H\text{Mor}(\mathcal{E}, N) \rightarrow \ker \beta_* \rightarrow 0$,

so the homology is finite exactly when $H(\text{Mor}^+)$ has no free summand, and f_X is the dimension of $H(\text{Mor}^X \otimes_{\mathbb{F}_2}(U_X))$ by flatness. When $H(\text{Mor}^+)$ is finite, the sequence gives dimension $(\tau - r) + (e - r)$. $\square$

Remark 6.6. The mapping cone is the algebraic step preceding [KWZ19, Thm. 5.22]; Lemma 6.5 records the ungraded form used here. The torsion condition in (c) is a genuine hypothesis: for the one-generator complex $(\iota_\circ, 0)$ the endomorphism complex $\iota_\circ \mathcal{B} \iota_\circ$ has $f_L = f_M = 1$, and for the complex N below $\text{End}(N)$ has $f_M = f_R = 1$. For every pair of complexes whose morphism homology is used in this paper, exact computation over $\mathbb{F}_2[U_X]$ shows $f_L = f_M = f_R = 0$ and $p_{X,i} = U_X$ for all i , so that τ is the number of torsion summands.

The two explicit twisted complexes. We send the corner $(0, \pi)$ to infinity. The curves encoded below, namely the arc N with ends at $(0, 0)$ and $(\pi, 0)$, its smoothings and closed earring curves, meet neither $(0, \pi)$ nor the corner (π, π) that no tangle character variety meets. CHKK's description applies to any corner, since translations of the pillowcase permute the corners, and in either case morphisms are computed in the wrapped category of the pillowcase with all four corners removed, so the morphism homology does not depend on which of the two corners is at infinity.

In this chart the $q = 7$ blue curve meets the arc system in 74 points of the torus cover, that is, in 37 points of the pillowcase. Three unit cancellations leave the 31-generator complex N whose $\iota_\circ$ generators are 1, 2, 5, 6, 15, 16, 20, 21, 32, 33; the remaining generators in

$$V_N = \{0, 1, \dots, 10, 13, 14, \dots, 23, 26, 27, 30, 31, \dots, 36\}$$

carry $\iota_\bullet$, and

$$\begin{aligned} \delta_N = & 17 \xrightarrow{R} 18 + 16 \xrightarrow{M} 17 + 16 \xrightarrow{L} 15 + 15 \xrightarrow{M} 14 + 14 \xrightarrow{R} 13 + 13 \xrightarrow{M^2} 10 + 10 \xrightarrow{R} 9 + 8 \xrightarrow{M^2} 9 \\ & + 7 \xrightarrow{R} 8 + 6 \xrightarrow{M} 7 + 6 \xrightarrow{L} 5 + 5 \xrightarrow{M} 4 + 4 \xrightarrow{R} 3 + 2 \xrightarrow{M} 3 + 2 \xrightarrow{L} 1 + 1 \xrightarrow{M} 0 \\ & + 0 \xrightarrow{R} 19 + 20 \xrightarrow{M} 19 + 20 \xrightarrow{L} 21 + 21 \xrightarrow{M} 22 + 22 \xrightarrow{R} 23 + 23 \xrightarrow{M^2} 26 + 26 \xrightarrow{R} 27 \\ & + 30 \xrightarrow{M^2} 27 + 31 \xrightarrow{R} 30 + 32 \xrightarrow{M} 31 + 32 \xrightarrow{L} 33 + 33 \xrightarrow{M} 34 + 34 \xrightarrow{R} 35 + 35 \xrightarrow{M^2} 36. \end{aligned}$$

Direct path multiplication gives $\delta_N^2 = 0$. The red earring curve meets the arc system in eight points, and one unit cancellation leaves a six-generator complex $\mathcal{E}$ with $\iota_\circ$ generators 5, 6, $\iota_\bullet$ generators 0, 3, 4, 7, and

$$\delta_{\mathcal{E}} = 3 \xrightarrow{M^2} 0 + 4 \xrightarrow{R} 3 + 5 \xrightarrow{M} 4 + 6 \xrightarrow{L} 5 + 6 \xrightarrow{M} 7 + 0 \xrightarrow{R} 7, \quad (6.2)$$

again with $\delta_{\mathcal{E}}^2 = 0$.

Local reconnections. A self-intersection of $L_{7,\text{PL}}$ is *connector-main* if one branch is a connector and the other an inherited strand, and *connector-connector* if both branches are connectors. The lift of the blue curve to the torus double cover is a single closed immersed curve C of homology class $(21, 10)$; the involution ι reverses its orientation, and each self-intersection orbit s of the model consists of two transverse double points of C interchanged by ι . Exactly one of the two smoothings of a double point is compatible with the orientation of C ; it splits off the loop of C between the two visits. The *census smoothing* at s performs it at both points of s , in disjoint disks; the pair is ι -invariant and defines a smoothing of the pillowcase curve. Performing the other smoothing at both points gives the *opposite smoothing*. At 38 of the 82 orbits the two branches lie on the same lift of the pillowcase arc, and there the census smoothing is also compatible with an orientation of that arc; the four orbits below are of this kind. A smoothing is

generator-preserving if, after the three unit cancellations, its encoding has the same idempotent module V_N ; its differential then differs from δ_N by an element $b_s \in \text{End}(N)$, the *switch* at s .

Proposition 6.7. *The wrapped morphism homology over $\mathbb{F}_2$ satisfies*

$$\dim_{\mathbb{F}_2} H \text{Mor}_{\text{Tw}(\mathcal{B})}(\mathcal{E}, N) = 7.$$

At each of the four self-intersection orbits $S_{18}, S_{25}, S_{69}, S_{74}$ of $L_{7, \text{PL}}$ tabulated below, the census smoothing produces a four-arrow switch $b_s \in \text{End}(N)$ with

$$D_N(b_s) = 0, \quad b_s^2 = 0, \quad \dim_{\mathbb{F}_2} H \text{Mor}_{\text{Tw}(\mathcal{B})}(\mathcal{E}, (N, b_s)) = 9.$$

The proposition holds to all orders inside $\text{Tw}(\mathcal{B})$, because the Maurer–Cartan equation there is the finite identity $(\delta_N + b_s)^2 = 0$. The four orbits and their switches are as follows; the last column records gr_q of the four terms and is not used in the proof.

s	(γ, θ)	old arrows	new arrows	gr_q
S_{18}	(0.0304313, 4.5024806)	$4 \xrightarrow{R} 3, 22 \xrightarrow{R} 23$	$22 \xrightarrow{R} 3, 4 \xrightarrow{R} 23$	$-8, 0, 0, 8$
S_{25}	(3.0828526, 1.7672244)	$6 \xrightarrow{L} 5, 20 \xrightarrow{L} 21$	$20 \xrightarrow{L} 5, 6 \xrightarrow{L} 21$	$-8, 0, 0, 8$
S_{69}	(0.0415966492, 5.4053953343)	$0 \xrightarrow{R} 19, 26 \xrightarrow{R} 27$	$0 \xrightarrow{R} 27, 26 \xrightarrow{R} 19$	$-8, 0, 0, 8$
S_{74}	(3.079988, 3.561056)	$20 \xrightarrow{L} 21, 32 \xrightarrow{L} 33$	$20 \xrightarrow{L} 33, 32 \xrightarrow{L} 21$	$-4, 0, 0, 4$

Each smoothing is disjoint from the arc system, so it keeps the generators, and b_s is the sum of the two old and the two new arrows. Every displayed arrow has relative delta degree -1 ; by the gr_q column none of the four deformations is bigraded, which is immaterial here.

Proof of Proposition 6.7. For each row, every composite of an arrow of b_s with an arrow of δ_N , in either order, is zero in $\mathcal{B}$, because the face labels differ or the endpoints do not match; and $b_s^2 = 0$ because no target of an arrow of b_s is the source of another. Direct encoding of the smoothed curve gives $\delta_{N_s} = \delta_N + b_s$ entry by entry, for smoothing disks of radii 0.003, 0.006 and 0.009. The census smoothing splits C into three torus components: an ι -invariant one, which projects to an arc, and two that project to one closed pillowcase component, the loop. With the orientations induced from C , their classes and the wrapped dimensions of arc and loop are

s	torus component classes	wrapped dimensions, arc + loop
S_{18}, S_{25}	$(17, 6), (2, 2), (2, 2)$	$5 + 4$
S_{69}	$(17, 8), (2, 1), (2, 1)$	$5 + 4$
S_{74}	$(15, 8), (3, 1), (3, 1)$	$7 + 2$

The calculation uses the full positive path algebra, with no word cutoff. For each pair the torsion condition of Lemma 6.5(c) holds by exact computation over $\mathbb{F}_2[U_X]$, and (6.1) gives the dimensions from the following data:

object or component	e	τ	r	$e + \tau - 2r$
N	104	105	101	7
$N_{S_{18}}, N_{S_{25}}$: arc; loop	80; 24	85; 20	80; 20	5; 4
$N_{S_{69}}$: arc; loop	84; 20	85; 20	82; 18	5; 4
$N_{S_{74}}$: arc; loop	76; 28	75; 30	72; 28	7; 2

Hence the undeformed dimension is seven and the four deformed dimensions are nine. $\square$

By Theorem 1.1, $\dim_{\mathbb{F}_2} I^{\natural}(K_7) = 9$. Under the CHKK correspondence the red curve carries the zero cochain: Smith proves this for the earring on Q_0 by comparison with the two-component unlink [Smi24, Lem. 5.3], and the argument applies verbatim to $\widehat{Q}_r$ for every slope r , with the unlink $\text{num}(Q_r + Q_{-r})$ in place of $\text{num}(Q_0 + Q_0)$; Smith uses it for $r = -1/2$ [Smi24, proof of Thm. 5.9]. Conditional on [CHKK22, Conj. D], Proposition 6.7 therefore implies that the object assigned to $Q_{1/3} + Q_{1/7}$ is not N with the zero cochain.

6.4. Selection by a second closure. Rank nine alone does not determine the object: among the four deformations of Proposition 6.7 it is attained by three homotopy classes (Corollary 6.9), and in the census of Proposition 6.10 it is attained at 11 orbits by 9 distinct switches. We therefore pair the same tangle with a second closure, whose instanton homology is computed by the squeeze of Theorem 1.1.

Proposition 6.8 (The second closure). *Set $K_* := \text{num}(Q_{-3/4} + Q_{1/3} + Q_{1/7})$. Then $I^{\natural}(K_*; \mathbb{Z}) \cong \mathbb{Z}^{31}$. Encoding the piecewise-linear curve of the earring tangle $\widehat{Q_{-3/4}}$ as above gives the 14-generator twisted complex $\mathcal{E}_*$ of (6.3). Its wrapped morphism dimensions with (N, b_s) are 31, 31, 23 and 25 for $s = S_{18}, S_{25}, S_{69}$ and S_{74} , and 25 with N .*

Corollary 6.9. *The four deformations of Proposition 6.7 represent exactly three homotopy-equivalence classes in $\text{Tw}(\mathcal{B})$. The S_{18} and S_{25} objects are strictly isomorphic; their common class, the S_{69} class and the S_{74} class are pairwise distinct.*

Proof. The idempotent-preserving permutation $(4\ 22)(5\ 21)$, extended by the identity on the other 27 generators, carries $\delta_N + b_{S_{18}}$ entry by entry to $\delta_N + b_{S_{25}}$. Pairing with $\mathcal{E}_*$ is a dg functor, so $\dim H \text{Mor}(\mathcal{E}_*, -)$ is a homotopy invariant; by Proposition 6.8 it takes the values 31, 23 and 25 on the S_{18}/S_{25} , S_{69} and S_{74} objects. The proof does not use the classification of curves by twisted complexes [CHKK22, §11.5], [HKK17, Thm. 4.3], [KWZ19, Thm. 1.5], which concerns graded and bigraded objects. $\square$

Proposition 6.10 (Census smoothings). *The model $L_{7,\text{PL}}$ has 82 self-intersection orbits, 52 connector–main and 30 connector–connector. Of the 82 census smoothings, 73 are generator-preserving. They determine 45 distinct, linearly independent switches $b_s \in \text{End}(N)$, and every one satisfies $D_N(b_s) = 0$ and $b_s^2 = 0$. Among all 73 occurrences, the simultaneous wrapped dimensions 9 against $\mathcal{E}$ and 31 against $\mathcal{E}_*$ occur only at S_{18} and S_{25} . These are also the only orbits whose shorter branch loop has torus class $(2, 2)$ up to sign.*

Proposition 6.11 (The remaining single smoothings). *The census smoothings at the nine orbits $S_7, S_8, S_{23}, S_{24}, S_{49}, S_{52}, S_{53}, S_{61}, S_{73}$ are not generator-preserving. Their encodings N_s are 31-generator twisted complexes whose wrapped dimensions against $(\mathcal{E}, \mathcal{E}_*)$ are $(3, 23)$ at S_7 , $(5, 23)$ at S_8, S_{53} and S_{73} , and $(7, 25)$ at the other five. Hence, among all 82 census smoothings, the pair $(9, 31)$ occurs only at S_{18} and S_{25} . Among the 82 opposite smoothings the pair $(9, 31)$ also occurs exactly at S_{18} and S_{25} ; there the smoothed curve is connected, its encoding has 21 generators, and its wrapped dimension against the earring $\widehat{\mathcal{E}_{3/4}}$ of $\widehat{Q_{3/4}}$ is 69, whereas that of $(N, b_{S_{18}})$ is 103. The knot $\text{num}(Q_{3/4} + Q_{1/3} + Q_{1/7})$ is alternating with determinant 103, so $\dim_{\mathbb{Q}} I^{\natural} = 103$.*

Proposition 6.12 (How far the two closures reach). *Every element $b = \epsilon_{18}b_{S_{18}} + \epsilon_{25}b_{S_{25}} + \epsilon_{69}b_{S_{69}} + \epsilon_{74}b_{S_{74}}$, $\epsilon_s \in \mathbb{F}_2$, is Maurer–Cartan, and among these 16 deformations the pair $(9, 31)$ occurs only for $b = b_{S_{18}}$ and $b = b_{S_{25}}$. The connector–main subcensus consists of 46 occurrences and 38 distinct switches; these are linearly independent, every sum of two of them is Maurer–Cartan, and hence all 2^{38} elements of their span are distinct and Maurer–Cartan. Of the $\binom{38}{2} = 703$ two-switch sums, 41 also have the pair $(9, 31)$, and none of these 41 presentations is carried to the S_{18} presentation by an idempotent-preserving permutation of the 31 generators.*

Hypothesis 6.13 ($q = 7$ local support). *If the CHKK tangle assignment is defined for $Q_{1/3} + Q_{1/7}$ with Smith’s perturbation for small $t < 0$, the object of $\text{Tw}(\mathcal{B})$ that it determines (an immersed curve with a bounding cochain) is homotopy equivalent to one of the following: (N, b_s) for one of the 45 distinct switches of Proposition 6.10; the encoding N_s of the census smoothing at one of the nine orbits of Proposition 6.11; or (N, b) for an element $b \in \text{span}_{\mathbb{F}_2}\{b_{S_{18}}, b_{S_{25}}, b_{S_{69}}, b_{S_{74}}\}$. Equivalently, it is homotopy equivalent to the encoding of the census smoothing at one of the 82 self-intersection orbits, or to (N, b) for b in this span.*

The hypothesis refers to the representative $L_{7, \text{PL}}$, perturbed off its tangential self-contacts; its set of self-intersections is not an invariant of the regular-homotopy class (Remark 6.15).

Corollary 6.14 (Conditional selection). *Assume the CHKK instanton-pillowcase correspondence and Hypothesis 6.13. Then the object assigned to $Q_{1/3} + Q_{1/7}$ is homotopy equivalent to the common S_{18}/S_{25} class.*

Proof of Propositions 6.8–6.12 and Corollary 6.14. The knot K_* has the 14-crossing planar diagram

$$(24, 13, 25, 14), (10, 25, 11, 26), (26, 11, 27, 12), (12, 27, 13, 0), (7, 14, 8, 15), (15, 8, 16, 9), \\ (9, 16, 10, 17), (23, 6, 24, 7), (5, 22, 6, 23), (21, 4, 22, 5), (3, 20, 4, 21), (19, 2, 20, 3), \\ (1, 18, 2, 19), (17, 0, 18, 1)$$

in the usual PD convention; in the census of SnapPy [CDGW] it is $K14n12201$. Seifert's algorithm gives

$$\Delta_{K_*}(t) = t^{-6} - t^{-5} + 2t^{-3} - 4t^{-2} + 5t^{-1} - 5 + 5t - 4t^2 + 2t^3 - t^5 + t^6,$$

so $\ell(K_*) = 31$ and $|\Delta_{K_*}(-1)| = 23$. The reduced Khovanov homology of K_* over $\mathbb{Z}$ is free, with ranks $(1, 0, 1, 1, 1, 2, 3, 3, 4, 5, 4, 3, 2, 1)$ in homological degrees 0 through 13, summing to 31 (computed with Khoca [Lew26]). The proof of Theorem 1.1 applies verbatim: the two bounds agree, the integral Kronheimer–Mrowka spectral sequence collapses on a free E_2 page, and $I^\natural(K_*; \mathbb{Z}) \cong \mathbb{Z}^{31}$. The knot K_* is not Khovanov-thin; the squeeze closes because $\ell(K_*)$ equals the Khovanov rank. The complex $\mathcal{E}_*$ has $\iota_\circ$ generators $1, 2, 7, 8, 11, 12, \iota_\bullet$ generators $0, 3, 4, 5, 6, 9, 10, 13$, and

$$\delta_{\mathcal{E}_*} = 1 \xrightarrow{M} 0 + 1 \xrightarrow{L} 2 + 2 \xrightarrow{M} 3 + 3 \xrightarrow{R} 4 + 5 \xrightarrow{M^2} 4 + 6 \xrightarrow{R} 5 + 7 \xrightarrow{M} 6 \\ + 8 \xrightarrow{L} 7 + 8 \xrightarrow{M} 9 + 10 \xrightarrow{R} 9 + 11 \xrightarrow{M} 10 + 12 \xrightarrow{L} 11 + 12 \xrightarrow{M} 13 + 0 \xrightarrow{R} 13. \quad (6.3)$$

For every pair used below the torsion condition of Lemma 6.5(c) holds by exact computation; for the pairing with $\mathcal{E}_*$ the data (e, τ, r) are $(228, 251, 227)$ for N and for $(N, b_{S_{74}})$, $(228, 251, 224)$ for $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$, and $(228, 251, 228)$ for $(N, b_{S_{69}})$, giving 25, 31 and 23. Under the correspondence the cochains on $\widehat{Q_{-1/2}}$ and $\widehat{Q_{-3/4}}$ vanish, by the argument following Proposition 6.7.

For the census, the provenance of each segment partitions the 82 orbits into 52 connector–main and 30 connector–connector orbits, and the census smoothing is encoded directly at every orbit. The nine orbits of Proposition 6.11 change the cancelled module; at every other orbit the two differentials differ by two old and two new arrows, and exact path multiplication gives $D_N(b_s) = b_s^2 = 0$. Duplicate presentations leave 45 distinct elements, linearly independent in the basis of arrows. Applying (6.1) to both earrings gives the following census of the pairs $(\dim H \text{Mor}(\mathcal{E}, (N, b_s)), \dim H \text{Mor}(\mathcal{E}_*, (N, b_s)))$:

pair	(5, 23)	(5, 25)	(7, 23)	(7, 25)	(7, 27)	(9, 23)	(9, 25)	(9, 31)
orbits	1	2	12	39	8	1	8	2
distinct switches	1	2	6	23	4	1	6	2

whose last column consists of S_{18} and S_{25} . For the class label, the two arcs of C between the two visits of a torus point of an orbit close up to complementary branch loops with classes summing to $(21, 10)$; the class of smaller ℓ^1 norm, up to sign, is the same at both torus points, and $(2, 2)$ occurs only at S_{18} and S_{25} .

For Proposition 6.11, the smoothed curves at the nine excluded orbits and the opposite smoothings at all 82 orbits are encoded as standalone twisted complexes and paired with $\mathcal{E}$ and $\mathcal{E}_*$, and at S_{18} and S_{25} with $\widehat{\mathcal{E}}_{3/4}$, with the torsion condition verified in each case. The knot

$\text{num}(Q_{3/4} + Q_{1/3} + Q_{1/7})$ is the numerator closure of three rational tangles of the same sign, hence alternating, with determinant $|4 \cdot 3 \cdot 7 (3/4 + 1/3 + 1/7)| = 103$, and for quasi-alternating knots the rank of $I^\natural$ is the determinant [KM11b, Cor. 1.6].

For Proposition 6.12, exact multiplication verifies the Maurer–Cartan equation for the sum of each pair of the 38 connector–main switches; since each switch is closed and square-zero, these checks say that all mixed anticommutators vanish, so the whole span is Maurer–Cartan. The two reductions applied to all 703 two-switch sums give 41 with the pair (9, 31). Regarding each differential as a directed graph on the 31 generators, labelled by idempotents, faces and path lengths, an exhaustive search over label-preserving bijections finds none carrying the S_{18} differential to any of the 41, and does find the S_{18} – S_{25} permutation; this distinguishes strict presentations only. Every sum of the four named switches is Maurer–Cartan; eight of the 16 have dimension seven against $\mathcal{E}$, and the other eight have dimension nine and auxiliary dimension 31 for $b_{S_{18}}$ and $b_{S_{25}}$; 23 for $b_{S_{69}}$, $b_{S_{18}} + b_{S_{25}} + b_{S_{69}}$, $b_{S_{25}} + b_{S_{69}} + b_{S_{74}}$ and $b_{S_{18}} + b_{S_{25}} + b_{S_{69}} + b_{S_{74}}$; and 25 for $b_{S_{74}}$ and $b_{S_{18}} + b_{S_{74}}$.

Under the correspondence, pairing the assigned object with the two earrings must give dimensions 9 and 31, which are homotopy invariants. Among the objects allowed by Hypothesis 6.13, the census, the pairs (3, 23), (5, 23) and (7, 25) of Proposition 6.11, and the sixteen sums leave only $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$, which are strictly isomorphic by Corollary 6.9. $\square$

Remark 6.15 (The traced analytic curve). A numerical trace of Smith’s perturbed curve $\mathcal{R}_t(Q_{1/3} + Q_{1/7})$ at $t = -0.015$ and at $t = -0.02$, computed through the symmetry $\mathfrak{s}$ of Remark 6.2 from the composition of $C_{3,-t}$ with the lines of slopes $1/3$ and $1/7$ through $(0, 0)$, has 50 transverse self-intersection orbits rather than 82, and its type-D encoding is strictly isomorphic to N . On the traced curve the census gives 43 generator-preserving census smoothings with 43 distinct switches, and the pair (9, 31) occurs at exactly two of them, the orbits whose shorter branch loop has class (2, 2), which correspond to S_{18} and S_{25} . The census smoothings at the other seven orbits change the cancelled module and were not paired. Among generator-preserving smoothings the selection is therefore not an artifact of the piecewise-linear model, although the trace is a floating-point computation and not a proof.

Remark 6.16 (Scope). Proposition 6.7 proves existence of four Maurer–Cartan elements, and Proposition 6.10 an exhaustive census in the stated geometric class; neither proves selection by the tangle. The ungraded delta-degree -1 endomorphism cohomology of N is much larger than its one-dimensional part in relative quantum grading zero, and the Maurer–Cartan class spanning that part changes the wrapped dimension seven to five, not to nine. Corollary 6.14 assumes the CHKK correspondence and that the assigned object lies in one of the stated classes, and the algebraic calculation proves neither assumption. The restriction on support is essential: Proposition 6.12 exhibits 41 two-switch sums outside the hypothesis with the same two closure dimensions. The elements b_s are not expressed in an ordered branch-jump complex, and they are not homogeneous in the only available grading.

Remark 6.17 (The opposite sign of the perturbation). For small $t > 0$ the curve, its self-intersections and their smoothings are the translates by $(\gamma, \theta) \mapsto (\gamma + \pi, \theta)$ of those used here (Remark 6.2), while the earring curves do not depend on t ; since the translation is a symplectomorphism of P^* of order two, we computed the pairings for $t > 0$ as those of the present objects with the translated earrings. The undeformed pairings remain 7 and 25, and the smoothings at S_{18} and S_{25} still give (9, 31), but those at S_{69} and S_{74} give (5, 25) and (7, 25). Dimension nine against the first earring is reached by seven of the 45 distinct switches: those at S_{10} , S_{18} , S_{22} , S_{25} , S_{29} , S_{30} , and the one shared by S_{26} , S_{27} , S_{28} , S_{32} , S_{33} . Among the 82 census smoothings, the 82 opposite smoothings and the 16 elements of the span of Proposition 6.12, the pair (9, 31) again occurs only at S_{18} and S_{25} , the opposite smoothings there are again separated

by the third closure (69 against 103), and in the span only $b_{S_{18}}$ and $b_{S_{25}}$ reach dimension nine. The two-closure selection therefore holds for either sign of t , whereas Proposition 6.7 does not: for $t > 0$ only two of the four objects have dimension nine. All other values in this section are stated for $t < 0$; the bigon statistics of Table 1 are the same for both signs.

6.5. A two-generator Yoneda reduction. A representability theorem would reach the assigned object without identifying the virtual chain of every figure-eight bubble: Fukaya, in the compact setting [Fuk25, Thms. 6.3 and 7.3], and Gao, under the hypotheses recalled in §6.2, obtain the composed object with its bounding cochain by a cyclic element and the Yoneda lemma. Let $\mathcal{C} \subset \mathcal{W}(P^*)$ be the CHKK full subcategory of unobstructed objects missing the corner (π, π) , and let $G = a^\circ \oplus a^\bullet$, so that $\mathcal{B} = \text{End}_{\mathcal{C}}(G)$. Then G generates $\mathcal{C}$ and $\text{Tw } \mathcal{C} \simeq \text{Tw } (\mathcal{B})$ [CHKK22, §11.5], by [Boc16, Thm. 7.6] and [HKK17, §3].

Proposition 6.18 (Yoneda recognition). *Assume that the quilted module $\mathcal{M}_{F_{1/3,t}}(A_{1/7})$ is defined and is represented by an object $X_{1/3,1/7} \in \mathcal{C}$. If there is an A_∞ quasi-isomorphism of right $\mathcal{B}$ -modules*

$$\mathcal{M}_{F_{1/3,t}}(A_{1/7})|_{\mathcal{B}} \simeq \text{Mor}_{\text{Tw}(\mathcal{B})}(G, (N, b_{S_{18}})),$$

then $X_{1/3,1/7} \simeq (N, b_{S_{18}}) \simeq (N, b_{S_{25}})$ in $\mathcal{C}$. The module must be given on a° and $a^\bullet$ as full chain-level complexes with every $\mathcal{B}$ -action and higher module map; homology dimensions of pairings with two test objects do not imply the conclusion.

Proof. Since G generates $\mathcal{C}$, restricted Yoneda is cohomologically fully faithful on $\mathcal{C}$, with values in perfect $\mathcal{B}$ -modules. Representability identifies the first module with the Yoneda module of $X_{1/3,1/7}$, so the assumed quasi-isomorphism is the image of a homotopy equivalence $X_{1/3,1/7} \simeq (N, b_{S_{18}})$; the second equivalence is Corollary 6.9. For the last assertion, the two-switch sum $(N, b_{S_{18}} + b_{S_{35}})$ is one of the 41 collisions of Proposition 6.12; but against the figure-eight curve Z built, as the earrings above, from the line of slope $-1/3$ through $(0, 0)$, its wrapped dimension is 11, against 9 for $(N, b_{S_{18}})$. That line passes through (π, π) , so Z is not the earring of $Q_{-1/3}$; the argument uses only that $\dim H \text{Mor}(Z, -)$ is a homotopy invariant. $\square$

By Proposition 6.4, the proposition becomes usable only after a wrapped representability theorem for immersed correspondences that covers the triple point, or an embedded correspondence inducing the same quilted module; neither is currently available. The full $\mathcal{B}$ -module comparison would then remain, and so would the identification of the symplectic assignment with the instanton tangle invariant, which CHKK state as [CHKK22, Conj. D].

6.6. Finite polygon counts and candidate supports. For $q = 5, 7, 11, 13$ we also record the output of a finite combinatorial test on the piecewise-linear curves $R_{\text{PL}} = \widehat{\mathcal{R}^b(Q_{-1/2})}$ and $L_{q,\text{PL}}$. De Silva, Robbin and Salamon identify holomorphic strips with combinatorial lunes for transverse embedded loops on a surface [dSRS14]; their hypotheses do not cover immersed curves in the orbifold pillowcase, nor the higher polygons of a bounding-cochain deformation.

The finite winding test. For intersection points x, y choose an arc α on red from x to y and an arc β on blue from y to x , and lift $\alpha \cdot \beta$ to the torus. The pair is a *candidate bigon* when (a) the lifted loop is null-homologous, (b) its winding function vanishes at the four corners, (c) that function is nonnegative on the sampled complementary regions, and (d) the local sectors at x and y are convex. The same conditions at every vertex define candidate triangles and quadrilaterals; we call their conjunction the *low-order test*, and the set of self-intersections at which the tabulated windows contain a complete polygon the *active set*. The matrices use one fixed lift of the red curve and sum over the two torus preimages of a blue self-intersection. Candidate arcs are restricted to explicit windows of segment indices, only single traversals are considered, and no

geometric bound proves the windows exhaustive. The parameters $(\varepsilon_{\text{blue}}, \varepsilon_{\text{red}}, r_{\text{red}})$ (connector amplitude, red perturbation amplitude, red pinch parameter) are $(0.05, 0.10, 0.25)$ at $q = 5$ and $(0.05, 0.16, 0.40)$ at $q = 7, 11, 13$ unless stated otherwise. With C_q the $\mathbb{F}_2$ -vector space on the transverse intersections and D_q^{big} the matrix of candidate bigons, the *bigon-rank statistic* is

$$h_q^{\text{big}} := \dim_{\mathbb{F}_2} C_q - 2 \operatorname{rank}_{\mathbb{F}_2} D_q^{\text{big}}; \quad (6.4)$$

this is a definition for a finite matrix, not the homology of a Floer complex.

At $q = 5$ the curves meet in nine points (one interior, two clusters of four near $\gamma = \pi$), and the test accepts exactly two candidate bigons with distinct vertices, so $h_5^{\text{big}} = 9 - 2 \cdot 2 = 5$; this agrees with the nine intersections and two bigons in Smith's proof of [Smi24, Thm. 5.9]. The generator counts depend on the parameters (19 against 23 at $q = 11$, 27 against 25 at $q = 13$), while the statistic agrees at two independent parameter choices, at both of which the raw bigon matrices for $q = 5, 7, 11, 13$ square to zero. One non-generic earring parameter choice at $q = 7$, which collapses the 13 generators to nine, was excluded.

Computation 6.19 (Finite bigon-rank pattern). For the specified representatives and parameters,

$$h_5^{\text{big}} = 5, \quad h_7^{\text{big}} = 7, \quad h_{11}^{\text{big}} = 15, \quad h_{13}^{\text{big}} = 17,$$

and the raw bigon matrices square to zero over $\mathbb{F}_2$. The value 5 at $q = 3$ in Table 1 is the computation of Hedden, Herald and Kirk [HHK18, §11.6] from a different tangle presentation; it is not an output of this family.

q	$\det K_q$	bigon statistic	$\dim_{\mathbb{F}_2} I^\natural$	difference	required rank change
3	3	5 (external)	5	0	0
5	1	5	7	-2	-1
7	1	7	9	-2	-1
11	5	15	13	+2	+1
13	7	17	15	+2	+1

TABLE 1. A finite numerical pattern. The instanton column is Theorem 1.1. The bigon-statistic column for $q \geq 5$ concerns the specified finite matrices; the $q = 3$ entry is external. The last column is the change in matrix rank that a deformed differential on the same generators would require.

At the five displayed rows $h_q^{\text{big}} - (q + 2) = 2 \operatorname{sgn}(|q - 6| - 3)$, so the sign of the difference changes as $\det K_q$ crosses 3; at $q = 11, 13$ the statistic is $q + 4$. This is a finite-sample observation, not a theorem and not an identification of h_q^{big} with a Floer homology.

A genuine deformation is a different object. An immersed Floer theory has two ordered branch-jump generators at each transverse self-intersection, of complementary degrees in a graded setup [AJ10]; a bounding cochain must satisfy the full Maurer–Cartan series, with repeated inputs at every order; and square-zero of the deformed differential is necessary, though not sufficient, for a proposed finite matrix. The search below labels the underlying double point instead. For a finite set S of self-intersections, a *support*, let $D_q^{\text{tab}}(S)$ add to D_q^{big} the tabulated triangle and quadrilateral terms through *distinct* crossings of S , and put

$$h_q^{\text{tab}}(S) := \dim_{\mathbb{F}_2} C_q - 2 \operatorname{rank}_{\mathbb{F}_2} D_q^{\text{tab}}(S). \quad (6.5)$$

Repeated insertions and higher operations are absent, and the accompanying Maurer–Cartan test uses only the tabulated monogon, self-bigon and distinct-input self-triangle terms.

Computation 6.20 (Low-order candidate supports). (i) At $q = 5$, among the singleton and two-element subsets of the active set, the unique smallest support with $h_5^{\text{tab}} = 7$ is $S = \{s_A, s_B\}$, with $(\gamma, \theta) \approx (0.03, 1.27)$ and $(3.06, 4.98)$. One tabulated quadrilateral changes the (x_4, y_6) bigon entry from 1 to 0; the resulting matrix has the sole entry (x_1, y_0) and squares to zero.

(ii) At $q = 7$, among the tested singleton crossings passing the low-order test, one crossing near $(0.042, 5.405)$ has $h_7^{\text{tab}} = 9$. Its tabulated triangle changes one bigon entry from 1 to 0; the resulting matrix has entries $(6, 4)$ and $(7, 5)$ and squares to zero.

(iii) At the default $q = 11$ parameters, 62 singleton crossings pass the low-order test and give matrix rank three. Exactly 56 of the resulting matrices square to zero; the square of each of the other six has a single nonzero entry, namely $(9, 15)$, $(9, 16)$, $(0, 14)$, and $(9, 14)$ for the remaining three. At the parameters $(0.03, 0.10, 0.25)$ the same test returns 66 singletons of the target rank, of which 16 fail square-zero and 50 pass.

At $q = 5$ the raw bigon matrix has rank two, and the pair s_A, s_B of Figure 3 is the only tested support that lowers the tabulated rank to one, so $h_5^{\text{tab}} = 9 - 2 = 7$; minimality holds only within this search. At $q = 7$ the raw rank three drops to two and $h_7^{\text{tab}} = 13 - 4 = 9$; at the parameters $(0.07, 0.16, 0.40)$ the crossing is near $(0.058, 5.413)$ and gives the same entries. This crossing is S_{69} , whose type-D deformation is the four-arrow switch $b_{S_{69}}$ rather than the single untyped support of the table. At $q = 11$ the raw matrix has 19 generators and rank two at the default parameters, and 23 generators and rank four at the second; the model has 240 self-intersections, some less than 10^{-3} apart. Resolving both local sectors at the selected $q = 5$ and $q = 7$ crossings and recounting candidate bigons reproduces the tabulated matrices for one sector choice; at $q = 7$ Proposition 6.7 replaces this experiment by an all-orders calculation. Computation 6.20 proves neither existence nor uniqueness of a bounding cochain: the candidates are not ordered branch jumps, the full Maurer–Cartan series is never tested, and neither 56 nor 50 counts bounding cochains.

7. THE KHOVANOV ARC

Throughout this section q is odd, $T_q = Q_{1/3} + Q_{1/q}$, and for $r \in \mathbb{Q} \cup \{\infty\}$

$$K_r(q) = \text{num}(Q_r + Q_{1/3} + Q_{1/q}),$$

so that $K_{-1/2}(q) = P(-2, 3, q)$ and $K_{-3/4}(7) = K_*$. For twisted complexes X, Y over $\mathcal{B}$ we write $HF(X, Y)$ for the homology of $\text{Mor}(X, Y)$, as in Lemma 6.5; for curves in pairing position it is their Lagrangian Floer homology [KWZ19, Thm. 5.22]. The object $(N, b_{S_{18}})$ of Corollary 6.14 is the sum of a rational arc and a closed curve carrying a two-dimensional local system, and one further Maurer–Cartan term, supported near the corner $(0, 0)$, carries it to the Bar-Natan invariant of T_q (Theorem 7.1). No rational closure distinguishes the two objects (Corollary 7.6).

7.1. Khovanov invariants in Smith’s coordinates. Let G be the group of isometries of $\mathbb{R}^2$ generated by the translations by $2\pi\mathbb{Z}^2$ and the reflection $v \mapsto -v$, so that the pillowcase is $\mathbb{R}^2/G$ in the coordinates (γ, θ) of §6.1. The group G acts freely on the complement of the lattice $\pi\mathbb{Z}^2$, and the quotient map $\mathbb{R}^2 \setminus \pi\mathbb{Z}^2 \rightarrow P^*$ is a covering, which we call the *planar cover*. The lattice point $(m\pi, n\pi)$ lies over the corner whose coordinates are the residues of $m\pi$ and $n\pi$ modulo 2π . A *lattice line* is a line in $\mathbb{R}^2$ through two lattice points. The lattice lines parallel to a given one form two G -orbits: the lines of one orbit pass over two of the corners, and those of the other over the remaining two. Every straight arc between two corners, such as the line of a rational tangle (§6.1), is the image of a segment joining consecutive lattice points of a lattice line.

The translation $\Phi(\gamma, \theta) = (\gamma + \pi, \theta)$ normalizes G and therefore descends to an involution of the pillowcase. It is a symplectomorphism of P^* , it exchanges the corners $(0, 0) \leftrightarrow (\pi, 0)$ and $(\pi, \pi) \leftrightarrow (0, \pi)$, and it is the translation of Remark 6.2.

Kotelskiy, Watson and Zibrowius assign to a pointed four-ended tangle T two immersed multicurves with local systems in the four-punctured sphere: the Bar-Natan invariant $\widetilde{\text{BN}}(T)$, and the Khovanov invariant $\widetilde{Kh}(T)$, which is the mapping cone of $H \cdot \text{id}$ on $\widetilde{\text{BN}}(T)$ [KWZ19, Def. 6.5]. Both are objects of $\text{Tw}(\mathcal{B})$, and their arc components end at the three non-special punctures. Under the identification of [KWZ19, §8.2] the special puncture is the corner $(0, \pi)$, which is the corner that holonomy-perturbed tangle character varieties miss in the coordinates of CHKK [CHKK22, Prop. 11.1 and §11.6]. In Smith's coordinates that corner is (π, π) (§6.1). We therefore draw an invariant Y of [KWZ19] in Smith's coordinates as $\Phi(Y)$, and we set

$$\text{BN}_q := \Phi(\widetilde{\text{BN}}(T_q)).$$

For a rational tangle Q , the arc $\widetilde{\text{BN}}(Q)$ and the figure-eight curve $\widetilde{Kh}(Q)$ around it agree, under this identification, with the character varieties $\mathcal{R}_\pi(Q)$ and $\mathcal{R}_\pi^h(Q)$ [KWZ19, §1.8 and §8.2]. When Q_r is glued to T_q , its mirror Q_{-r} enters the pairing theorem [KWZ19, Thm. 1.9], and in Smith's coordinates the curve $\Phi(\widetilde{Kh}(Q_{-r}))$ is the earring curve E_r of $\widehat{Q}_r$ used in Section 6, a Herald–Kirk figure-eight [HK24, Def. 6.7]; we have verified this identification directly, by comparing encodings, for 49 slopes. We call the straight arc that E_r surrounds the *arc of* E_r . For $r = m/n$ in lowest terms it joins $(\pi, 0)$ to $(0, \pi)$ if m and n are odd, $(0, 0)$ to $(0, \pi)$ if n is even, and $(0, 0)$ to $(\pi, 0)$ if m is even (§6.1). The complexes $\mathcal{E}$ and $\mathcal{E}_*$ of Section 6 encode $E_{-1/2}$ and $E_{-3/4}$. Since Φ is a symplectomorphism, the pairing theorem therefore reads

$$\dim_{\mathbb{F}_2} HF(E_r, \text{BN}_q) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2), \quad r \in \mathbb{Q} \cup \{\infty\}. \quad (7.1)$$

We have compared the two sides of (7.1) at the 340 slopes used for $q = 5$ and the 342 slopes used for $q = 7$ in Corollary 7.6, computing the right side with Khoca [Lew26]; they agree at every slope.

7.2. The structure theorem. Put $r_q = (1 - q)/(3q - 4)$. Its numerator is even and its denominator is odd, so the arc of E_{r_q} joins $(0, 0)$ to $(\pi, 0)$. We use three curves.

- α_q is the arc of E_{r_q} , an embedded arc from $(0, 0)$ to $(\pi, 0)$.
- c is the boundary of a regular neighborhood of the arc of E_{-1} , which joins $(\pi, 0)$ to $(0, \pi)$. Thus c is a simple closed curve separating $\{(\pi, 0), (0, \pi)\}$ from $\{(0, 0), (\pi, \pi)\}$, and its lifts are lines parallel to the lifts of the arc of E_{-1} that pass through no lattice point.
- (c, J_2) is c with the indecomposable two-dimensional local system $J_2 = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. As a twisted complex it is the doubled word of c : the doubled word represents c with the transposition local system, which over $\mathbb{F}_2$ is similar to J_2 [KWZ19, §5.3], [AS21, Lem. 2.24].

Up to isotopy, Φ carries each of these curves to itself. The arc α_q and the arc BN_q are bigraded, as are all invariants of [KWZ19]. The curve c admits a δ -grading in the sense of [Zib20, Def. 4.44] but no bigrading, and neither does (c, J_2) . Indecomposable two-dimensional local systems also occur on components of $\widetilde{\text{BN}}$ of other tangles [KWZ26, Rem. 3.20].

Write N_q for the reduced twisted complex that encodes $L_{q, \text{PL}}$ in the chart of §6.3, so that $N_7 = N$; the complex N_5 is displayed in (A.1). Each is an arc from $(0, 0)$ to $(\pi, 0)$, and its end at $(0, 0)$ is the generator 36 of N_7 , respectively 0 of N_5 . In that chart the faces R, M, L contain the corners $(0, 0), (\pi, 0), (\pi, \pi)$, so an arrow labelled R is a chord around the corner $(0, 0)$. Two reduced twisted complexes are *strictly isomorphic* if an idempotent-preserving bijection of their generators carries one differential to the other entry by entry. The four-arrow element $b_5 \in \text{End}(N_5)$ of (A.2) satisfies $D_{N_5}(b_5) = b_5^2 = 0$, and (N_5, b_5) is strictly isomorphic to the

encoding of the census smoothing of $L_{5, \text{PL}}$ at each of two of its self-intersection orbits; it raises the pairing with $E_{-1/2}$ from 5 to $7 = q + 2$.

To describe BN_q , fix a lift p_0 of $(0, 0)$ and let p_M be the lift of $(\pi, 0)$ for which the segment $[p_M, p_0]$ is a lift of α_q . Let ℓ be the lattice line through p_0 parallel to the lifts of c ; its lattice points lie alternately over $(0, 0)$ and (π, π) . Number them p_j , $j \in \mathbb{Z}$, consecutively and so that the angle between $p_1 - p_0$ and $p_0 - p_M$ is acute, and let ℓ^+ be the open half-plane bounded by ℓ that does not contain p_M . The translation by $p_2 - p_0 \in 2\pi\mathbb{Z}^2$ generates the stabilizer in G of each lift of c .

Theorem 7.1 (Theorem 1.4). (a) *Let $q = 7$ and $b \in \{b_{S_{18}}, b_{S_{25}}\}$, or $q = 5$ and $b = b_5$. Then (N_q, b) is strictly isomorphic to $\alpha_q \oplus (c, J_2)$.*
 (b) *Let $5 \leq q \leq 21$ be odd. Then BN_q is the arc whose lift starts at p_M , runs along $[p_M, p_0]$, crosses ℓ between p_{-1} and p_0 , and then runs in ℓ^+ , parallel to ℓ and close to it, until it ends at p_4 .*
 (c) *Let q and b be as in (a), and let*

$$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19 \in \text{End}(N_7), \quad \kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11 \in \text{End}(N_5).$$

Then $(\delta_{N_q} + \kappa_q)^2 = (\delta_{N_q} + b + \kappa_q)^2 = 0$, and $(N_q, b + \kappa_q)$ is strictly isomorphic to BN_q .

Part (b) says that BN_q arises from α_q by sliding its end at $(0, 0)$ twice around that corner, parallel to c : beyond p_0 the lift covers two periods of a lift of c , ending at the lift $p_4 = p_0 + 2(p_2 - p_0)$ of $(0, 0)$. In (c), the arrows $0 \xrightarrow{R} 19$ and $12 \xrightarrow{R} 11$ are terms of δ_{N_7} and δ_{N_5} , so adding κ_q deletes one arrow and adds an arrow labelled R at the end of N_q at $(0, 0)$. Both terms of κ_q are chords around the corner $(0, 0)$.

Proof. Each statement asserts that two explicit reduced twisted complexes are strictly isomorphic, or that an explicit matrix over $\mathcal{B}$ squares to zero. The complexes of α_q , of c , of its doubled word and of the arc in (b) are obtained by encoding their lifts in the chart of §6.3; $\widetilde{\text{BN}}(T_q)$ is computed with Zibrowius's program [Zib] and drawn in Smith's coordinates by Φ . All complexes involved are of loop type, that is, every generator meets at most two arrows, so a strict isomorphism is decided by comparing the cyclic or linear words of the components, matching idempotents, labels and the directions of arrows.

For (a), $(N_7, b_{S_{18}})$ and $(N_7, b_{S_{25}})$ have components of 23 and 8 generators, and (N_5, b_5) of 15 and 8; in each case the first component is strictly isomorphic to α_q and the second to the doubled word of c , which represents (c, J_2) . For (b), the encoding of the arc described there is strictly isomorphic to BN_q for $q = 5, 7, \dots, 21$; it is the only one among the 24 arcs obtained from α_q by sliding one of its ends along c , in either direction and on either side, through one, two or three periods. For (c), direct multiplication in $\mathcal{B}$ gives the two Maurer–Cartan identities, and the words of $(N_q, b + \kappa_q)$ and of BN_q agree. $\square$

Remark 7.2 (The corner term). The arrow that κ_q deletes is a chord of the closed component of (N_q, b) around the corner $(0, 0)$. Deleting it opens that component into an arc with two ends at $(0, 0)$, and the added arrow joins one of them to the end of α_q . Thus κ_q changes which two of the three arc ends at $(0, 0)$ are joined. An exhaustive search over the complexes obtained from (N_q, b) by deleting one arrow and adding one arrow between two generators that meet at most one arrow finds, for each of $b_{S_{18}}$, $b_{S_{25}}$ and b_5 , exactly two that are strictly isomorphic to BN_q (566 candidates for each b at $q = 7$, and 402 at $q = 5$). In each, the added arrow is labelled R and starts at the end of N_q at $(0, 0)$; κ_7 is the one common to $b_{S_{18}}$ and $b_{S_{25}}$, and the others are listed in §A.2.

Call an element of $\text{End}(X)$ an *algebraic switch* of a reduced twisted complex X if it replaces two arrows $i \xrightarrow{Y^k} j$ and $i' \xrightarrow{Y^k} j'$ of δ_X , with j and j' of the same idempotent, by $i \xrightarrow{Y^k} j'$ and $i' \xrightarrow{Y^k} j$, and if the result satisfies the Maurer–Cartan equation. Of the 45 distinct switches of census smoothings at $q = 7$ (§6.3), 41 are algebraic switches; each of the other four exchanges the heads of an arrow labelled M and an arrow labelled M^2 . The term κ_q is not a sum of switches: no sum of at most three algebraic switches of N_q is strictly isomorphic to BN_q (there are 110 algebraic switches of N_7 and 221 925 such sums, and 58 switches of N_5 and 32 567 sums), and no sum of at most three of the 45 census switches is strictly isomorphic to BN_7 .

Remark 7.3 (The opposite sign of the perturbation). By Remark 6.17, the pairings for small $t > 0$ are those of the present objects with the curves $\Phi(E_r)$. Since Φ is an involution, $\Phi(E_r) = \widetilde{Kh}(Q_{-r})$, and the comparison object becomes $\Phi(\text{BN}_q) = \widetilde{\text{BN}}(T_q)$. The curves α_q and c are Φ -invariant, so part (a) is unchanged; in (b) the slide takes place at the end $(\pi, 0)$; and (c) holds with

$$\kappa'_7 = 2 \xrightarrow{M} 3 + 18 \xrightarrow{M^2} 3, \quad \kappa'_5 = 10 \xrightarrow{M} 11 + 22 \xrightarrow{M^2} 11$$

in place of κ_7 and κ_5 , where 18 and 22 are the ends at $(\pi, 0)$. The two corner terms are not interchangeable. For $t < 0$, the object $(N_7, b_{S_{18}} + \kappa'_7)$ pairs to 13 with E_{-1} , whereas $\text{rank } \widetilde{Kh}(K_{-1}(7)) = 11$; under the CHKK correspondence this pairing would compute $\dim I^\natural(K_{-1}(7))$, contradicting $\dim I^\natural \leq \text{rank } \widetilde{Kh}$ [KM11b]. At $q = 5$ the corresponding values are 9 and 7.

7.3. Pairing with linear curves. For curves X, Y in P^* , at least one of them compact, let $i(X, Y)$ be the minimal number of intersection points of curves homotopic to X and Y , where arcs are homotoped through arcs with the same ends.

Definition 7.4. Let $w \in \mathbb{Z}^2$ be primitive. A compact curve $R \subset P^*$ is *linear of direction w* if every lift of R lies within a small distance of a line of direction w and is a graph over it, and either the line meets no lattice point, in which case R is the simple closed curve of direction w , or the line is a lattice line and the lift passes each of its lattice points on a prescribed side, the choices of side being invariant under G .

Every simple closed curve is linear. So is every earring E_r , whose lifts pass the consecutive lattice points of the lattice line containing the lifts of its arc on alternating sides, and more generally every figure-eight around a straight arc between two corners.

Theorem 7.5 (Theorem 1.5). *Let $5 \leq q \leq 21$ be odd, and let R be a linear curve whose direction is parallel neither to α_q nor to c .*

- (a) $i(R, \text{BN}_q) = i(R, \alpha_q) + 2i(R, c)$.
- (b) *If R is bigraded, then*

$$\dim HF(R, \text{BN}_q) = \dim HF(R, \alpha_q) + 2 \dim HF(R, c) = \dim HF(R, \alpha_q \oplus (c, J_2)).$$

Proof. Part (b) from part (a). For two δ -graded compact curves that are not parallel, with local systems X and X' , the dimension of HF is $(\dim X)(\dim X')$ times their minimal intersection number [Zib20, Thm. 4.45]; this applies to R and (c, J_2) and gives $\dim HF(R, (c, J_2)) = 2i(R, c) = 2 \dim HF(R, c)$. For bigraded curves one of which is an arc, the same formula is [KWZ19, Thm. 5.25], applied after moving all curves back by Φ ; Kotelskiy, Watson and Zibrowius write out the proof for compact curves, which is the proof of [Zib20, Thm. 4.45], and state that the remaining cases are treated similarly. It applies to R and the bigraded arcs α_q and BN_q . Hence (a) implies (b).

Geodesic representatives. Fix a small $\varepsilon > 0$ and remove from $\mathbb{R}^2$ the open ε -discs $B(p)$ about the lattice points p . Their boundary circles are locally concave, so the flat surface that remains

is locally CAT(0) by the theorem of Alexander, Berg and Bishop [AKP26, Thm. 7.28], and its universal cover is CAT(0) by the globalization theorem [AKP26, Thm. 5.6]. Two geodesics in a CAT(0) space meet in a connected set. Hence geodesic representatives in the quotient by G , in which an arc ends on the boundary circle of its corner, realize the minimal intersection numbers, provided a common arc of tangency along a boundary circle is counted once if the four ends of the two curves near it interleave and not at all otherwise. The representatives are: for R , the lines of direction w tangent to the discs of their lattice points on the prescribed sides; for α_q , the part of $[p_M, p_0]$ outside the two discs; for c , the line in ℓ^+ parallel to ℓ at distance ε ; and for BN_q , the segment from $\partial B(p_M)$ to a tangent point of $\partial B(p_0)$, an arc of $\partial B(p_0)$, and then the same parallel line up to $\partial B(p_4)$, where it ends. The segment $[p_M, p_0]$ meets no other disc because its direction is primitive.

Reduction to one lattice point. A lift of an arc has trivial stabilizer, so for an arc X the number $i(R, X)$ counts the intersections of the full preimage of R with one lift of X . A fundamental domain for the stabilizer of the lift of c is one period, so $2i(R, c)$ counts the intersections with the half-open stretch of the parallel line from p_0 to p_4 , the end near p_0 included. Compare the lift of BN_q with the union of the lift of α_q and this stretch. Outside small neighborhoods of $p_M, p_0, \dots, p_4$ the two coincide. Near p_M both end along the same segment, and near p_1, p_2, p_3 both pass straight. The translation by $p_4 - p_0 \in 2\pi\mathbb{Z}^2$ belongs to G , so it carries the lifts of R through p_4 bijectively onto those through p_0 , preserving the sides. It therefore suffices to show, for every lift λ of R through p_0 , that

$$\text{corner}(a, b') + \text{end}(-b') = \text{end}(a) + 1, \quad (7.2)$$

where a and b' are the directions from p_0 to p_M and to p_1 ; $\text{end}(x)$ is the number of intersections of λ with an arc that ends on $\partial B(p_0)$ along the direction x ; $\text{corner}(a, b')$ is the number of intersections of λ with the part of BN_q near p_0 ; and the summand 1 counts the straight passage of the stretch of c .

Local counts. Let D be the open half-plane bounded by the lattice line of λ into which λ is displaced near p_0 , and D' the other one. The directions a and b' are not parallel to w , so each points into D or into D' . The line λ is tangent to $\partial B(p_0)$ on the side of D . An arc ending orthogonally on $\partial B(p_0)$ along x meets it once if x points into D and not at all otherwise. A straight passage meets it once, since two distinct tangent lines of a circle meet in one point. The angle between a and b' is less than π , and the part of BN_q near p_0 runs in along a , around the far side of $B(p_0)$ from that angle, and out along b' ; it meets λ once if a and b' point into different half-planes, twice if both point into D , and not at all if both point into D' . Since $\text{end}(-b') = 1$ exactly when b' points into D' , the two sides of (7.2) are

	$a \in D, b' \in D'$	$a \in D', b' \in D$	$a, b' \in D$	$a, b' \in D'$
left side	$1 + 1$	$1 + 0$	$2 + 0$	$0 + 1$
right side	$1 + 1$	$0 + 1$	$1 + 1$	$0 + 1$

and they agree in every case. This proves (a). $\square$

Corollary 7.6 (Corollary 1.6). *Let $q \in \{5, 7\}$ and let b be as in Theorem 7.1(a). For every $r \in \mathbb{Q} \cup \{\infty\}$,*

$$\dim HF(E_r, (N_q, b)) = \dim HF(E_r, (N_q, b + \kappa_q)) = \text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2).$$

Proof. The second equality is Theorem 7.1(c) together with (7.1). The earring E_r is linear, and after moving all curves back by Φ it becomes the bigraded curve $\widetilde{Kh}(Q_{-r})$. If its direction is parallel neither to α_q nor to c , Theorems 7.1(a) and 7.5(b) give $\dim HF(E_r, (N_q, b)) = \dim HF(E_r, \text{BN}_q)$.

The two remaining earrings are E_{-1} , parallel to c , and E_{r_q} , parallel to α_q , and for them all three numbers are computed directly:

q	E_{-1}			E_{r_q}		
	(N_q, b)	$(N_q, b + \kappa_q)$	$\text{rank } \widetilde{Kh}$	(N_q, b)	$(N_q, b + \kappa_q)$	$\text{rank } \widetilde{Kh}$
5	7	7	7	30	30	30
7	11	11	11	46	46	46

The undeformed complex N_q pairs to 7 and 28 at $q = 5$, and to 11 and 44 at $q = 7$. $\square$

We have also computed the three numbers directly at 340 slopes for $q = 5$ and 342 slopes for $q = 7$: every slope with $|r| \leq 2$ and denominator at most 14, every slope with $|r| \leq 5$ and denominator at most 6, the integers with $|r| \leq 10$, $\pm r_q$ and ∞ . These include 108 slopes with odd numerator and odd denominator, for which the arc of E_r joins $(\pi, 0)$ to $(0, \pi)$. The three numbers agree at every slope. The undeformed complex N_q pairs to less than $\text{rank } \widetilde{Kh}$ at 40 of these slopes for $q = 5$ and at 43 for $q = 7$, and never to more. Under the CHKK correspondence the pairing of the assigned object with E_r computes $\dim I^\sharp(K_r(q))$, which is at most $\text{rank } \widetilde{Kh}(K_r(q))$ [KM11b]. Both (N_q, b) and $(N_q, b + \kappa_q)$ attain this bound at every rational closure: at $r = -1/2$ the common value is $q + 2 = \dim I^\sharp(P(-2, 3, q))$ (Theorem 1.1), and at $r = -3/4$ and $q = 7$ it is $31 = \dim I^\sharp(K_*)$ (Proposition 6.8).

Remark 7.7 (The direction of c). The hypothesis that the direction is not parallel to c cannot be dropped. Among figure-eights around straight arcs and simple closed curves, exactly two curves pair differently with BN_q and with $\alpha_q \oplus (c, J_2)$, and both are parallel to c : the curve c itself, with 11 against 13 at $q = 7$ and 7 against 9 at $q = 5$, and the figure-eight around the arc parallel to c that joins $(0, 0)$ to (π, π) , with 13 against 11 and 9 against 7. The latter surrounds the corner (π, π) , which no tangle character variety meets, and is not the earring of a rational tangle; the earring E_{-1} , around the arc joining $(\pi, 0)$ to $(0, \pi)$, pairs equally. For general side patterns we have computed all linear curves with direction (m, n) , $|m|, |n| \leq 6$, or parallel to α_q , on lattice lines of both orbits and with every side pattern of period 2, 4 or 6, together with the simple closed curves: 1887 curves for each $q = 5, 7, \dots, 13$. For each q , exactly 22 of them pair differently, all parallel to c : all 19 side patterns on the lattice line through $(0, 0)$ and (π, π) , the two constant patterns on the line through $(\pi, 0)$ and $(0, \pi)$, and the simple closed curve. The constant patterns on either line and the simple closed curve are isotopic to c . No curve parallel to α_q pairs differently, although the proof does not cover that direction.

Remark 7.8 (Curves parallel to c). Curves parallel to c separate (N_q, b) from $(N_q, b + \kappa_q) \cong \text{BN}_q$. At $q = 7$ the curve c pairs to 11 with BN_7 and to 13 with $(N_7, b_{S_{18}})$. The curve (c, J_2) pairs to 22 and to 26, the difference being $\dim HF((c, J_2), (c, J_2)) = 4$. The image under Φ of the special component of $\widetilde{Kh}(T_7)$ in the sense of [KWZ19], the component other than the figure-eight $\widetilde{Kh}(Q_{-r_7})$, is parallel to c and pairs to 52 and to 44. At $q = 5$ the corresponding values are 7, 9; 14, 18; and 36, 28.

7.4. The instanton object of T_q . Theorem 7.1 and Corollary 7.6 suggest, without deciding, the following question.

Question 7.9 (Question 1.7). Let $q \geq 5$ be odd with $\gcd(q, 3) = 1$. Suppose that the CHKK tangle assignment is defined for T_q with Smith's perturbation for small $t < 0$, and let X_q be the object of $\text{Tw}(\mathcal{B})$ that it determines, an immersed curve with a bounding cochain, drawn in Smith's coordinates. Is X_q homotopy equivalent to BN_q ?

For $q = 5$ and $q = 7$ the evidence is as follows. The object BN_q arises from the piecewise-linear model of Smith's curve by the explicit Maurer–Cartan element $b + \kappa_q$ (Theorem 7.1). By Corollary 7.6, (N_q, b) and $(N_q, b + \kappa_q)$ have the same pairing with every rational earring, equal to the rank of reduced Khovanov homology of the closure; both are compatible with $\dim I^\natural \leq \text{rank } \widetilde{Kh}$ at every rational closure, and rational closures cannot distinguish them. They are distinguished by curves parallel to c (Remark 7.8), and by gradings: BN_q is bigraded and (N_q, b) is not, but no quantum grading is known on the instanton side, so this difference is not a test.

Proposition 7.10. *The object BN_7 is not homotopy equivalent to any object allowed by Hypothesis 6.13, although $\dim HF(E_r, \text{BN}_7) = \dim HF(E_r, (N, b_{S_{18}}))$ for every $r \in \mathbb{Q} \cup \{\infty\}$.*

Proof. The dimension of $HF(Z, -)$ for a fixed object Z is invariant under homotopy equivalence. By (7.1), BN_7 pairs to 9 with $E_{-1/2}$ and to 31 with $E_{-3/4}$, the ranks of reduced Khovanov homology of $P(-2, 3, 7)$ and K_* (Section 3 and Proposition 6.8). Among the objects allowed by Hypothesis 6.13, only $(N, b_{S_{18}})$ and $(N, b_{S_{25}})$ have this pair (proof of Corollary 6.14), and they are strictly isomorphic (Corollary 6.9). By Theorem 7.1(a),

$$\dim HF((c, J_2), (N, b_{S_{18}})) = \dim HF((c, J_2), \alpha_7) + \dim HF((c, J_2), (c, J_2)) = 22 + 4 = 26,$$

whereas $\dim HF((c, J_2), \text{BN}_7) = 22$. The last statement is Corollary 7.6. $\square$

Consequently an affirmative answer to Question 7.9 for $q = 7$ would show that Hypothesis 6.13 fails, and no rational closure could detect the failure: the two closures of Corollary 6.14, like every other rational closure, assign the same dimension to BN_7 as to the class of $(N, b_{S_{18}})$ that the hypothesis selects.

Remark 7.11 (The cases $q \geq 11$). For $q = 11$ and 13 the encoding of a numerical trace of Smith's curve is strictly isomorphic to N_q , with 47 and 55 generators, and its pairing with $E_{-1/2}$ is $q + 4$, two more than $\text{rank } \widetilde{Kh}(P(-2, 3, q)) = q + 2$; for $q = 17$ and 19 the pairing of the traced curve is $q + 8$. At $q = 5, 7$ the pairing is q , and one switch raises it to $q + 2$. The deformation of Theorem 7.1 does not persist in the following finite sense. Call the minimal number of arrows that must be deleted from δ_{N_q} , and the same number added, to reach a complex strictly isomorphic to BN_q the *edit distance*. It is 2 for $q = 5, 7$, 4 for $q = 11, 13$, and at least 6 for $q = 17, 19$. At $q = 5, 7$ one algebraic switch followed by one replacement of an arrow reaches BN_q ; at $q = 11$ and 13 no chain of at most three algebraic switches, each Maurer–Cartan for the complex it deforms, ends one replacement away from BN_q . At $q = 11$ the first such chains have four switches, and they end at the sum of an arc and a closed curve other than $\alpha_{11} \oplus (c, J_2)$; the latter is more than five switches away from N_{11} , against one at $q = 5, 7$. None of the 35 004 complexes (6 331 up to strict isomorphism) obtained from N_{11} by at most two algebraic switches has α_{11} as a summand, and each of them either pairs with $E_{-1/2}$ to a number other than $\dim I^\natural(P(-2, 3, 11)) = 13$, or pairs with some earring of a panel of 25 slopes to more than the rank of reduced Khovanov homology of the closure. These are finite searches within the class of algebraic switches, and they do not exclude a larger bounding cochain.

8. DISCUSSION

8.1. Relation to earlier work. Hedden, Herald and Kirk verify their Conjecture 6.5 for two-bridge knots [HHK18, §7] and for a list of torus knots [HHK18, §11]; the torus knots $T(3, n)$ are treated in [Wue26c]. For the decomposition of Theorem 1.2, Fukumoto, Kirk and Pinzón-Caicedo describe the unperturbed traceless character variety and count nine generators for $P(-2, 3, 7)$ [FKP17, §5]; they do not perturb or compute a Floer complex. Theorems 1.2 and 1.3 supply

the perturbed curves, their Floer complexes and the gradings. For $q \geq 9$ the differential in Theorem 1.2 is nonzero, so the agreement with $I^\natural$ is not a count of generators. The twists act on the pillowcase by the shear of [HHK18, (28)–(30)].

Smith proves that for $q = 5$ his decomposition $\widehat{Q_{-1/2}} \cup (Q_{1/3} + Q_{1/5})$ with zero bounding cochains gives a Floer homology of rank 5 rather than $\dim I^\natural = 7$ [Smi24, Thm. 5.9], so that under CHKK’s correspondence a nonzero bounding cochain is required [Smi24, proof of Thm. 5.9]; in his closing paragraph he proposes finding the bounding cochains that correct the rank. Sections 6 and 7 give two answers at $q = 5$ and $q = 7$ that no rational closure distinguishes: the smoothings (N_q, b) , and their deformations $(N_q, b + \kappa_q) \cong \text{BN}_q$.

For rational tangles the traceless character varieties coincide with the invariants of Kotelskiy, Watson and Zibrowius [KWZ19, §1.8]. Hedden, Herald, Hogancamp and Kirk assemble Khovanov’s tangle invariant from the traceless character varieties of the crossingless resolutions [HHHK20, Thm. 1.1 and Cor. 1.2], with a result that agrees with $\widehat{Kh}(T)$ [KWZ20], and they ask whether higher terms can be added so as to recover the Kronheimer–Mrowka spectral sequence [HHHK20, Question 1.4]. Cazassus, Herald, Kirk and Kotelskiy conjecture curve-valued instanton invariants $I(T)$ and $I^\natural(T)$ of tangles, and predict that the relevant algebra is that of [KWZ19] [CHKK22, Conj. E and §11.4]. They do not conjecture $I(T) = \widehat{\text{BN}}(T)$, which cannot hold for all tangles because $\dim I^\natural < \text{rank } \widehat{Kh}$ for some knots, for instance $T(4, 5)$ [CHKK22, §10.3]. Question 1.7 concerns one tangle, T_q , for whose closures $P(-2, 3, q)$ and K_* the two dimensions agree (Theorem 1.1 and Proposition 6.8). Kotelskiy, Watson and Zibrowius suggest that the differentials of the spectral sequences from Khovanov homology to Heegaard Floer and instanton homology might be realized by smoothing self-intersections of $\widehat{\text{BN}}(T)$ and $\widehat{Kh}(T)$ [KWZ19, §1.8]. Sections 6 and 7 proceed in the opposite direction: a smoothing of the curve on the instanton side reaches the Khovanov rank, and one further term carries it to the Khovanov curve.

Smoothing a self-intersection of an immersed curve by a degree-one bounding cochain is Lagrangian surgery. Palmer and Woodward prove that immersed Floer cohomology is invariant under surgery for Lagrangians of dimension at least three, and in dimension two under an additional assumption [PW26, Thm. 1.3]; for curves on surfaces, Auroux and Smith compare surgery with the mapping cone, which involves a bounding cochain [AS21, Lem. 2.25 and Rem. 2.26]. The corner term κ_q is of a different kind: it is supported on chords around a corner (Remark 7.2), which are morphisms of the wrapped category of [CHKK22, §11.5] and not self-intersection points of the curve.

8.2. Open problems. *The instanton object.* Question 1.7 cannot be decided by rational closures at $q = 5, 7$ (Corollary 7.6); curves parallel to c , and gradings, separate the candidates (Remark 7.8). For $q \geq 11$ a deformation of Smith’s curve to BN_q , if it exists, is not small in the sense of Remark 7.11.

The local support hypothesis. Proposition 7.10 shows that Hypothesis 6.13 is not implied by closure data. A direct proof would have to identify the rigid figure-eight boundary strata of the quilted moduli spaces, exclude simultaneous support at several nodes, and intertwine the output with the reconnection morphisms of §6.3. In the strip-shrinking model the zero-width boundary stratum is predicted to contain quilted trajectories with any positive number of figure-eight bubbles, and the zero-input correction a count of figure-eight bubble trees [BW18, §§4.2–4.4], so neither zero input nor index zero forces support at a single node. A proof must use the geometry near the six internal edge circles.

The formal route. Proposition 6.18 reduces the identification of the assigned object to a wrapped representability theorem of the kind Gao proves [Gao17, Thm. 7.24 and Cor. 7.25], and Gao’s theorem requires an embedded correspondence; by Proposition 6.4 the correspondence

induced by the line of $Q_{1/3}$ has a triple point. An immersed version of Gao's theorem covering this triple point, or an embedded replacement inducing the same quilted module, would be needed, and neither is available. Independently, every selection statement of Sections 6 and 7 is conditional on the correspondence between instanton and pillowcase invariants [CHKK22, Conj. D]; neither Fukaya's nor Gao's correspondence theorem [Fuk25, Gao17] compares the two.

Other families. Theorem 1.3 covers $T(3, n; 2, m)$ for $n \equiv 1, 2 \pmod{6}$. For $n \equiv 4, 5 \pmod{6}$ and $n \geq 10$ the curve of [Wue26c] has a central circle, and the shear argument of Section 4 needs that circle to be a straight line, which we prove only for $n = 5$; the signature argument also fails for $n \equiv 5 \pmod{6}$.

Gradings of bounding cochains. The deformations at $q = 7$ are not bigraded (§6.3), and we know of no grading of bounding cochains in the pillowcase that would select among them. A $\mathbb{Z}/4$ -graded selection principle compatible with HHK's absolute grading would replace the closure arguments of Section 6 by a structural one.

APPENDIX A. DATA AND COMPUTATIONS

A.1. The complex N_5 . Encoding $L_{5, \text{PL}}$ in the chart of §6.3 and performing the unit cancellations gives the reduced twisted complex N_5 on the generators $0, 1, \dots, 22$. Its $\iota_\circ$ generators are $3, 4, 9, 10, 13, 14, 19, 20$, the others carry $\iota_\bullet$, and

$$\begin{aligned} \delta_{N_5} = & 1 \xrightarrow{M^2} 0 + 2 \xrightarrow{R} 1 + 3 \xrightarrow{M} 2 + 4 \xrightarrow{L} 3 + 4 \xrightarrow{M} 5 + 5 \xrightarrow{R} 6 + 6 \xrightarrow{M^2} 7 + 8 \xrightarrow{R} 7 \\ & + 9 \xrightarrow{M} 8 + 10 \xrightarrow{L} 9 + 10 \xrightarrow{M} 11 + 12 \xrightarrow{R} 11 + 13 \xrightarrow{M} 12 + 14 \xrightarrow{L} 13 + 14 \xrightarrow{M} 15 \\ & + 16 \xrightarrow{R} 15 + 17 \xrightarrow{M^2} 16 + 18 \xrightarrow{R} 17 + 19 \xrightarrow{M} 18 + 20 \xrightarrow{L} 19 + 20 \xrightarrow{M} 21 + 21 \xrightarrow{R} 22. \end{aligned} \tag{A.1}$$

No arrow ends where an arrow with the same face label begins, so $\delta_{N_5}^2 = 0$. The complex is a single arc with ends 0, at $(0, 0)$, and 22, at $(\pi, 0)$. The encoding of a numerical trace of Smith's perturbed curve at $q = 5$, computed as in Remark 6.15, is strictly isomorphic to N_5 . Its pairings with $E_{-1/2}$ and $E_{-3/4}$ are 5 and 15; the first agrees with Smith's rank 5 for this decomposition [Smi24, Thm. 5.9].

The smoothing used in Theorem 7.1 is the four-arrow element

$$b_5 = 8 \xrightarrow{R} 7 + 16 \xrightarrow{R} 15 + 8 \xrightarrow{R} 15 + 16 \xrightarrow{R} 7, \tag{A.2}$$

which exchanges the targets of the arrows $8 \xrightarrow{R} 7$ and $16 \xrightarrow{R} 15$. The complex (N_5, b_5) consists of the arc $0, 1, \dots, 7, 16, 17, \dots, 22$ and the closed component $8, 9, \dots, 15$, whose word $9 \xrightarrow{M} 8, 10 \xrightarrow{L} 9, 10 \xrightarrow{M} 11, 12 \xrightarrow{R} 11, 13 \xrightarrow{M} 12, 14 \xrightarrow{L} 13, 14 \xrightarrow{M} 15, 8 \xrightarrow{R} 15$ is the doubled word of c . Adding $\kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11$ opens the closed component between 11 and 12 and attaches 11 to the end 0; the result is the single arc $22, 21, \dots, 16, 7, 6, \dots, 0, 11, 10, 9, 8, 15, 14, 13, 12$. Its pairings with $E_{-1/2}$ and $E_{-3/4}$ are 7 and 21, the ranks of reduced Khovanov homology of $P(-2, 3, 5)$ and $K_{-3/4}(5)$.

A.2. Corner terms. The search of Remark 7.2 finds the following one-arrow replacements carrying (N_q, b) to a complex strictly isomorphic to BN_q , written as elements of $\text{End}(N_q)$ (the first term is the deleted arrow):

object	first solution	second solution
$(N_7, b_{S_{18}})$	$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19$	$22 \xrightarrow{R} 3 + 36 \xrightarrow{R} 3$
$(N_7, b_{S_{25}})$	$\kappa_7 = 0 \xrightarrow{R} 19 + 36 \xrightarrow{R} 19$	$4 \xrightarrow{R} 3 + 36 \xrightarrow{R} 3$
(N_5, b_5)	$\kappa_5 = 12 \xrightarrow{R} 11 + 0 \xrightarrow{R} 11$	$8 \xrightarrow{R} 15 + 0 \xrightarrow{R} 15$

The arrows $22 \xrightarrow{R} 3$ and $8 \xrightarrow{R} 15$ are arrows of $b_{S_{18}}$ and b_5 rather than of δ_{N_q} , so the second solutions in the first and third rows delete an arrow of the smoothing. The term κ_7 is the only one that serves both smoothings at $q = 7$. The analogous search with $\widehat{\text{BN}}(T_q)$ in place of BN_q , which is the comparison for $t > 0$ (Remark 7.3), finds for both smoothings at $q = 7$ the terms $2 \xrightarrow{M} 3 + 18 \xrightarrow{M^2} 3$ and $20 \xrightarrow{M} 19 + 18 \xrightarrow{M^2} 19$, and at $q = 5$ the terms $10 \xrightarrow{M} 11 + 22 \xrightarrow{M^2} 11$ and $14 \xrightarrow{M} 15 + 22 \xrightarrow{M^2} 15$. The exact edit distance from N_q to BN_q is 2 for $q = 5, 7$, with two minimal edits in each case; at $q = 7$ both consist of arrows labelled R and touch the end 36.

A.3. Methods. All statements of Sections 6 and 7 that rest on computation are finite computations in exact arithmetic over $\mathbb{F}_2$, with the exceptions named below.

Morphism homology. Morphism complexes in $\text{Tw}(\mathcal{B})$ are computed in full, without truncating word length, by the mapping-cone reduction of Lemma 6.5, whose torsion condition is checked for each pair. A second, independent implementation computes $H\text{Mor}(X, Y)$ from the structure of $\text{Mor}(X, Y)$ as a module over the polynomial ring $\mathbb{F}_2[H]$ of [KWZ19]; it reproduces the dimensions of Propositions 6.7 and 6.8, and the dimensions of Section 7 were computed with it.

Encodings. Curves given by their lifts to the planar cover are encoded as twisted complexes by recording their crossings with the lifts of the arc system and the faces between consecutive crossings, as in §6.3. This encoder was compared with the invariants of 216 rational tangles computed by Zibrowius’s program [Zib]: in every case $\widehat{\text{BN}}(Q)$ is the straight lattice arc and $\widetilde{Kh}(Q)$ the figure-eight around it. The invariants $\widehat{\text{BN}}(T_q)$ and $\widetilde{Kh}(T_q)$ for $q = 3, 5, \dots, 21$ were computed with the same program, and the ranks $\text{rank } \widetilde{Kh}(K_r(q); \mathbb{F}_2)$ with Khoca 1.5 [Lew26].

Traced curves. The perturbed curves of Smith for $q = 5, 7, 11, 13, 17, 19$ were traced numerically from the equations of [Smi24, Prop. 4.1], composed with the lines of slopes $1/3$ and $1/q$ through $(0, 0)$, for several perturbation parameters in $[0.005, 0.02]$ and several step sizes, and transported to negative perturbation parameter by the symmetry $\mathfrak{s}$ of Remark 6.2. For $q \leq 13$ all traces at a given q have strictly isomorphic encodings, isomorphic to N_q . These are floating-point computations and serve as corroboration only.

Searches. Strict isomorphism of loop-type complexes is decided by comparing the words of the components up to rotation and reversal. Edit distances are computed by an exact search over the ways of cutting the word of one complex into intervals that reassemble to the word of the other; for $q = 17, 19$ the search excludes edit distances at most 5 and was stopped at 6. Chains of algebraic switches are enumerated breadth first, up to strict isomorphism, from both ends, and the two sides are compared. Each search is exhaustive within the class it states, and no conclusion is drawn beyond that class.

The sheared curves. The proofs in Sections 4 and 5 do not use computation. The following computations corroborate them.

- (i) *Lemma 4.7.* Each displayed identity in Steps 1–7 of its proof holds identically in the indeterminates v and ζ , as verified by symbolic computation. Without the assumption $E = 0$ one has $\mathbf{Vec} \rho(g) \cdot \mathbf{Vec} \rho(a) = \frac{1}{2} \zeta \sin 2v E^2$ and $\text{tr}(\rho(g)\rho(a)) = -2\zeta \sin 2v E^2$. In the Fricke coordinates $x = \text{tr} \rho(A)$, $y = \text{tr} \rho(B)$, $z = \text{tr} \rho(AB)$, the condition $\text{tr} \rho(b) = 0$

reads $x = yz$; then $E = y^2 + z^2 - 3$,

$$\mathrm{tr} \rho(a) = -yE, \quad \mathrm{tr} \rho(g) = -zE, \quad \mathrm{tr} \rho(ga) = yzE^2,$$

and $\mathrm{tr} \rho(w) - 2$ lies in the ideal generated by $x - yz$ and E . For the word $(ba)^{-2}c(ba)^2a$ the same reduction leaves $-y^2 - 1$, so that word is not trivial on C_0 . At three points of C_0 with algebraic coordinates, exact quaternion arithmetic gives $\rho(w) = 1$, and a numerical trace of $\Pi(C_0)$ at 50 significant digits satisfies $|\theta - 6\gamma + \pi| \leq 1.4 \cdot 10^{-47}$.

- (ii) *Combinatorics.* For every odd $q \leq 401$ we verified in exact rational arithmetic: the crossings of the sheared polygons of Corollary 4.8 with the strands, for both resolutions; the value of N_q ; the order of the crossing abscissae in $(\frac{\pi}{6}, \frac{5\pi}{6})$ and the absence of a crossing at $\gamma = \frac{\pi}{2}$; the bookkeeping of the graded homology in Theorem 4.18; the count of Remark 4.19; and the number of generators on the remnant circle.
- (iii) *Groups.* Besides the reduced-word check of Lemma 4.2 recorded after its proof, the boundary relation $ba = cd$ holds for the words (4.2), and for all $4 \leq n \leq 26$ prime to 3 and $m \leq 6$ the Alexander polynomial obtained by Fox calculus from the presentation of Lemma 4.1 with $h = -m$ equals that of the closure of $(\sigma_1\sigma_2)^n\sigma_1^{2m}$ computed by the Burau representation. The closed form of the Alexander polynomials in Theorem 1.3 was checked as an identity of rational functions and exactly for 1092 knots, and the determinants for $n \leq 50$ and $m \leq 20$.
- (iv) *The shear.* Applying t_e^h to the boundary values of points of the traced variety W_ε reproduces S_h to within 10^{-12} for $h = -6, \dots, 2$, whereas the shear of the opposite sign is off by about π .
- (v) *Numerical Floer complexes.* The variety W_ε is traced with the continuation code of [Wue26c]. Independent code applies S_{-m} , locates the generators on the strands, computes their degrees from (2.3), and enumerates the bigons as embedded Jordan domains in $\mathbb{R}^2$. Maslov indices are computed on the unsheared curve against the line field spanned by $(1, 6 - q)$, which by Lemma 4.6(f) gives the index of the sheared curve against ℓ_1 ; an index obtained by unwrapping tangent angles along the sheared polygonal curve is unreliable near the junctions when ε is small or q is large. The results, including the scan of the resolution type in 24 directions, are listed in Remark 4.21. For the twisted torus knots with $n = 7, 8, 13$ and $m = 1, 2, 3$, at two perturbations each, the graded homology agrees with Theorem 1.3 in all 18 runs.

A.4. Reproducibility. The programs that carry out the computations of this paper are included as ancillary files with its arXiv submission, and they are also available in the public repository

<https://github.com/bwuebben/pillowcase-bounding-cochain>.

In the ancillary files the programs for Section 6 are in the directory `pillowcase/`, and those for Sections 4, 5 and 7 and for this appendix are in `code/`; in the repository these are the directories `pillowcase/` and `paper2/code/`. Each program prints every number that the paper states next to the computed value and reports whether the two agree, and the file `README.md` in each directory lists the statements that each program checks. The programs for Sections 6 and 7 use only the Python standard library; those for Sections 4 and 5 also use `numpy`, `sympy` and `mpmath`. The invariants computed with Zibrowius's program [Zib] and the ranks computed with Khoca [Lew26] are included as data, together with their input files, so the checks run without these programs. The instanton dimensions used throughout are Theorem 1.1 and Proposition 6.8, and they are independent of all of this code.

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BERND JOHANNES WUEBBEN, NEW YORK, NY, wuebben@gmail.com